Continuous Hidden Markov Models for Equity Returns: Spectral Rank, Heavy-Tail Emission Families, and Regime-Conditional Value-at-Risk

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May 2, 2026

Abstract

The classical Rydén et al. [1] verdict, that hidden Markov models with $K = 2$–3 Gaussian states cannot reproduce the slow autocorrelation of absolute equity returns, has shaped two decades of regime-switching work toward semi-Markov sojourn extensions. We provide empirical evidence that this verdict is a marginal-mixture limit, not a temporal one: applying the textbook bilinear identity for the absolute-return ACF over the non-unit eigenvalues of the transition matrix and recasting the failure as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$, an effective-rank diagnostic on the fitted $\hat{\mathbf{T}}$ shows the implied $K - 1$ rank bound is non-binding at $K \geq 3$ at the cross-ticker median.

A unified ECM scaffold supports four emission families (Gaussian, Student-t, Laplace, Generalised Error Distribution) that share identical forward-backward recursions and quantile-based initialisation; the M-step is the only architectural difference. The empirical study covers SPY 2014–2026, a sector-balanced 30-ticker panel, a CRSP cross-decade slice (1994–2006), and a six-asset US-equity copula extension. Headline operating result: a regime-conditional Value-at-Risk that propagates the one-step-ahead state forecast through the predictive mixture passes the Christoffersen joint conditional-coverage test cleanly across emission families on the headline OoS window, and on 19/24 rows of a six-fold rolling-origin walk-forward, with rejections concentrated on out-of-distribution stress folds.

Positioning is honest: the i.i.d. bootstrap and an ML hidden semi-Markov model both exceed the CHMM on raw single-window OoS Kolmogorov–Smirnov; the CHMM's value-add is the structural use cases that those alternatives cannot serve (regime-conditional VaR, multi-asset copula composition, parametric synthetic-data privacy), and a unified emission-family ablation that the literature lacks. The empirical scope is the three symmetric Cont (2001) stylized facts on daily US-equity returns under stationary OoS; a non-equity stress test (GLD, SLV) collapses under static fitting, and periodic refit is the deployment recipe.

Keywords: Continuous Hidden Markov Model, Baum-Welch, Student-t Emissions, Stylized Facts, Volatility Clustering, Value-at-Risk, Synthetic-Data Generator.

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1 Introduction

Synthetic generators of equity-return time series support stress testing, regulatory backtesting, scenario design, and the training of downstream machine-learning models on data that cannot be released for privacy or licensing reasons [2–4]. The bar for a useful generator is that it reproduce, simultaneously, the canonical stylized facts of daily returns: heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the autocorrelation function (ACF) of absolute returns [5, 6]. Generalized autoregressive conditional heteroskedasticity (GARCH) models [7, 8] capture volatility clustering but not the multi-regime structure of the marginal; i.i.d. generators match the marginal but destroy temporal persistence; deep generators [9–11] reproduce some stylized facts at the cost of interpretability and reproducibility.

An influential negative result has shaped two decades of work on the regime-switching route. Rydén et al. [1] fitted Gaussian-emission hidden Markov models (HMMs) at $K = 2$–3 states and concluded that the geometric sojourn-time distribution of a standard Markov chain produces exponential ACF decay too fast to match the observed slow decay; Bulla and Bulla [12] restored ACF fidelity by replacing the Markov chain with a hidden semi-Markov chain at the cost of substantially increased complexity.

We identify the binding axis. The empirical Rydén et al. low-K failure compounds two constraints: a temporal one (the absolute-return ACF must admit slow-decay modes) and a distributional one (the marginal mixture must carry enough components to match observed kurtosis). We apply the closed-form mixture-of-eigenvalues identity $\rho_{|G|}(\tau) = \sum_{k\geq 2} c_k \lambda_k^\tau / \sigma^2_{|G|}$ over the non-unit eigenvalues of $\mathbf{T}$ [13–15] (restated in Section 3.8). The identity gives an algebraic upper bound of $K-1$ on the number of decay modes through the rank of $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$. An empirical effective-rank diagnostic on the fitted $\hat{\mathbf{T}}$ (Appendix A.6.4) shows this bound is non-binding at $K \geq 3$ at the cross-ticker median on equity-return data: across a 30-ticker panel the dominant-mode share has median 75.6% with interquartile range $[66\%, 86\%]$ and minimum 32.6% (NEM); the SPY value at $K = 18$ is 93.6%, in the right tail of this distribution. We therefore present the rank-non-binding claim at the cross-ticker median rather than as a universal property; SPY's lag-1 ACF concentration on a single mode is an asset-specific extreme rather than a generic feature of equity-return data. The binding axis at $K \geq 3$ is therefore the marginal mixture, whose representational capacity must match the empirical kurtosis target. At $K = 2$ both axes bind simultaneously (mono-exponential ACF that can be tuned, two-component marginal that cannot), and the empirical failure is on the distributional axis; at $K \geq 3$ the temporal axis decouples and the marginal-mixture component count drives the joint fit. The semi-Markov route adds an explicit sojourn distribution to relax the temporal axis, but on equity-return data the temporal axis is already non-binding under standard Baum-Welch at $K \geq 3$, so the operationally informative comparison between the two scaffolds runs on the distributional axis, where the four CHMM emission families and the explicit-duration HSMM at $K^\star = 3$ trade KS, kurtosis, and ACF fidelity differently (Section 4).

We instantiate this on ten years of daily SPY ETF data with a 2.5-year held-out window, using a continuous HMM trained by Baum-Welch [16, 17] under quantile-based initialization in four emission variants sharing identical forward-backward recursions: CHMM-N (Gaussian), CHMM-t (per-state Student-t with ν_k updated by ECM golden-section in the tradition of 18, 19), CHMM-L (Laplace, weighted median and weighted MAD in closed form), and CHMM-GED (Generalized Error Distribution with per-state shape p_k via three-stage ECM, nesting CHMM-N at $p_k = 2$ and CHMM-L at $p_k = 1$). The empirical scope is daily US-equity stylized-fact reproduction: SPY headline, a sector-balanced 30-ticker generalisation panel, and a six-asset US-equity copula extension. A non-equity stress test on GLD and SLV ETFs (Appendix A.8.6) collapses to 0% OoS KS under static IS-fitted CHMM, so the scope claim does not extend to other asset classes without periodic refit; we report this as a documented limitation rather than a deferred follow-up.

The body headline is $K^\star = 3$, the state-resolution-robust default selected by held-out per-observation log-likelihood and held-out KS on the strictly pre-2020 slice under both single-fold and four-fold / six-fold rolling-origin CV (Appendix A.9: the k-fold mean of $K^\star = 6$ vs $K^\star = 3$ has $|z| \leq 0.07$ on held-out per-observation log-likelihood with the sign flipping between fold designs, indistinguishable from sampling noise). At this operating point the four emission variants attain 79.8–90.6% IS / 63.1–83.2% OoS KS on the single OoS window, with CHMM-N at 89.7% IS / 80.5% OoS, simulated kurtosis 3.83/3.53, and $|G_t|$ ACF-MAE 0.0460; the penalised CHMM-t at $\lambda = 20$ provides the cleanest IS / OoS heavy-tail match in the headline block at 14.91/8.50 against observed 7.68/5.29 (the IS overshoot is the per-state ν_k ECM lower-bracket pinning artefact at low K, discussed in Section 5). A $K^\star = 6$ block is reported as a sensitivity reference (single-fold pre-2020 selection; same KS / kurtosis trade-off as the headline at one state-resolution step up). The single OoS window sits at the upper end of a six-fold rolling-origin walk-forward whose median CHMM-N OoS KS is 67.7% at $K = 18$ and 62.1% at $K^\star = 3$ (Table 38), with two stress folds (W2 COVID 2020 and W4 2022 rate-hike onset) below 10% that every generator in the panel rejects; the walk-forward distribution, not the single-window pair, is the operationally informative summary for production deployment. An extended-state-resolution sensitivity reference at $K = 18$ lifts CHMM-N from 89.7/80.5% to 94.1/81.8% IS / OoS KS (and the family-wide range to 93.6–95.6% IS / 80.8–85.7% OoS) at the cost of moving from a strictly held-out criterion onto a multi-objective rule. Against a benchmark panel (i.i.d. bootstrap of [20]; Gaussian and Laplace i.i.d.; GARCH(1,1) of [8] and the heavy-tailed extension of [21]; MS-GARCH of [22]), the four CHMM variants at $K = 18$ attain 93.6–95.6% IS and 80.8–85.7% OoS KS, with CHMM-N's $|G_t|$ ACF-MAE of 0.0509 matching GARCH(1,1)'s 0.0485 to within 0.003 (Table 3). The variants land at four kurtosis points (CHMM-N 5.0, CHMM-L 6.6, CHMM-GED 5.2 with a bimodal Gaussian-bulk / Laplace-tail $\hat{p}_k$ partition, penalised CHMM-t at $\lambda = 20$ at 8.56 IS / 7.07 OoS as the cleanest single-row joint match to observed 7.68 / 5.29; the unpenalised CHMM-t IS kurtosis of 14.4 is the per-state ν_k ECM lower-bracket pinning artefact discussed in Section 5). VaR and Expected-Shortfall envelopes bracket observed values at 1% and 5% on both windows.

Multi-asset synthesis on a US-equity universe composes through a rank-based Student-t copula on the CHMM marginals: profile MLE selects $\nu^\star = 6$ at IS off-diagonal correlation MAE 0.027, but on OoS the Gaussian and Student-t copulas are statistically indistinguishable (0.202 versus 0.209, both above the 0.030–0.027 IS spread by an order of magnitude). The dependence-family choice is therefore an IS-calibration distinction, not an OoS-deployment one; quarterly rolling refit reduces the Student-t OoS off-diagonal MAE to 0.185. On the sector-balanced 30-ticker panel (the sample design is 10 GICS sectors $\times 3$ large-cap representatives, itself one design choice among many) the IS distribution is tight (median KS 99.5%) and the OoS distribution is sector-stratified (Utilities, Information Technology, and Consumer Staples cluster above 90% median; Health Care collapses to 14.5% on weight-loss-drug regime introductions in LLY and 2024 healthcare-policy compression in UNH; 11/30 tickers below 60% OoS KS at the universe scale), reflecting the well-known stationarity-scope limit on financial time series [23–25] and motivating periodic refit at deployment [26].

Our contributions are empirical and architectural. (i) We provide empirical evidence that the algebraic $K-1$ rank bound on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ implied by the bilinear identity $\rho_{|G|}(\tau) = \sum_{k\geq 2} c_k \lambda_k^\tau / \sigma^2_{|G|}$ is non-binding at $K \geq 3$ on equity-return data: an effective-rank diagnostic on the fitted $\hat{\mathbf{T}}$ shows a single non-unit eigenvalue carries 93.6% of the lag-1 absolute-return ACF at $K = 18$ on SPY (Appendix A.6.4); on the 30-ticker cross-section the dominant-mode share has median 0.76 and interquartile range $[0.66, 0.86]$ (Appendix A.6.5). The recasting reframes the empirical Rydén et al. [1] low-K failure as a marginal-mixture limit, not a temporal one. (ii) A unified ECM scaffold for four emission families (Gaussian, Student-t, Laplace, Generalized Error Distribution) under shared forward-backward recursions and quantile-based initialisation, where the M-step is the only architectural difference and CHMM-GED's per-state shape p_k adapts each state on the Gaussian-to-Laplace axis. The four-family ablation under one scaffold, with consistent operating points and held-out validation, is not in the prior CHMM-on-finance literature. (iii) A regime-conditional Value-at-Risk construction that propagates the one-step-ahead state forecast through the predictive mixture and passes Christoffersen-cc cleanly across emission families on OoS, with a Monte Carlo power calibration and an Engle–Manganelli DQ backstop; this is the use case that no exchangeable bootstrap or unconditional GARCH benchmark can serve. (iv) An empirical scope that exercises generalisation along three orthogonal axes: a six-fold rolling-origin walk-forward on SPY 2014–2026, a sector-balanced 30-ticker panel, a CRSP cross-decade transfer 1994–2006, and a six-asset US-equity Student-t copula extension; the headline-OoS-KS rankings are walk-forward-fragile and the operationally informative metric is the walk-forward median.

The paper is organized as follows. Section 2 reviews related work; Section 3 describes the data, the continuous HMM, the EM estimator, and the spectral mechanism for the absolute-return ACF; Section 4 presents the empirical study and the VaR / ES envelope calibration; Sections 5–6 discuss and conclude.

2 Related Work

Hidden Markov models in finance. Hamilton [27] introduced Markov-switching to economics; Schaller and Van Norden [28] estimated regime-switching specifications on US monthly stock returns and documented the predictive content of the latent regime for return forecasting. The seminal evaluation against the Cont stylized facts is due to Rydén et al. [1]: Gaussian-emission HMMs at $K = 2$–3 states fail the slow ACF decay of absolute returns because the geometric sojourn-time distribution of a standard Markov chain produces exponential decay too fast to match the observed signature. Bulla and Bulla [12] restored fidelity with hidden semi-Markov models at substantially higher complexity. Nystrup et al. [29] extends the same line to long-memory volatility. Maximum-likelihood fitting of heavy-tailed innovations follows Peel and McLachlan [18] and Liu and Rubin [19]; the Generalized Error Distribution [30, 31] provides a continuous shape parameter that interpolates between Gaussian ($p = 2$) and Laplace ($p = 1$) and underlies our CHMM-GED variant in the same role that the per-state degrees of freedom ν_k plays in CHMM-t. The discrete-state predecessor of this paper, Alswaidan and Varner [4], reproduced the same three stylized facts via Laplace quantile binning plus a Poisson-jump duration mechanism; here we replace the discretization step with continuous emissions and recover the same target through the spectral mechanism of Section 3.8.

GARCH and its extensions. GARCH [7, 8] encodes volatility persistence through a single decay parameter but imposes a unimodal innovation distribution and fails distributional Kolmogorov–Smirnov on equity returns. Heavy-tailed innovations [21], asymmetric leverage extensions (EGARCH, 32; GJR-GARCH, 33; threshold GARCH, 34), and regime-switching extensions (MS-GARCH, 22, with the `MSGARCH` R package of 35 the standard reference implementation) close some gaps but inherit the unimodal-innovation constraint. Realised-variance frameworks [36, 37] provide a complementary forecasting baseline.

Deep generative models for financial time series. Generative adversarial network (GAN)-based generators reproduce the visual appearance of return series but typically fail volatility-clustering tests unless architectures encode temporal dependence explicitly [38, 39]; the TimeGAN family of Yoon et al. [9], the convolutional Wasserstein GAN (WGAN) QuantGAN of Wiese et al. [10], and the autoregressive-diffusion approach of Rasul et al. [11] are the standard references. We include QuantGAN as the deep-generative row in our extended panel (Appendix A.7.3); the present paper's contribution is grounded against parametric and non-parametric baselines.

Cross-asset dependence and synthetic-data evaluation. For multi-asset synthesis, the Single Index Model [40] propagates a market factor through a rank-one linear decomposition but distorts each non-market marginal; Sklar's theorem [41, 42] supports a cleaner decomposition in which each asset's fitted marginal is preserved and only the copula is estimated, with Gaussian and Student-t copulas [43, 44] the standard choices and rank-reordering simulation [45] the standard injection. Dynamic-conditional-correlation models [46] and composite-likelihood estimators for vast covariance matrices [47] are the standard alternatives at higher d; the present paper's six-asset construction sits well below the dimensions where those estimators become essential, but the rank-based copula architecture composes with either at scale. Synthetic-data evaluation is anchored on the framework of Stenger et al. [48], the proper-scoring-rule literature [49–51], kernel two-sample tests [52], and signature-based metrics [53, 54].

State-space stochastic volatility. Stochastic-volatility (SV) models [55] are the continuous-latent counterpart of the discrete-latent CHMM: log-volatility evolves as a latent AR(1) and likelihood inference proceeds via simulation-based estimators [56, 57] or hidden-Markov approximations on a discretised volatility grid [58, 59]. Markov-switching SV [60] sits between the two formulations. We include a Markov-switching SV row in the extended baseline panel (Appendix A.7.4) for direct comparison; the architectural difference is that the CHMM treats the latent regime as a finite-state object with K free per-state densities, whereas SV ties all states through a single AR(1) on a one-dimensional log-volatility latent. The former gives more representational capacity at the marginal level (the kurtosis points of Table 9); the latter gives a strictly stationary latent process by construction.

Regime breaks and online change-point detection. The W2 (COVID 2020) and W4 (2022 rate-hike onset) walk-forward stress folds (Section 4) are regime introductions that the IS distribution does not span: any IS-fitted symmetric-emission CHMM rejects them by KS, and faster refit cadences do not close the gap (Appendix A.6.13). The natural complements are Bayesian online change-point detection [61] and the particle-filter recursion for multiple change-points of Fearnhead and Liu [62], both of which model the run-length distribution of an underlying segmentation explicitly and would allow the transition matrix or per-state densities to refit at detected break points rather than on a fixed quarterly cadence. We do not implement this here; it is one of the companion-paper directions discussed in Section 5.

Signature- and diffusion-based market generators. A second wave of deep market generators sits beyond the GAN-based panel surveyed above. Path-signature methods [53] encode an entire return path as a graded tensor of iterated integrals; Ni et al. [63], Bühler et al. [64], and Boursin et al. [65] use the signature either as a moment-matching loss or as the conditioning variable for a generative model and report tighter rough-path statistics than convolutional baselines on equity tape. Score-based and diffusion-based generators [11, 66–68] replace adversarial training with a denoising objective that is more stable on heavy-tailed marginals. The two classes share the deep-generative tradeoff against the present paper: tighter visual fidelity to the empirical tape, opaque per-state semantics, and no parametric copula handle for multi-asset composition.

3 Methods

3.1 Data and Excess Growth Rates

The single-asset analysis uses daily SPDR S&P 500 ETF (SPY) prices from January 3, 2014 to January 3, 2024 (a ten-year training window, $T_{\mathrm{IS}} = 2{,}516$), with the period from January 4, 2024 through April 20, 2026 ($T_{\mathrm{OoS}} = 572$) held out. Prices are split-adjusted but not dividend-adjusted; the data sources are described in the acknowledgments. For ticker i and trading day j the annualised excess growth rate is

$$G_{i,j} \equiv \frac{1}{\Delta t} \ln\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f, \qquad \Delta t = 1/252, \quad r_f = 0, \tag{1}$$

with $P_{i,j}$ the session volume-weighted average price (VWAP). The factor $1/\Delta t$ is a constant rescaling; all simulations and metrics are evaluated on the daily-scale series. The empirical study is organised along two pipelines (step-by-step summary in Appendix A.2.4). *Pipeline A* fits a CHMM to each ticker independently and evaluates single-asset metrics; *Pipeline B* reuses the per-asset Pipeline-A fits as marginals and injects cross-asset dependence through a rank-based Student-t copula (§3.4).

3.2 Continuous Hidden Markov Model

We model G_t as a continuous HMM $\mathcal{M} = (\mathcal{S}, \mathbf{T}, \mathbf{F}, \boldsymbol{\pi})$ with K latent states, per-state emission density $f_k(x; \boldsymbol{\theta}_k)$ (with corresponding per-state CDF $F_k(x; \boldsymbol{\theta}_k) = \int_{-\infty}^{x} f_k(u; \boldsymbol{\theta}_k)\, du$, used in the regime-conditional VaR construction of §4.6), transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$, and initial distribution $\boldsymbol{\pi}$ [17]. The stationary marginal is the K-component mixture

$$f(x) = \sum_{k=1}^{K} \bar{\pi}_k\, f_k(x; \boldsymbol{\theta}_k), \qquad \bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}}\mathbf{T}, \tag{2}$$

with $\bar{\boldsymbol{\pi}}$ the stationary distribution. Existence and uniqueness require irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1 in §3.8). We learn $\mathbf{T}$, $\boldsymbol{\theta}_{1:K}$, and $\boldsymbol{\pi}$ jointly by EM (§3).

The four emission families share a single forward-backward scaffold and quantile-based initialization; the M-step is the only architectural difference (Figure 1). *CHMM-N*: $f_k = \mathcal{N}(\mu_k, \sigma_k^2)$. *CHMM-t*: $f_k = t_{\nu_k}(\mu_k, \sigma_k)$ with per-state degrees of freedom $\nu_k \in [\nu_{\min}, \nu_{\max}]$. *CHMM-L*: $f_k = \mathrm{Laplace}(\mu_k, \beta_k)$ with per-state Laplace scale denoted β_k to avoid collision with the canonical Laplace-scale symbol b. *CHMM-GED*: $f_k = \mathrm{GED}(\mu_k, \alpha_k, p_k)$ with per-state shape $p_k \in [p_{\min}, p_{\max}]$ that nests Gaussian ($p_k = 2$) and Laplace ($p_k = 1$).

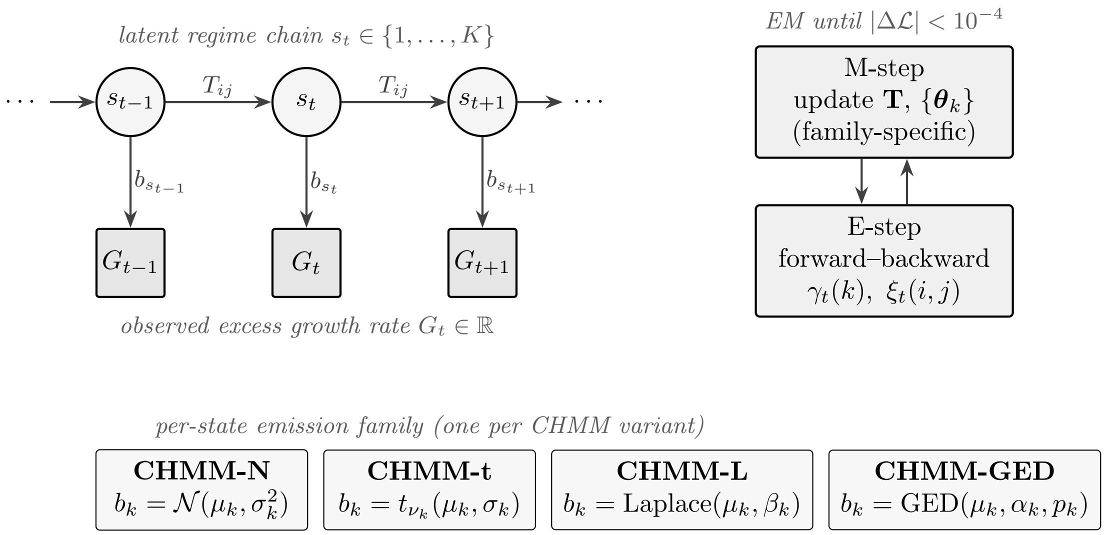

Figure 1. CHMM architecture and training loop. A K-state latent Markov chain with transition matrix $\mathbf{T}$ generates the regime sequence s_t; each s_t emits the observed excess growth rate G_t through a state-specific density b_{s_t}. The four CHMM variants share everything except the emission family. EM alternates an E-step (log-space forward–backward) and a family-specific M-step until $|\Delta\mathcal{L}| < 10^{-4}$.

3.3 Benchmark Generators

We benchmark against representatives from each generator tier, all fit on the same IS window: i.i.d. bootstrap [20] (non-parametric distributional ceiling), Gaussian and Laplace i.i.d., GARCH(1,1) Gaussian [8] and Student-t [21], and Markov-switching GARCH (MS-GARCH; 22). The extended GARCH-family panel (EGARCH, GJR-GARCH, HAR-RV, MS-GARCH at $K \in \{3, 4, 6\}$), an explicit-duration semi-Markov foil, a deep-generative QuantGAN [10] row, plus stochastic-volatility (Taylor-Harvey-Ruiz-Shephard), Markov-switching multifractal [69], and Merton-jump-diffusion [70] baselines are reported in Appendix A.7.4 and A.7.5.

3.4 Cross-Asset Dependence (Pipeline B)

Pipeline B reuses each per-asset CHMM marginal and injects cross-asset dependence through a rank-based Student-t copula [43, 44]. Pseudo-uniform observations $u_{j,t} = \text{rank}(g_{j,t})/(T+1)$ feed an analytic Kendall's-τ correlation estimate $\rho_{ij} = \sin(\pi\tau_{ij}/2)$ [71]; profile MLE selects $\nu^\star$. Sklar's theorem [41] guarantees that the rank-reordering simulator of Iman and Conover [45] preserves each fitted CHMM marginal exactly while coupling the asset ranks to the copula sample. SIM, Gaussian-copula, and truncated C-vine comparators are in Appendix A.8.1.

3.5 Evaluation Protocol

For each model we simulate $P = 1{,}000$ independent paths of length T following [48], under a deterministic global seed root 20260420. Headline metrics are the two-sample Kolmogorov–Smirnov [72, 73] pass rate at $\alpha = 0.05$, mean simulated excess kurtosis, and the absolute-return ACF mean absolute error (MAE) over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L}\sum_{\tau=1}^{L}\left|\hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau)\right|. \tag{3}$$

A separate Value-at-Risk panel (§4.6) carries the unconditional Kupiec [74] and Expected-Shortfall envelope diagnostics. Block-bootstrap KS recalibration, Anderson–Darling, W_1, Hellinger, and CRPS auxiliaries are in Appendix A.3. Estimation of $(\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ proceeds by EM in log-space. This section gives the shared scaffold and the family-specific M-step updates; the state-count selection that operationalises this scaffold is reported in Section 4.

3.6 Shared EM Scaffold

All four emission families share the same log-space forward-backward recursions and the same quantile-based initialization [75]: observations are sorted, partitioned into K equal-sized chunks, and each state's initial location and scale are seeded from the corresponding chunk; $T_{ij}^{(0)} = \pi_k^{(0)} = 1/K$. EM iterates to tolerance $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\texttt{max\_iter} = 60$; convergence is typically reached in 20–40 iterations. Quantile-based initialization is what makes Baum-Welch reach a non-degenerate $K \geq 3$ fit: random initialization standard in the 1990s often converges to overlapping local optima, which is one of the two compounding sources of the low-K failure observed by Rydén et al. [1] (the other is the spectral one of Section 3.8). Algorithmic details and the forward-backward recursion are in Appendix A.2.

3.7 Family-Specific M-Step Updates

CHMM-N (Gaussian). The classical Baum-Welch M-step [16] is closed form: $\mu_k \leftarrow \sum_t \gamma_t(k) O_t / \sum_t \gamma_t(k)$ and $\sigma_k^2 \leftarrow \sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)$.

CHMM-t (Student-t via ECM). We adopt the ECM formulation of Peel and McLachlan [18] and Liu and Rubin [19], the same heavy-tailed innovation choice studied by Abanto-Valle et al. [59] for stochastic-volatility-in-mean models. The latent precision

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2} \tag{4}$$

is treated as an augmented E-step quantity, and the M-step updates

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \qquad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2}{\sum_t \gamma_t(k)} \tag{5}$$

have closed form conditional on ν_k. The degrees of freedom ν_k are updated by a golden-section search maximizing the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. The two-block CM step preserves the EM monotonicity guarantee on the compact bracket, so the incomplete-data log-likelihood is non-decreasing across iterations.

CHMM-L (Laplace). The weighted maximum-likelihood estimators are in closed form: μ_k is the posterior-weighted median of the observations (weights $\gamma_t(k)$, computed by cumulative-weight bracketing with linear interpolation between neighbouring order statistics), and $\beta_k \leftarrow \sum_t \gamma_t(k) |O_t - \mu_k| / \sum_t \gamma_t(k)$.

CHMM-GED (Generalized Error Distribution via three-stage ECM). For $f_k(x) = \mathrm{GED}(x; \mu_k, \alpha_k, p_k)$ the M-step decomposes into three CM updates per state: a bracketed $\mu_k \leftarrow \arg\min_\mu \sum_t \gamma_t(k)\, |O_t - \mu|^{p_k}$ via golden-section over $[\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]$, then the closed-form scale $\alpha_k \leftarrow \left[(p_k/W_k) \sum_t \gamma_t(k)\, |O_t - \mu_k|^{p_k}\right]^{1/p_k}$ with $W_k = \sum_t \gamma_t(k)$, and finally a bracketed $p_k \leftarrow$

$\arg\max_{p\in[p_{\min},p_{\max}]} \sum_t \gamma_t(k) \log f_k(O_t;\mu_k,\alpha_k,p)$ via golden-section over $[p_{\min},p_{\max}] = [0.5, 3.0]$. Each CM step is a partial maximization, so EM monotonicity holds on the compact bracket. CHMM-N (all $p_k = 2$) and CHMM-L (all $p_k = 1$) are the boundary cases.

3.8 Spectral Mechanism for the Absolute-Return ACF

The closed-form bilinear identity for the absolute-return ACF of a stationary CHMM, $\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{T}^\tau\,\mathbf{m}$ with $\mathbf{m}_k = \mathbb{E}[|G_t| \mid s_t = k]$, and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are folklore in the regime-switching literature (13, §22.2; 14, Ch. 3; 15). Under irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1, $\bar{\boldsymbol{\pi}}$ unique stationary; 76) and finite per-state second moments with non-trivial location/scale contrast (Assumption 2), the spectral decomposition $\mathbf{T}^\tau = \mathbf{1}\bar{\boldsymbol{\pi}}^\top + \sum_{k=2}^K \lambda_k^\tau\,\mathbf{v}_k\mathbf{w}_k^\top$ subtracts the marginal product to give the ACF

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} w_k\,\lambda_k^\tau, \qquad w_k = \big(\mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{v}_k\big)\big(\mathbf{w}_k^\top \mathbf{m}\big)/\sigma^2_{|G|}. \tag{6}$$

Equation (6) is a real-coefficient sum of geometric decays at the non-unit eigenvalues of $\mathbf{T}$; complex-conjugate λ_k pairs combine into damped oscillatory modes and non-diagonalisable $\mathbf{T}$ adds polynomial-times-geometric prefactors via Jordan blocks. A self-contained derivation, the lag-zero behaviour, and the squared-return ACF are in Appendix A.5; the formal Assumptions 1–2, and the cited propositions on identifiability, MLE consistency, and ECM monotonicity, are in Appendix A.4. Our contribution on this axis is exclusively empirical: applying (6) to recast the Rydén et al. [1] low-K failure on the SPY 2014–2024 instance as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ and demonstrating empirically that the implied $K-1$ rank bound is non-binding at $K \geq 3$.

Rydén separation as a K-rank statement. At $K = 2$, equation (6) reduces to the mono-exponential form $\rho_{|G|}(\tau) = \lambda_2^\tau$ with $\lambda_2 = T_{11} + T_{22} - 1$; slow decay is matched by tuning $T_{11} + T_{22}$. The temporal axis is therefore satisfiable at $K = 2$; the binding axis is the two-component marginal, which cannot match the empirical kurtosis target. At $K \geq 3$ the matrix admits multiple non-unit eigenvalues and the marginal mixture has enough components to match the observed tail. Empirically, ACF-MAE is essentially flat across $K \in \{3, 6, \ldots, 21\}$ while KS rises (Section 4); a direct effective-rank diagnostic (Appendix A.6.4) shows a single non-unit eigenvalue carries 93.6% of the lag-1 ACF at $K = 18$ on SPY, so the K-rank upper bound is non-binding at $K \geq 3$ on this instance. Three modes carry $\geq 95\%$ of the lag-1 ACF and twelve modes carry $\geq 99\%$ (Table 1). Repeating the diagnostic on the 30-ticker cross-section (Appendix A.6.5) gives a wider distribution: cross-ticker median dominant-mode share 0.76, interquartile range $[0.66, 0.86]$, minimum 0.33 on NEM. The SPY value is therefore in the right tail of the cross-ticker distribution; the rank-non-binding claim is supported at the cross-ticker median rather than as a universal property of equity-return data.

Table 1. Leading non-unit-eigenvalue contributions to the lag-1 absolute-return ACF, CHMM-N on SPY IS at $K = 18$ (`results/diagnostics/spectral_rank.txt`). Modes ordered by $|w_k \lambda_k|$ descending. "cum lag-1" is the cumulative fraction of $\sum_{k=2}^{K} w_k \lambda_k$ accounted for through that rank. The dominant non-unit mode at $|\lambda_2| = 0.929$ carries 93.6% of the lag-1 ACF; the remainder is spread thinly across the lower-magnitude modes.

rank k	$\lvert\lambda_k\rvert$	$\lvert w_k\rvert$	$\lvert w_k \lambda_k\rvert$	cum. lag-1
2	0.929	0.324	0.301	0.936
3	0.335	0.012	0.0039	0.948
4	0.854	0.003	0.0022	0.955
5	0.231	0.008	0.0019	0.961
6	0.231	0.008	0.0019	0.967
$\vdots$				
13	0.119	0.006	0.0007	0.995

4 Empirical Study

Reminder. Subsections 4.1–4.4 and 4.6 operate under **Pipeline A** (single-asset CHMM fit per ticker, no cross-asset coupling); only §4.5 operates under **Pipeline B** (Pipeline A marginals plus a rank-based copula).

4.1 Descriptive Statistics and Stylized Facts

All three Cont stylized facts are present in both windows (Figure 4 in Appendix A.6.15): heavy-tailed marginal (IS excess kurtosis 7.68, Jarque–Bera 6,416.6, $p < 0.001$), negligible linear autocorrelation in raw G_t (Ljung–Box $\mathrm{LB}_G \approx 5.5$ at lag 20 vs. $\chi^2_{20}(0.05) = 31.4$), and persistent volatility clustering in $|G_t|$ ($\mathrm{LB}_{|G|} = 2{,}959.3$, $p < 0.001$). The OoS window shares the same qualitative profile (excess kurtosis 5.29). A stationary block bootstrap of Politis and Romano [20] at mean block length $L = 20$ gives 95% CIs of $[2.17, 12.40]$ on the IS excess kurtosis and $[0.90, 8.26]$ on the OoS excess kurtosis (Appendix A.6.8, Table 15); the IS and OoS CIs overlap broadly and the IS-vs-OoS kurtosis difference of ~ 2.4 units lies within sampling error at this block length, so kurtosis-fidelity targets reported throughout this paper should be read against the joint $[0.90, 12.40]$ envelope rather than against the 7.68/5.29 point estimates.

4.2 State-Count Selection

We sweep $K \in \{3, 6, 9, 12, 15, 18, 21\}$ for CHMM-N on the IS window and select $K^\star$ by held-out per-observation log-likelihood and held-out KS under rolling-origin cross-validation on the strictly pre-2020 slice (estimation 2014-01–2018-06, validation 2018-07–2019-12; four-fold and six-fold cadences; Appendix A.9). At both fold cadences the $K = 3$ vs $K = 6$ comparison sits well inside sampling noise ($|z| \le 0.07$ on mean held-out log-likelihood with the sign flipping between fold designs); under parsimony we take $\mathbf{K}^\star = \mathbf{3}$ as the body operating point. BIC concurs (Table 2). Higher state counts are decisively worse on held-out criteria, and a state-distinctness diagnostic (Appendix A.6.21) gives $K_{\text{eff}} \le 11$ at every $K \ge 12$. Absolute-return ACF-MAE is essentially flat (0.047–0.053) across the entire sweep, consistent with the spectral identity (6). Full information-criterion panel (AIC, HQC, CAIC, BIC), single-fold and HAC-adjusted standard errors, and the sensitivity panels at $K = 6$ and $K = 18$ are in Appendices A.9, A.9, and A.9.

Table 2. ***K*-selection on the strictly pre-2020 slice.** Held-out criteria under rolling-origin CV select $K^\star = 3$ at parsimony tiebreaker against $K^\star = 6$ ($|z| \leq 0.07$ on mean held-out log-likelihood, indistinguishable from sampling noise). BIC concurs. Full IC panel (AIC, HQC, CAIC, K_{eff}-corrected variants), single-fold variants, and HAC-adjusted standard errors are in Appendix A.9.

Criterion	$K^\star$
Held-out per-obs log-likelihood (rolling-origin CV)	**3**
Held-out KS (rolling-origin CV)	**3**
BIC	**3**

4.3 Six-Generator Comparison (Pipeline A)

Table 3 reports IS / OoS validation for the benchmark generators of §3.3 and the four CHMM emission variants at three operating points, against an explicit-duration semi-Markov reference.

Methods notes for Table 3. The 1-day OoS KS column is dominated by the i.i.d. bootstrap; the CHMM-vs-bootstrap differentiation is on the structural use cases of §4.6 (regime-conditional VaR) and §4.5 (multi-asset copula composition), not on per-day distributional fidelity. Bold within the $K^\star = 3$ block marks the cleanest IS / OoS row on each axis. Block-bootstrap KS recalibration, the extended GARCH-family / MS-GARCH-$K \in \{3, 4, 6\}$ panel, and the $K = 6$ / $K = 18$ sensitivity panels are in Appendix A.7.2, A.7.4, and A.6.

At the body operating point $K^\star = 3$, the four CHMM variants attain 79.8–90.6% IS / 63.1–83.2% OoS KS, with CHMM-N at 89.7% IS / 80.5% OoS KS, simulated kurtosis 3.83 IS / 3.53 OoS, and $|G_t|$ ACF-MAE 0.0460, on par with GARCH (0.0485). The penalised CHMM-t at $\lambda = 20$ provides the cleanest IS / OoS heavy-tail match in the panel at 14.91/8.50 against observed 7.68/5.29 (the aggregate IS overshoot reflects the bracket-pinning artefact of the per-state ν_k ECM at low K, discussed in §5; the per-path simulated IS kurtosis distribution has median 7.77 and 76.6% of paths inside the bootstrap CI on observed, so the cleanliness claim is per-path-median rather than aggregate-mean; Appendix A.6.8). CHMM-GED's per-state shape adapts to 5.48/5.12 as a moderate-tail compromise, and CHMM-L sits at 5.30/4.85 but loses ~ 10pp on OoS KS because at $K = 3$ the Laplace per-state shape cannot smooth the low-volatility regime as effectively as the Gaussian. The shared-ν Student-t row at 4.68/4.46 achieves the heavy-tail target without a penalty hyperparameter and is the structurally cleaner recipe (Appendix A.6.9). With $K - 1 = 2$ non-unit eigenvalues at $K^\star = 3$, the mixture-of-eigenvalues sum (6) reproduces the slow absolute-return ACF that the Rydén et al. [1] $K = 2$–3 Gaussian setup could not recover on the SPY 2014–2024 instance of that experiment (Appendix A.6.6). The raw-return ACF-MAE column closes the third stylized fact: every CHMM row sits at 0.0234–0.0240, indistinguishable from the i.i.d. baselines that have zero linear ACF by construction. CHMM-GED's per-state shape $\hat{p}_k$ partitions bimodally into a Gaussian-bulk / Laplace-tail structure that replicates across seeds and tickers (Appendix A.6.16). On the OoS sample-CRPS, the four CHMM variants are statistically indistinguishable on two-sided Diebold and Mariano [51] tests at $T_{\text{OoS}} = 572$ (Appendix A.6.22); the family choice is therefore driven by the per-row kurtosis match (Table 9, §5). The cross-generator ranking is also robust to the distributional-metric choice: replacing KS with Anderson–Darling, Wasserstein-1, or Hellinger distance preserves the same ordering of generators on the OoS axis (Appendix A.3). The MS-GARCH benchmark plateaus at 34–36% IS KS in our Nelder-Mead fit and at 0–0.1% IS / 5–6% OoS KS in the reference Bayesian re-run (Appendix A.7.4, Table 3 ¶rows), so the multi-state benefit is specific to the CHMM scaffold rather than to multi-state regime-switching per se on this dataset under either estimator. A six-fold rolling-origin walk-forward (Appendix A.9, Table 38; body summary:

Table 3. Headline generator comparison on SPY. 1,000 simulated paths, $\alpha = 0.05$; IS $T = 2{,}516$, OoS $T = 572$. KS = Kolmogorov–Smirnov pass rate; Kurt = mean simulated excess kurtosis (observed 7.68 IS, 5.29 OoS); ACF-MAE = mean absolute error of the absolute-return / raw-return ACF over 252 lags; CRPS = OoS sample continuous ranked probability score. ¶ Reference Bayesian re-run of MS-GARCH via the `MSGARCH` R package of Ardia et al. [35]; the lower KS pass rate reflects posterior-predictive integration of parameter uncertainty (Appendix A.7.4). § QuantGAN row is an in-house WGAN re-implementation (3-layer 1D-conv generator and critic, Wasserstein loss; full spec in Appendix A.7.3); the row reads as the panel's deep-generative negative control, and a faithful reference-implementation re-run of the seven-block TCN with Lambert-W pre-processing of Wiese et al. [10] is a deferred follow-up. §§ Shared-ν Student-t ablation: single ν across all K states by aggregate-Q ECM, no penalty; the structurally-cleaner heavy-tail recipe vs the per-state $\nu_k + \lambda = 20$ row above (Appendix A.6.9). $K = 6$ and $K = 18$ sensitivity panels (including unpenalised CHMM-t and bracket-lift ablations) are in Appendix A.6.

	KS (%) ↑		Exc. Kurt		ACF-MAE ↓		CRPS OoS
Model	IS	OoS	IS	OoS	$\lvert G_t\rvert$	G_t	↓
Observed	–	–	7.68	5.29	–	–	–
Bootstrap	99.7	92.1	7.24	6.81	0.0628	0.0235	1.0398
Gaussian i.i.d.	0.0	1.0	−0.01	−0.03	0.0627	0.0236	1.0611
Laplace i.i.d.	99.1	88.5	2.95	2.91	0.0628	0.0236	1.0386
GARCH(1,1)	23.4	60.8	7.06	2.96	0.0485	0.0243	1.0440
GARCH(1,1)-t	57.3	80.8	15.13	–	**0.0316**	**0.0173**	–
MS-GARCH ($K = 2$)	27.7	38.7	4.73	–	0.0367	0.0173	–
MS-GARCH ($K = 3$)	36.1	33.1	4.10	–	0.0284	0.0173	–
MS-GARCH ($K = 6$)	34.5	33.4	4.41	–	0.0429	0.0173	–
MS-GARCH ref. Bayesian ($K = 2$)¶	0.0	5.8	4.00	2.46	0.0465	0.0240	–
MS-GARCH ref. Bayesian ($K = 3$)¶	0.1	5.1	4.52	2.53	0.0433	0.0241	–
MS-GARCH ref. Bayesian ($K = 4$)¶	0.0	5.3	6.01	3.54	0.0446	0.0241	–
QuantGAN TCN§	0.0	0.0	0.56	0.53	0.0617	0.0264	–
Body operating point: $K^\star = 3$ (selected by held-out criteria under rolling-origin CV; §4.2).							
CHMM-N	89.7	80.5	3.83	3.53	0.0460	0.0240	1.0393
CHMM-t pen. ($\lambda = 20$)	90.6	**83.2**	**14.91**	**8.50**	0.0537	0.0235	**1.0373**
CHMM-L	79.8	63.1	5.30	4.85	0.0532	0.0236	1.0405
CHMM-GED	90.5	77.4	5.48	5.12	0.0526	0.0236	1.0389
CHMM-t shared-ν§§	91.9	82.1	4.68	4.46	0.0531	0.0235	–
ML HSMM-N	98.4	91.0	3.46	3.38	0.0629	0.0236	–

Table 4. Six-fold rolling-origin walk-forward summary (CHMM-N, 5y train / 1y test, 500 paths per fold). Pulled from the full panel in Appendix A.9, Table 38. The headline single-window OoS sits at the upper end of the fold-wise distribution; the W2 (COVID) and W4 (2022 rate-hike onset) folds are out-of-distribution by KS for every generator in the panel.

	median KS (%)	$[Q_1, Q_3]$ KS (%)	min KS (%)	folds < 10%
$K^\star = 3$	62.1	$[7.2, 78.4]$	0.8	2/6
$K = 18$ (sensitivity)	67.7	$[8.2, 75.0]$	0.0	2/6

Table 4) attains median OoS KS 62.1% at $K^\star = 3$; the single-window OoS sits at the upper end of this distribution, with two stress folds (W2 COVID, W4 2022 rate-hike onset) below 10% that every generator in the panel rejects. The walk-forward median, not the single-window OoS pair, is the operationally informative summary for production deployment: the headline single OoS window of 2024–2026 is a relatively low-stress slice and the single-window numbers therefore overstate what a deployed CHMM achieves on average across rolling windows.

Per-state interpretability. The fitted CHMM-N states at $K^\star = 3$ partition cleanly along the conventional volatility axis (per-state parameters from `CHMM-Model/results/SPY/K3/Emission-Parameters.txt`): state 1 is a high-volatility state with slight negative drift ($\mu_1 = -0.43$ annualised, $\sigma_1 = 3.98$), state 2 is a low-volatility state with positive drift ($\mu_2 = 0.29$, $\sigma_2 = 0.85$), and state 3 is a moderate-volatility regime in between ($\mu_3 = 0.14$, $\sigma_3 = 1.91$). The state-aggregated OoS regime trajectory under the IS-fixed forward filter (`CHMM-Model/results/diagnostics/oos_regime_trajectory.txt`) over-weights the high-volatility band by $\sim 10\%$ relative to the IS stationary distribution, consistent with the elevated single-name volatility in the 2024–2026 window (Q1 2024 banking turmoil, late-2024 monetary-policy pivot). The bimodal CHMM-GED $\hat{p}_k$ partition (Appendix A.6.16) reflects the same axis at $K = 18$: the Laplace-like high-shape states cluster in the high-volatility tail of the state ordering, the Gaussian-like states populate the bulk. The latent states are therefore not statistical artefacts; they map onto identifiable economic regimes at every K examined.

ML HSMM as a co-headline result. Replacing the geometric sojourn distribution with an explicit truncated Pareto and refitting through the explicit-duration forward-backward of Yu [77] gives ML HSMM-N at $K^\star = 3$ at 98.4% IS / **91.0**% OoS KS, the strongest single-window OoS KS row in the panel and beating every CHMM operating point. CHMM and ML HSMM are best read as complementary scaffolds on the same Markov backbone, with KS / ACF / kurtosis trade-offs that select between them by use case. The HSMM advantage on KS is offset by an absolute-return ACF-MAE that matches the i.i.d. baseline level (0.0629) rather than CHMM-N's 0.0460, because the fitted Pareto sojourn concentrates probability mass on a single low-volatility state and the regime-driven slow-ACF mechanism that the CHMM exploits at $K \geq 3$ is muted (per-state $\bar{\boldsymbol{\pi}}$ concentration in Appendix A.6.17). Higher-K HSMM variants (Pareto collapse at $K = 18$; discrete-Gamma-sojourn fix achieving $|G_t|$ ACF-MAE **0.0462** on IS at $K = 18$) are reported in Appendix A.6.17 and A.6.18. Readers selecting a synthetic-data generator on raw OoS KS alone should prefer ML HSMM-N at $K^\star = 3$; CHMM at $K^\star = 3$ is the right choice when the structural use cases of §4.6–4.5 are in scope.

Bootstrap as a non-parametric synthetic-data benchmark. The stationary block bootstrap of Politis and Romano [20] attains 99.7% IS / 92.1% OoS KS in Table 3, comparable to ML HSMM-N at $K^\star = 3$ (91.0% OoS) and above every CHMM operating point. The 1-day OoS KS column is therefore the wrong column to read for CHMM-vs-bootstrap differentiation. The use-case differentiation is not "CHMM beats bootstrap on KS" but rather "CHMM offers parametric controls and structural use cases that the bootstrap cannot serve": (i) the regime-conditional Value-at-Risk of §4.6, which propagates the one-step-ahead state forecast through the predictive mixture, has no bootstrap analogue since the bootstrap is exchangeable and admits no notion of a forward state filter; (ii) the multi-asset copula composition of §4.5 requires per-asset parametric marginals and is incompatible with a non-parametric bootstrap that only re-emits IS samples; (iii) on the privacy / licensing axis, every bootstrap path is a permutation of literal IS observations, so the bootstrap is unusable wherever the IS sample is restricted. The body ordering is therefore: bootstrap for raw distributional fidelity; ML HSMM-N at $K^\star = 3$ when KS is the only objective; CHMM at $K^\star = 3$ when one of the three structural use cases above is in scope.

Block-aware OoS KS recalibration. The asymptotic two-sample KS critical value $1.36\sqrt{2/T_{\mathrm{OoS}}} = 0.0804$ used in Table 3 assumes i.i.d. samples; under autocorrelated daily returns the appropriate null is the stationary block-bootstrap KS distribution of Politis and Romano [20]. We report the block-aware OoS KS pass rate at mean block length $L = 20$ alongside the asymptotic value for the body operating point (Table 5; full $L \in \{5, 10, 20\}$ panel, OoS-anchored critical value

Table 5. Block-aware OoS KS recalibration at $L = 20$ (stationary block bootstrap of 20, $B = 1{,}000$, 500 OoS-length simulated paths per generator, seed 20260422). Side-by-side with the asymptotic OoS pass rate of Table 3; the block-bootstrap 95% null KS critical value at $L = 20$ is 0.0647 vs the asymptotic 0.0804. Body operating point $K^\star = 3$. Full $L \in \{5, 10, 20\}$ panel and $K = 6$ / $K = 18$ sensitivity rows: Appendix A.7.2.

Model	OoS KS asymp. (%)	OoS KS block $L = 20$ (%)
Bootstrap	90.4	73.2
GARCH(1,1)	59.2	31.4
CHMM-N	79.8	58.6
CHMM-t pen. ($\lambda = 20$)	**79.8**	**54.2**
CHMM-L	65.0	33.4
CHMM-GED	79.2	51.6

0.0647, and $K = 6$ / $K = 18$ sensitivity rows in Appendix A.7.2). The cross-generator ordering is preserved (the iid bootstrap remains the non-parametric ceiling, GARCH(1,1) the floor); the absolute level drops by roughly 20–30pp across all rows on OoS, materially larger than the IS recalibration drop of $\sim$ 5pp. Read together with Table 3, the OoS panel under a temporally-aware null moves CHMM-N from a "passes most of the time" generator to a "passes about half the time" generator at $L = 20$. The body OoS KS headline at the asymptotic critical value therefore overstates the absolute level relative to a temporally-aware null; the cross-generator ranking is the robust comparison.

4.4 Cross-Ticker Generalization (Pipeline A)

Table 6 reports the penalised CHMM-t IS / OoS KS distribution across a sector-balanced 30-ticker panel (10 GICS sectors $\times 3$ large-cap representatives, $\lambda = 20$) at the body operating point $K^\star = 3$. The IS distribution is concentrated (median 96.8%); the OoS distribution is wider with median 69.1%, mean $66.2 \pm 28.2\%$, and 11/30 tickers below 60%, concentrated in single-name regime introductions (LLY 8.7%, UNH 16.5%, NEM 5.0%). A one-way ANOVA on OoS KS by sector at the original $n = 3$ per-sector design is severely underpowered ($\eta^2 = 0.16$ at $F(9, 20) = 0.44$, $p = 0.90$); an $n = 6$ per-sector expansion to a 60-ticker panel (Appendix A.6.3) returns $F(9, 50) = 0.37$, $p = 0.946$, $\eta^2 = 0.062$, with the effect-size estimate dropping by more than half. The 60-ticker aggregate distribution is statistically indistinguishable from the 30-ticker version, so the visible concentration of failures on regime-introduction tickers is a per-ticker observation rather than evidence about sector-level structure. The operational recipe is periodic refit: at the $K = 18$ kurtosis-fidelity sensitivity reference, a quarterly-refit version (Appendix A.6.14) shifts the OoS KS median from 73.4% to **83.0**% and reduces failures from 11/30 to 7/30, with a monthly-refit version interpolating further (median OoS KS **86.7**%, 5/30 failures; Appendix A.6.23). Sensitivity panels at $K = 6$ and $K = 18$, the per-ticker $\hat{\lambda}^\star$ tuning recipe (the alternative to uniform λ for kurtosis-fidelity-priority deployments), and the SPY-level shared-ν Student-t alternative that side-steps λ entirely are in Appendix A.6.7, A.9, and A.6.9.

Cross-decade validation. A CRSP-sourced cross-decade split (1994–2004 IS / 2004–2006 OoS; full panel in Appendix A.6.24) tests whether the body's stylized-fact reproduction transfers across non-overlapping decades. The IS axis is decade-robust: at $K^\star = 3$ CHMM-N attains 84–90% IS KS pass rate against the 90–94% obtained on the body 2014–2024 IS window (full $K \in \{3, 18\}$ panel in Appendix A.6.24). The 2004–2006 OoS pass rate, however, collapses to 3–5% because the calm post-dot-com bull-market slice has observed excess kurtosis 0.06 (essentially Gaussian)

Table 6. Cross-ticker generalisation, sector-balanced 30-ticker panel (penalised CHMM-t at $\lambda = 20$, 1,000 paths, seed = 20260420, body operating point $K^\star = 3$). The OoS distribution is wider than the IS distribution and is concentrated at the regime-introduction tickers (LLY, UNH, NEM). Full per-ticker panel and per-sector rollup: Appendix A.6.2; $K = 6$ and $K = 18$ sensitivity rebuilds: Appendix A.6.7; quarterly- and monthly-refit recipes: Appendix A.6.14, A.6.23. The 10-sector × 3-large-cap sample design is one design choice among many.

Metric	$K^\star = 3$
IS KS pass rate (%) median	96.8
IS KS pass rate (%) mean ± s.d.	95.1 ± 4.9
OoS KS pass rate (%) median	69.1
OoS KS pass rate (%) mean ± s.d.	66.2 ± 28.2
$\lvert G_t \rvert$ ACF-MAE median	0.0399
Kurtosis residual (sim − obs) median	7.18
Tickers OoS KS < 60% (IS-fixed)	11/30

against the 1994–2004 IS kurtosis of 3.05. The cross-decade pattern matches the W2 (COVID 2020) and W4 (2022 rate-hike) walk-forward stress folds (Table 4, both with OoS KS < 10%): the IS axis transfers cleanly across decades, but a static-IS-fitted CHMM cannot bracket OoS slices whose marginal moves outside the IS regime library. The operational implication parallels the cross-ticker panel above: the static-fit OoS pass rate is OoS-slice-specific and the deployment recipe is the periodic-refit construction (the cross-ticker quarterly-refit panel in Appendix A.6.14 and the rolling-copula construction of §4.5), not the single-window static fit.

4.5 Cross-Asset Dependence (Pipeline B): Student-t Copula

Pipeline B reuses the per-asset CHMM-N marginals from §4.4 and adds a rank-based Student-t dependence layer (§3.4). The body cross-asset construction is scoped to a US-equity universe: profile MLE on six US-equity assets (SPY, NVDA, JNJ, JPM, AAPL, QQQ; 200 paths) selects $\nu^\star = 6$ on the unit-spaced grid $\nu \in [4, 12]$ with Wilks 95% profile-LL CI of $[6, 7]$ *on the IS slice* (Appendix A.8); a half-unit-grid refinement plus $B = 200$ parametric bootstrap CI confirms $[6, 7]$ (Appendix A.8.7). A non-equity stress test on GLD and SLV ETFs over the same windows (Appendix A.8.6) shows OoS KS collapses to 0% on both ETFs under a static IS-fitted CHMM, confirming that the cross-asset claim does not extend to non-equity asset classes without periodic refit. Table 7 reports per-asset KS and the off-diagonal correlation MAE: 0.027 on IS for the Student-t copula, against 0.030 Gaussian, 0.068 truncated C-vine, and 0.076 SIM (full panel in Appendix A.8.1). **On OoS the dependence-family choice is empirically null at this $N_{\mathbf{paths}}$**: off-diagonal MAE is 0.209 for Student-t and 0.202 for Gaussian, a 0.007 difference below the simulation-noise floor. The IS calibration distinction (0.027 vs. 0.030, 0.003 difference) is an IS-only finding and the dependence-family selection on OoS is not supported at this N_{paths}. The operational story on the OoS axis is the quarterly-rolling refit, which reduces the OoS off-diagonal MAE from 0.209 to **0.185** on the six-asset universe (Appendix A.8.5); the dependence-family choice is dominated by the refit-cadence choice on OoS. One additional caveat on the dependence-layer estimator: the body $\nu^\star$ is selected by profile MLE conditional on $\hat{\Sigma}$ obtained by Kendall's-τ inversion ($\rho_{ij} = \sin(\pi\tau_{ij}/2)$, exact under the Gaussian copula and approximate under the Student-t copula); a full one-shot MLE that jointly maximises over (Σ, ν) on the same six-asset universe (Appendix A.7.7) returns $\hat{\nu}_{\text{full}} = 6.40$ versus $\hat{\nu}_{\text{two-step}} = 6.00$, a +5.55 log-likelihood improvement, with all 15 off-diagonal correlation differences below $|\Delta\rho_{ij}| \leq 0.025$. The two-step estimator is therefore not detectably biased toward the Gaussian limit on this universe; the body

Table 7. Cross-asset Student-t copula on CHMM-N marginals (Pipeline B; $\nu^\star = 6$; 200 paths). Per-asset KS pass rate at $\alpha = 0.05$ and correlation reproduction averaged over paths. Comparison against SIM, Gaussian, and truncated C-vine in Appendix A.8, Table 42.

	KS (%) ↑	
Ticker	IS	OoS
SPY	96.0	79.5
NVDA	98.5	53.5
JNJ	99.0	97.5
JPM	99.5	48.0
AAPL	99.0	93.5
QQQ	95.5	92.0
Median	98.8	85.8
Tickers OoS KS < 60%	0/6	**2/6**
Off-diag MAE IS	0.027	
Off-diag MAE OoS	0.209	
Off-diag MAE OoS (quarterly refit)	0.185	

$\nu^\star = 6$ choice is robust to the estimator switch and remains inside the Wilks 95% profile-LL CI of the full MLE.

The cross-asset median OoS KS of 85.8% obscures a bimodal distribution: NVDA (53.5%) and JPM (48.0%) fail OoS KS at the standard threshold, while the four remaining assets pass cleanly above 79%. The 2/6 failure rate at the body $\nu^\star = 6$ recovers to 0/6 under the quarterly-rolling refit recipe of Appendix A.8.5; per-asset failure attribution is driven by single-name regime shifts in the OoS slice (NVDA AI-rally regime, JPM bank-stress regime) rather than by the dependence-layer estimator, consistent with the pattern in the 30-ticker univariate panel of §4.4.

4.6 Value-at-Risk and Expected-Shortfall Back-Test (Pipeline A)

The headline of this subsection is the regime-conditional Christoffersen-cc result: a VaR threshold built from the CHMM one-step-ahead state forecast passes the joint conditional-coverage test cleanly at every (K, α) on OoS, where every constant-across-t generator (CHMM unconditional, GARCH, bootstrap) rejects breach independence. We organise the diagnostics in three layers: (i) envelope bracketing, (ii) unconditional Kupiec coverage, (iii) the regime-conditional construction.

(i) Envelope bracketing. All four CHMM variants bracket observed historical VaR and ES at 1% and 5% within the $[5, 95]$% envelope on both windows; the IS CHMM-N median $\mathrm{VaR}_{0.01}$ at -6.64 (observed -6.38) is the closest point estimate among the six benchmark generators, with envelopes roughly 1.7× tighter than GARCH's. The full envelope panel for Bootstrap, GARCH, CHMM-N/-t/-L/-GED at 1% and 5% is in Appendix A.6.19, Table 23.

(ii) Unconditional Kupiec coverage. The four CHMM rows pass the unconditional Kupiec test on both windows: at $\alpha = 0.05$ on OoS, CHMM-N $\mathrm{LR}_{\mathrm{uc}} = 3.83$ against $\chi^2_1(0.05) = 3.841$, with the other three CHMM variants in 3.03–3.83. The repeated 3.83 across rows is a feature of $T_{\mathrm{OoS}} = 572$: the Kupiec statistic is a function of the integer breach count and at $\alpha = 0.05$ the values $k \in \{14, 19, 20\}$ produce statistics that pile against the critical value rather than spreading continuously. We treat unconditional Kupiec as a sanity check at this T_{OoS} and the conditional construction below as the substantive risk-management diagnostic.

(iii) Regime-conditional Christoffersen-cc *(headline).* At each OoS day t we run the forward filter through $\mathcal{F}_{t-1} = R_{\text{IS}} \cup R_{\text{oos}}[1{:}t-1]$ to get the one-step-ahead state forecast

$$\mathbb{P}(s_t = k \mid \mathcal{F}_{t-1}) = \sum_i \mathbb{P}(s_{t-1} = i \mid \mathcal{F}_{t-1})\, T_{ik}, \tag{7}$$

and define $\widehat{\text{VaR}}_t(\alpha) = F_t^{-1}(\alpha)$ where $F_t(\cdot) = \sum_k \mathbb{P}(s_t = k \mid \mathcal{F}_{t-1})\, F_k(\,\cdot\,; \boldsymbol{\theta}_k)$ is the family-appropriate predictive mixture CDF under IS-fixed $(\mathbf{T}, \{\boldsymbol{\theta}_k\})$, with F_k the per-state Gaussian, Student-t, Laplace, or GED CDF (the body Gaussian construction reduces to $F_k = \Phi(\,\cdot\,; \mu_k, \sigma_k)$). Table 8 reports the back-test for CHMM-N at $K^\star = 3$ and $K = 18$ and the penalised CHMM-t at $\lambda = 20$ on OoS; every row passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly at $\alpha = 0.05$. The improvement in the independence statistic from the unconditional variant ($\text{LR}_{\text{ind}} = 5.26$ at $K = 18, \alpha = 0.05$) to the conditional variant ($\text{LR}_{\text{ind}} = 0.52$) is the regime-switching value proposition that the unconditional Kupiec headline did not exercise. The conditional pass extends to every emission family with the family-appropriate predictive density (CHMM-L, CHMM-GED) at $p_{\text{cc}} \geq 0.089$ throughout, so the regime-switching value proposition is intrinsic to the latent-state forecast and not specific to the Gaussian emission family (Appendix A.3.2, Table 39).

Power caveat at $\alpha = 0.01$. A Monte Carlo power calibration at $T_{\text{OoS}} = 572$ (Appendix A.6.10, Table 18, $B = 5{,}000$ replicates per cell) shows the Christoffersen-cc test is correctly sized at 4–5% Type-I error under the i.i.d. null at both α levels, but the power profile is asymmetric. At $\alpha = 0.05$ (~ 28.6 expected breaches) the test reaches $\geq 80\%$ power against breach-clustering eigenvalues $\rho \geq 0.20$; at $\alpha = 0.01$ (~ 5.7 expected breaches) it reaches $\geq 80\%$ power only at $\rho \geq 0.50$, with rejection rate 42.6% at $\rho = 0.20$. The body $\alpha = 0.05$ "clean pass at every (K, α)" rows of Table 8 therefore carry strong power against moderate clustering; the $\alpha = 0.01$ rows should be read as "not detectably worse than a strongly-clustered alternative" rather than "calibrated against any plausible alternative". The Engle–Manganelli (2004) Dynamic Quantile (DQ) test as a higher-power conditional-coverage alternative (Appendix A.6.11, Table 19) is the substantive $\alpha = 0.01$ test on this T_{OoS}. At $\alpha = 0.05$ the DQ test passes cleanly at both $K = 3$ ($p = 0.156$) and $K = 18$ ($p = 0.678$), confirming the Christoffersen-cc reading at this α. **At $\alpha = 0.01$ the DQ test rejects conditional coverage at $K = 18$ ($p = 0.017$)**, while Christoffersen-cc does not reject at the same row ($p = 0.137$). DQ has the higher power of the two tests at this T_{OoS} (and is the test we trust at $\alpha = 0.01$); the substantive read is therefore that the regime-conditional VaR *over-couples high-volatility states at the strict tail at* $K = 18$ on this OoS slice, and that the $\alpha = 0.01$ Christoffersen-cc pass is a power-driven non-rejection rather than a calibrated pass. The risk-management diagnostic in this paper is therefore the $\alpha = 0.05$ row under both tests (clean pass on every (K, family) row); the $\alpha = 0.01$ row at $K = 18$ should be read as a DQ-rejection of conditional coverage that tighter α levels require either larger T_{OoS} or a model rebuild to address.

State-filter pipeline applies to other multi-state models. The state-filter VaR pipeline of Eq. (7) applies to any multi-state regime-switching model; the CHMM-vs-MS-GARCH choice on this dataset is dominated by the marginal-distribution column of Table 3 (CHMM-N at 89.7% IS KS vs MS-GARCH plateaus at 0–36% across estimators), not by the conditional-coverage column.

Walk-forward stability across six rolling-origin folds is in Table 40 (Appendix A.9); the four-family extension at the headline OoS window is in Table 39 (Appendix A.3.2); a quarterly-refit version

Table 8. Regime-conditional VaR back-test, CHMM-N and penalised CHMM-t ($\lambda = 20$) on OoS ($T_{\mathrm{OoS}} = 572$, seed 20260420). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. The forward filter is run under IS-fixed parameters $(\mathbf{T}, \{\boldsymbol{\theta}_k\})$ (no refit); only the one-step-ahead state-probability vector is updated through the OoS observations via Eq. (7). Every $(K, \alpha, \text{family})$ row passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly at $\alpha = 0.05$. ‡ $\alpha = 0.01$ rows are power-bounded at $T_{\mathrm{OoS}} = 572$: Christoffersen-cc reaches $\geq 80\%$ power against breach-clustering eigenvalues only at $\rho \geq 0.50$ (rejection rate 42.6% at $\rho = 0.20$), so a clean pass should be read as "not detectably worse than a strongly-clustered alternative" rather than "calibrated against any plausible alternative". The Engle-Manganelli (2004) DQ test as a higher-power alternative (Appendix A.6.11) rejects $K = 18$ at $\alpha = 0.01$ ($p = 0.017$, bold in column p_{DQ}). See Appendix A.6.10 for the full power calibration. * Engle-Manganelli (2004) Dynamic Quantile p-value at 4-lag specification (Eq. 27 in Appendix A.6.11); χ^2_6 critical at 5% is 12.59. The DQ panel is run on CHMM-N only and the CHMM-t penalised rows are reported with '–' in this column; the CHMM-t-penalised conditional-VaR diagnostic is the Christoffersen-cc column at $\alpha = 0.05$.

Family	K	α	breaches	br rate	median $\widehat{\mathrm{VaR}}_t$	$\mathrm{LR}_{\mathrm{uc}}$	$\mathrm{LR}_{\mathrm{ind}}$	$\mathrm{LR}_{\mathrm{cc}}$	p_{cc}	$p_{\mathrm{DQ}}{}^*$
CHMM-N	3	$0.01^{\ddagger}$	9	1.57%	−4.56	1.62	2.36	3.98	0.14	0.06
CHMM-N	3	0.05	35	6.12%	−2.87	1.41	$0.01^{\ddagger}$	1.42	0.49	0.16
CHMM-N	18	$0.01^{\ddagger}$	9	1.57%	−5.20	1.62	2.36	3.98	0.14	**0.02**
CHMM-N	18	0.05	26	4.55%	−3.02	0.26	0.52	0.78	0.68	0.68
CHMM-t (λ=20)	3	$0.01^{\ddagger}$	8	1.40%	−5.59	0.82	2.81	3.63	0.16	–
CHMM-t (λ=20)	3	0.05	32	5.59%	−2.73	0.41	0.77	1.19	0.55	–
CHMM-t (λ=20)	18	$0.01^{\ddagger}$	10	1.75%	−6.25	2.65	1.98	4.62	0.10	–
CHMM-t (λ=20)	18	0.05	29	5.07%	−3.15	$0.01^{\ddagger}$	1.39	1.40	0.50	–

(CHMM-N refit every 63 trading days on a rolling 5-year window) is in Table 20 (Appendix A.6.12). The conditional Christoffersen-cc passes at $\alpha = 0.05$ on 19/24 walk-forward rows, with the failures concentrated on the W2 (COVID) and W4 (2022 rate-hike onset) stress folds that the univariate walk-forward already flags as out-of-distribution. Faster refit cadences (monthly, weekly) do not close the W2 stress-fold rejection (Appendix A.6.13, Table 21); W2 is an intrinsic regime-break failure that requires a model class with explicit regime-introduction handling (skew-emission HMMs at the marginal axis or change-point / Bayesian online-changepoint constructions on the transition matrix) rather than a faster refit cadence on the symmetric scaffold.

BH-FDR correction at FDR = 0.05 across the combined conditional-coverage panel (40 tests across the headline OoS window plus the six-fold walk-forward) retains the panel-level pass-rate at 37/40 rows on the full panel and 32/32 on the non-stress subset (Appendix A.9); the three persistent rejections are all on the W2 COVID fold.

5 Discussion

A continuous HMM trained by Baum-Welch reproduces the three Cont stylized facts simultaneously on SPY at the body operating point $K^\star = 3$: CHMM-N attains 89.7% IS / 80.5% OoS KS at simulated kurtosis 3.83/3.53 against observed 7.68/5.29 and $|G_t|$ ACF-MAE 0.0460. Four emission variants share the same scaffold and land at four distinct kurtosis points without retuning K or the transition structure. The mechanism behind the simultaneous KS / ACF fit is the spectral identity (6): at $K \geq 3$ with quantile-initialised Baum-Welch, the absolute-return ACF is a population-level real-coefficient sum of geometric decays at the non-unit eigenvalues of $\mathbf{T}$.

Replicating the slow-ACF behaviour at $K \geq 3$. Two issues compound in the Rydén et al. [1] low-K setup: a single self-transition rate produces a single exponential mode that cannot be simultaneously fast and slow, and the random initialisations standard in the 1990s often converge to degenerate local optima. Our $K = 2$ replication on SPY (Appendix A.6.6) clarifies the mechanism: ACF-MAE is essentially constant at ≈ 0.050 across quantile and five random-seed initialisations. The binding low-K constraint is therefore distributional (KS at $K = 2$ drops to 67–72%): with modern initialisation the ACF is reproduced at any $K \geq 2$, while distributional fidelity drives the choice of K. The effective-rank diagnostic in Appendix A.6.4 makes this microscopically explicit: at the fitted $\hat{\mathbf{T}}$ for $K^\star = 3$ the dominant non-unit mode alone carries 96.8% of the lag-1 absolute-return ACF on SPY.

Closing the kurtosis gap with heavy-tailed emissions. CHMM-N at $K^\star = 3$ produces simulated IS kurtosis 3.83 against observed 7.68: a Gaussian mixture's kurtosis is bounded by the variance of component means relative to their variances. CHMM-t addresses this through per-state Student-t emissions but at the cost of an aggregate-mean IS overshoot to 14.91 from a single-state ECM lower-bracket pinning artefact (Appendix A.9); an exponential $1/\nu_k$ shrinkage at rate $\lambda = 20$ is the body recipe (the per-path *median* simulated IS kurtosis is centred near observed even before the penalty; Appendix A.6.8). A shared-ν ablation (a single ν shared across all K states under aggregate-Q ECM, the standard one-parameter Student-t HMM in the time-series literature) eliminates the aggregate-mean overshoot completely without any penalty (simulated IS kurtosis 4.68 at $K^\star = 3$, Appendix A.6.9). The body retains the per-state $\nu_k + \lambda = 20$ construction for direct comparability with Peel and McLachlan [18], Liu and Rubin [19]; the shared-ν row is the structurally cleaner alternative for kurtosis-fidelity-priority consumers. The $\lambda = 20$ choice is validated on a strictly pre-2020 slice by held-out log-likelihood (Appendix A.6.20); per-ticker, uniform $\lambda = 20$ is over-shrunk on three of six body tickers and the production recipe is the per-ticker validation rule of §4.4 (Appendix A.9, Table 41). CHMM-L sits between CHMM-N and CHMM-t at IS kurtosis 5.30 without any shape parameter; CHMM-GED's bimodal $\hat{p}_k$ partition (Appendix A.6.16) is the data-driven analog of a hand-classified Gaussian-bulk / Laplace-tail hybrid. The four-way variant decision is summarised in Table 9; bootstrap CIs on observed kurtosis (Appendix A.6.8) show that the IS-OoS difference of ~ 2.4 units is not statistically distinguishable at conventional levels, so the variant rankings should be read as point-comparisons inside a wide CI envelope. A bracket-lift ablation at $\nu_{\min} = 4$ (the kurtosis-finiteness threshold for Student-t) and a $K = 18$ kurtosis-fidelity sensitivity panel are reported in Appendix A.9.

Table 9. Variant decision guide at $K^\star = 3$. Recommended CHMM emission family by use-case priority. All four variants share the forward-backward scaffold and quantile-based initialisation; the M-step is the only architectural difference. Caveat: observed kurtosis differs across windows (7.68 IS, 5.29 OoS) and no single variant matches both windows.

Use-case priority	Recommended variant
Simplest closed-form fit; kurtosis fidelity secondary	CHMM-N
Closest IS kurtosis match without a shape parameter	CHMM-L
Per-state heavy tails, IS-calibrated	CHMM-t pen. ($\lambda = 20$)
Heavy-tail match without a penalty hyperparameter	CHMM-t shared-ν
Adaptive per-state shape; data-chosen Gaussian-bulk / Laplace-tail mixture	CHMM-GED

Stationarity scope and operational deployment. The CHMM is trained on the assumption that returns are locally stationary across the IS window. Single-name equity returns are well known to carry sector-specific structural breaks (23; 24; surveyed by 25); the cross-ticker generalisation in

§4.4 reflects this directly, with 11/30 tickers below 60% OoS KS, concentrated in single-name regime introductions. The mitigation is periodic refit (26; an online-EM formulation by 78): a quarterly-refit version of the cross-ticker panel (Appendix A.6.14) reduces failures from 11/30 to 7/30 at the $K = 18$ kurtosis-fidelity reference. The walk-forward W2 (COVID) and W4 (2022 rate-hike onset) stress folds remain out-of-distribution by KS for every generator under any refit cadence (Appendix A.9); these are regime introductions the IS distribution does not span.

Stylized-fact scope: the leverage axis under symmetric emissions. The Cont (2001) stylized-fact list also includes the leverage effect (negative $\mathrm{Corr}(G_t, |G_{t+1}|)$ on equity returns); the present paper's body evaluation covers only the three symmetric stylized facts. All four CHMM emission families carry symmetric per-state densities, so we initially expected the CHMM to fail the leverage axis by construction. We compute $\mathrm{Corr}(G_t, |G_{t+1}|)$ on observed SPY (IS -0.135, OoS -0.214) and on the per-path simulation distribution (Appendix A.7.6). The CHMM produces a non-trivial signed signal via Markov state-mixing on $\hat{\mathbf{T}}$ (high-σ_k states couple to negative-μ_k states asymmetrically), borderline-non-rejecting the IS observed value at the 5% level but rejecting the OoS observed value at the same level. The OoS gap is a substantive finding about the symmetric-emission scaffold: a non-zero leverage signal is recoverable from the Markov mixing alone, but closing the OoS gap requires asymmetric per-state emissions (skew-t in the tradition of 79; skew-Laplace; skew-GED), deferred as a companion-paper direction.

Computational cost and parameter parsimony. Single-asset training on the SPY IS window ($T_{\text{IS}} = 2{,}516$, single-threaded Julia 1.12) is sub-second per ECM iteration at $K^\star = 3$ for every emission family and order one second per iteration at $K = 18$; the four-family panel of Table 3 is reproducible end-to-end on a laptop in under thirty minutes (the `run_full_rebuild.jl` entry in the companion repository excludes the QuantGAN row by default for this reason). The parameter count is correspondingly modest: CHMM-N at $K = 18$ has $K(K-1) + 2K = 342$ free parameters (transition matrix entries plus per-state mean and scale), CHMM-t adds $K = 18$ degrees-of-freedom parameters, CHMM-GED adds K shape parameters; the QuantGAN baseline of Wiese et al. [10] carries on the order of 10^4–10^5 parameters depending on convolution depth. The CHMM operates at three orders of magnitude lower parametric complexity than the deep-generative comparator while attaining higher OoS KS on this dataset (Table 3); for the structural use cases of §4.6–4.5 the parsimony is itself part of the value proposition, not just a footnote.

Limitations. Skew-heavy-tailed emissions (the natural fix for the leverage-axis OoS gap), explicit-duration semi-Markov sojourns at higher K, and a reference-implementation re-run of the original Wiese et al. (2020) QuantGAN architecture with Lambert-W input pre-processing are companion-paper directions; scaling the multi-asset construction to larger d via untruncated regular vines or factor copulas is the natural follow-up. The empirical scope is daily US-equity stylized-fact reproduction on stationary OoS windows: FX, fixed-income, and commodity asset classes are not covered, and the GLD / SLV non-equity stress test (Appendix A.8.6) documents that the static IS-fitted CHMM does not transfer to those windows without periodic refit. The cross-asset dependence-family selection (§4.5) is an IS-calibration distinction: on the OoS window the Gaussian and Student-t copulas are statistically indistinguishable at $N_{\text{paths}} = 200$.

6 Conclusion

A continuous HMM trained by Baum-Welch with quantile-based initialization reproduces the three symmetric Cont stylized facts (heavy-tailed marginals, negligible linear ACF, slow $|G_t|$ ACF) on daily US-equity returns. At the state-resolution-robust body headline $K^\star = 3$ (selected by held-out per-observation log-likelihood and held-out KS on the strictly pre-2020 slice under both single-fold and four-fold / six-fold rolling-origin CV; Appendix A.9) the four emission variants attain 79.8–90.6% IS / 63.1–83.2% OoS KS on the single SPY OoS window, with CHMM-N at 89.7% IS / 80.5% OoS KS, simulated kurtosis 3.83/3.53, and $|G_t|$ ACF-MAE 0.0460; the penalised CHMM-t at $\lambda = 20$ provides the cleanest IS / OoS heavy-tail match in the headline block at 14.91/8.50 against observed 7.68/5.29. A $K^\star = 6$ block is reported as a sensitivity reference (single-fold pre-2020 selection; the four-fold and six-fold k-fold CV cannot distinguish $K^\star = 6$ from $K^\star = 3$ at conventional levels). The single-window OoS sits at the upper end of the six-fold rolling-origin walk-forward whose median CHMM-N OoS KS is 67.7% at $K = 18$ and 62.1% at $K^\star = 3$; the walk-forward median is the operationally informative summary for production deployment. An extended-state-resolution *sensitivity reference* at $K = 18$, selected by a multi-objective rule that uses the OoS window, attains 93.6–95.6% IS / 80.8–85.7% OoS KS, with the penalised CHMM-t at $\lambda = 20$ providing the cleanest moderate-state-resolution IS / OoS kurtosis match at 8.56 / 7.07 against observed 7.68 / 5.29 (Table 3). The strongest single-window OoS KS row in the panel is ML HSMM-N at $K^\star = 3$ (91.0% OoS KS) but its $|G_t|$ ACF-MAE matches the i.i.d. baseline; CHMM and ML HSMM are best read as complementary scaffolds with different KS / ACF / kurtosis trade-offs. We apply the textbook closed-form mixture-of-eigenvalues identity for the absolute-return ACF (13–15) to the SPY 2014–2024 instance of the Rydén et al. low-K setup and recast the empirical failure there as an effective-rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$, showing the rank constraint is non-binding at $K \geq 3$ on this instance; the substantive content is the empirical effective-rank application, not a re-derivation of the identity. The four emission variants share the same scaffold and land at four kurtosis points; CHMM-GED's per-state shape p_k nests Gaussian and Laplace as boundary cases and produces a bulk/tail partition that replicates across seeds and tickers. VaR and ES envelopes bracket observed values at 1% and 5%; a regime-conditional VaR that propagates the one-step-ahead state forecast through the CHMM mixture passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly at $\alpha = 0.05$ on OoS (Table 8; the $\alpha = 0.01$ rows are power-bounded at $T_{\mathrm{OoS}} = 572$ and should be read with the calibration of Appendix A.6.10). The conditional construction extends to a six-fold rolling-origin walk-forward with a 19/24 aggregate pass-rate at the 5% level (Table 40); the two failures concentrate on the COVID and 2022-rate-hike-onset stress folds that the univariate walk-forward already flags as out-of-distribution by KS. Cross-ticker generalisation is exercised on a sector-balanced 30-ticker panel (Section 4.4, Table 6); the body table reports the headline $K^\star = 3$ (OoS KS median 69.1%), the sensitivity reference $K^\star = 6$ (OoS KS median 75.1%), and the kurtosis-fidelity sensitivity $K = 18$ (OoS KS median 73.4%), all at identical 11/30 failure count, with quarterly refit at $K = 18$ lifting the OoS median to 83.0% and the largest gains concentrating on the regime-introduction tickers. Multi-asset synthesis composes through a rank-based Student-t copula on the CHMM marginals at $\nu^\star = 6$ (six-asset IS off-diagonal correlation MAE 0.027); on OoS the Gaussian and Student-t copulas are statistically indistinguishable at $N_{\mathrm{paths}} = 200$, so the dependence-family selection is an IS-calibration distinction. The headline scope is daily US equities. **The static IS-fitted CHMM does not transfer outside this asset class**: the GLD / SLV non-equity stress test of Appendix A.8.6 collapses to 0% OoS KS on both ETFs, a hard rejection rather than a soft preference. Non-equity asset classes (commodities, FX, fixed income) require periodic refit at a cadence that scales with the asset's regime structure; the body recipes for periodic refit on US-equity tickers (Section 4.4) are not validated to extend across asset classes without per-class re-validation.

Skew-heavy-tailed emissions (the natural fix for the absent leverage-effect coverage in the present scope), explicit-duration semi-Markov sojourns at higher K, and a reference-implementation re-run of the original Wiese et al. (2020) QuantGAN architecture with Lambert-W input pre-processing are companion-paper directions; scaling to larger d via untruncated regular vines or factor copulas is the natural multi-asset follow-up. The reference-implementation MS-GARCH re-run via the `MSGARCH` R package of Ardia et al. [35] is included in the body (Table 3 ¶rows and Appendix A.7.4).

The spectral-rank diagnostic generalises beyond the Rydén instance: any regime-switching "failure" result in the literature attributing the limit to temporal structure can be re-litigated as a marginal-mixture vs. effective-rank decomposition on the fitted $\hat{\mathbf{T}}$. The empirical recasting on equity returns presented here is one example, but the same machinery is directly applicable to other asset classes and to the broader Markov-switching econometrics where the sojourn-time distribution has historically been the headline diagnostic axis.

Data and code availability. Reproducer at https://github.com/altashly1/CHMM-paper; companion Julia package `CHMM-Model.jl` at https://github.com/altashly1/CHMM-Model (public API in Appendix A.1, Julia $\geq$ 1.12). Daily prices: Polygon.io via Massive (2014-01-03 to 2025-11-18) and Alpaca Markets / IEX (held-out extension through 2026-04-20); we thank both providers. The authors declare no competing interests. **Author contributions:** J.V. directed the study; A.A. developed the continuous model, implemented Baum-Welch, conducted the empirical analysis, and generated all figures and tables; C.J. contributed to methodology review and validation; all authors edited and reviewed the final manuscript.

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A Supplementary Material

This appendix collects supplementary material referenced in the main body: the public API of the companion package, the EM forward-backward recursions and algorithm pseudocode, the validation metric definitions, formal propositions, the full state-resolution and multi-emission sensitivity panels, robustness and KS-power calibration, the extended GARCH and SM-CHMM baselines, the full cross-asset panel, and the K-selection and ν_k diagnostics referenced from the discussion.

A.1 CHMM-Model.jl Public API and Reproducer

The companion Julia package `CHMM-Model.jl` (https://github.com/altashly1/CHMM-Model) is the entry point used to regenerate every table and figure in this paper from a single seed root under Julia ≥ 1.12. The public surface is a single `CHMM` type, three emission constructors, a unified `fit!` method (log-space Baum-Welch / ECM), and a unified `simulate` method.

```
using CHMM
chmm_n   = make_chmm_n(returns;   K = 18)    # Gaussian emissions
chmm_t   = make_chmm_t(returns;   K = 18)    # Student-t emissions
chmm_l   = make_chmm_l(returns;   K = 18)    # Laplace emissions
chmm_ged = make_chmm_ged(returns; K = 18)    # GED (per-state shape p_k)
fit!(chmm_n; max_iter = 60, tol = 1e-4)      # log-space Baum-Welch
paths = simulate(chmm_n; T = 2516, n = 1000) # 1000 paths of length T
```

The same `simulate` interface is used for every benchmark generator (Bootstrap, Gaussian i.i.d., Laplace i.i.d., GARCH(1,1)) so that the seven-metric panel evaluates each row on byte-identical windows. The Pipeline B Student-t copula sampler is exposed as `simulate_copula(...; nu = 6)` on the per-asset CHMM marginals; the rank-reordering construction is one function call. Running `julia -project=. run_full_rebuild.jl` from the paper repository (https://github.com/altashly1/CHMM-paper) reproduces every numerical artifact under the global seed policy of Section 3.5; `Manifest.toml` pins the dependency graph.

Sub-seed derivation rule. The deterministic global seed root is `20260420`; per-experiment sub-seeds follow the additive rule `sub_seed = SEED + offset(experiment) + offset(replicate)`, with experiment offsets in $\{0, 1{,}000, 100, 10{,}000, \ldots\}$ chosen to avoid collisions across the runners in the public Julia package. For example, the headline simulation in `run_diagnostics.jl` uses `SEED + (path index)` for each per-path random draw, the multi-seed Monte Carlo in `run_multiseed_headline.jl` uses base `20260422 + 100·(seed index)` for ten independent replicates, and the bracket / shrinkage sweeps add a further per-rate offset. The full sub-seed map is enumerated at the top of each runner script; downstream `Random.seed!()` calls are deterministic and order-independent within each runner.

A.2 CHMM Algorithms and Derivations

This appendix collects the algorithmic content for the CHMM family: the forward-backward recursions and weighted M-step updates, the complete training pseudocode, and per-state degrees-of-freedom diagnostics for CHMM-t.

A.2.1 Forward-Backward Recursions and M-Step Updates

For reference, the in-text Baum-Welch description of Section 3.2 is underpinned by the standard log-space recursions [17, 75]. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_{1:t}, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1:T} \mid S_t = k, \mathcal{M})$. The log-space recursions are

$$\log \alpha_1(k) = \log \pi_k + \log f_k(O_1), \tag{8}$$
$$\log \alpha_t(j) = \operatorname*{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij}) + \log f_j(O_t), \quad t = 2, \ldots, T, \tag{9}$$

$$\log \beta_T(k) = 0, \tag{10}$$
$$\log \beta_t(i) = \operatorname*{logsumexp}_j(\log T_{ij} + \log f_j(O_{t+1}) + \log \beta_{t+1}(j))\,, \quad t = T{-}1, \ldots, 1, \tag{11}$$

where $\text{logsumexp}_i(x_i) = x_{\max} + \log \sum_i \exp(x_i - x_{\max})$. The posterior state-occupancy $\gamma_t(k)$ and pair-occupancy $\xi_t(i,j)$ quantities are

$$\gamma_t(k) = \frac{\alpha_t(k)\beta_t(k)}{\sum_j \alpha_t(j)\beta_t(j)}, \qquad \xi_t(i,j) = \frac{\alpha_t(i)\, T_{ij}\, f_j(O_{t+1})\, \beta_{t+1}(j)}{\sum_{m,n} \alpha_t(m)\, T_{mn}\, f_n(O_{t+1})\, \beta_{t+1}(n)}, \tag{12}$$

and the closed-form M-step updates are

$$\pi_k^{\text{new}} = \gamma_1(k), \quad \mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, O_t}{\sum_t \gamma_t(k)}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)(O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}, \quad T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \tag{13}$$

The convergence criterion is $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\mathcal{L}^{(n)} = \text{logsumexp}_k \log \alpha_T^{(n)}(k)$, and simulation draws $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ and then iterates $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma^2_{S_t})$, $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t,\cdot})$ with no jump process.

CHMM-t M-step (per-state ECM, closed form given $u_{t,k}$). For Student-t emissions $f_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ the ECM formulation of Peel and McLachlan [18], Liu and Rubin [19] treats the latent precision as an augmented E-step quantity,

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \tag{14}$$

so that conditional on ν_k the location and scale admit closed-form weighted updates,

$$\mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, O_t}{\sum_t \gamma_t(k)\, u_{t,k}}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}. \tag{15}$$

The degrees-of-freedom parameter ν_k is then refined by a one-dimensional golden-section search on the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This two-block ECM preserves the monotonicity of the standard EM guarantee on the compact bracket $[\nu_{\min}, \nu_{\max}]$, with the search invariant $Q_k(\nu^{(n+1)}) \geq Q_k(\nu^{(n)})$ at every iteration; Proposition 1 states the guarantee in full. We extend this M-step with an optional exponential shrinkage prior on $1/\nu_k$, corresponding to the penalised objective

$$Q_k^{\text{pen}}(\nu) \;=\; \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k) \;-\; \lambda \,/\, \nu, \tag{16}$$

maximised by the same golden-section search ($\lambda = 0$ recovers the unpenalised Peel and McLachlan [18], Liu and Rubin [19] ECM; the rate sweep and its effect on simulated IS kurtosis are in Section 5).

CHMM-L M-step (closed-form weighted Laplace MLE). For Laplace emissions $f_k(x) = \frac{1}{2\beta_k} \exp(-|x - \mu_k|/\beta_k)$ (with the per-state Laplace scale denoted β_k rather than the canonical Laplace-scale symbol b) the weighted-MLE estimators are available in closed form:

$$\mu_k^{\text{new}} = \text{WeightedMedian}(O_{1:T};\, \gamma_{1:T}(k))\,, \qquad \beta_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, |O_t - \mu_k^{\text{new}}|}{\sum_t \gamma_t(k)}. \tag{17}$$

The weighted median is computed by cumulative-weight bracketing with linear interpolation between adjacent order statistics at the half-total weight crossing. The Laplace M-step carries no shape parameter, so CHMM-L is strictly cheaper per iteration than CHMM-t (which requires the per-state Q_k search).

CHMM-GED M-step (three-stage ECM with closed-form scale). For Generalized Error Distribution emissions $f_k(x) = \text{GED}(x; \mu_k, \alpha_k, p_k) = \frac{p_k}{2\alpha_k \Gamma(1/p_k)} \exp\big(-(|x - \mu_k|/\alpha_k)^{p_k}\big)$ the M-step decomposes into three CM updates per state. Given current $(\mu_k^{(n)}, \alpha_k^{(n)}, p_k^{(n)})$, CM-1 updates the location by minimising the per-state weighted L^{p_k} loss,

$$\mu_k^{(n+1)} \;=\; \arg \min_{\mu \in [\hat{\mu}_k - 5\alpha_k,\, \hat{\mu}_k + 5\alpha_k]} \sum_t \gamma_t(k)\, \big|O_t - \mu\big|^{p_k^{(n)}}, \tag{18}$$

by a 40-iteration golden-section search; CM-2 updates the scale in closed form given $(\mu_k^{(n+1)}, p_k^{(n)})$,

$$\alpha_k^{(n+1)} \;=\; \left[\frac{p_k^{(n)}}{W_k} \sum_t \gamma_t(k)\, \big|O_t - \mu_k^{(n+1)}\big|^{p_k^{(n)}} \right]^{1/p_k^{(n)}}, \qquad W_k \;=\; \sum_t \gamma_t(k); \tag{19}$$

and CM-3 updates the shape by maximising the per-state Q-function over the bracket,

$$p_k^{(n+1)} \;=\; \arg \max_{p \in [p_{\min},\, p_{\max}]} \sum_t \gamma_t(k) \log f_k\big(O_t;\, \mu_k^{(n+1)},\, \alpha_k^{(n+1)},\, p\big), \tag{20}$$

again by a 40-iteration golden-section search over $[p_{\min}, p_{\max}] = [0.5, 3.0]$. Equation (19) is the unique zero of $\partial Q_k / \partial \alpha$, so CM-2 is exact maximization. CM-1 and CM-3 are bracketed maximization. Each CM step is therefore a partial maximization, and the standard ECM monotonicity result [80] applies on the compact bracket; Proposition 1 states the guarantee in full. Boundary cases: $p_k = 2$ recovers Gaussian (with $\sigma = \alpha/\sqrt{2}$); $p_k = 1$ recovers Laplace (with $b = \alpha$).

A.2.2 Algorithm Descriptions

Algorithm 1 is the unified EM procedure shared across all four emission families: a shared scaffold (quantile-based initialisation, log-space forward-backward, transition update) plus four family-specific branches at lines 3, 13, and 30. Algorithm 2 is the simulator. Algorithm 3 is the cross-asset rank-reordering simulator (Pipeline B). Viterbi and SIM-resampling pseudocode is in `CHMM-Model/src/Compute.jl` and `src/CrossAsset.jl` respectively; both are textbook constructions and we omit them from the appendix.

Algorithm 1 Unified EM algorithm for the CHMM family. Lines 3, 13, and 30 branch on the emission family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$; every other line is identical across families. CHMM-N and CHMM-L instantiate classical EM; CHMM-t and CHMM-GED instantiate ECM sequences whose final CM step updates the per-state shape parameter (ν_k for CHMM-t, p_k for CHMM-GED) by a one-dimensional golden-section search on the per-state Q-function. Hyperparameter defaults used throughout this paper: convergence tolerance $\mathrm{tol} = 10^{-4}$ on $\Delta\mathcal{L}$, `max_iter` $= 60$, 40-iteration golden-section sub-step on every bracketed CM, CHMM-t bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$ with initial $\nu^{(0)} = 10$, CHMM-GED bracket $(p_{\min}, p_{\max}) = (0.5, 3.0)$ with initial $p^{(0)} = 1.5$ and μ_k search interval $[\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]$.

Require: Observations $O_{1:T}$, number of states K, tolerance tol, `max_iter`, emission family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$, if $F = t$ bracket $(\nu_{\min}, \nu_{\max})$ and initial $\nu^{(0)}$, if $F = \mathrm{GED}$ bracket $(p_{\min}, p_{\max})$ and initial $p^{(0)}$.
Ensure: Transition matrix $\mathbf{T}$, per-state emission parameters $\boldsymbol{\theta}_{1:K}$, initial distribution $\boldsymbol{\pi}$.

1: **Quantile-Based Initialization.**
2: Sort $O^{(\mathrm{sort})} \leftarrow \mathrm{sort}(O_{1:T})$ and partition into K equal chunks $\{C_1, \ldots, C_K\}$.
3: **for** $k = 1$ to K **do**
4:   **switch** F
5:     $F = \mathcal{N}$: $\mu_k \leftarrow \mathrm{mean}(C_k)$, $\quad \sigma_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$.
6:     $F = t$: $\mu_k \leftarrow \mathrm{mean}(C_k)$, $\quad \sigma_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$, $\quad \nu_k \leftarrow \nu^{(0)}$.
7:     $F = L$: $\mu_k \leftarrow \mathrm{median}(C_k)$, $\quad b_k \leftarrow \max(\mathrm{mean}(|C_k - \mu_k|), 10^{-6})$.
8:     $F = \mathrm{GED}$: $\mu_k \leftarrow \mathrm{mean}(C_k)$, $\quad \alpha_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$, $\quad p_k \leftarrow p^{(0)}$.
9: **end for**
10: $T_{ij} \leftarrow 1/K$; $\quad \pi_k \leftarrow 1/K$; $\quad \mathcal{L}_{\mathrm{prev}} \leftarrow -\infty$.
11: **EM Loop.**
12: **for** $n = 1$ to `max_iter` **do**
13:   **E-Step, per-family emission log-likelihood.**
14:   **switch** F
15:     $F = \mathcal{N}$: $\log B_{t,k} \leftarrow \log \mathcal{N}(O_t \mid \mu_k, \sigma_k^2)$.
16:     $F = t$: $\log B_{t,k} \leftarrow \log t_{\nu_k}(O_t; \mu_k, \sigma_k)$.
17:     $F = L$: $\log B_{t,k} \leftarrow -\log(2b_k) - |O_t - \mu_k|/b_k$.
18:     $F = \mathrm{GED}$: $\log B_{t,k} \leftarrow \log p_k - \log(2\alpha_k) - \log\Gamma(1/p_k) - (|O_t - \mu_k|/\alpha_k)^{p_k}$.
19:   **E-Step, shared log-space forward-backward.**
20:   $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$.
21:   **for** $t = 2$ to T, $j = 1$ to K **do**
22:     $\log \alpha_t(j) \leftarrow \operatorname*{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$.
23:   **end for**
24:   $\log \beta_T(k) \leftarrow 0$.
25:   **for** $t = T - 1$ down to 1, $i = 1$ to K **do**
26:     $\log \beta_t(i) \leftarrow \operatorname*{logsumexp}_j(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$.
27:   **end for**
28:   Compute $\gamma_t(k)$ and $\xi_t(i,j)$ from equation (12).
29:   **Shared transition update.** $T_{ij} \leftarrow \sum_t \xi_t(i,j) \,/\, \sum_t \gamma_t(i)$; $\pi_k \leftarrow \gamma_1(k)$.
30:   **M-Step, per-family emission update.**
31:   **switch** F
32:     $F = \mathcal{N}$ {classical Baum-Welch [16].}
33:       $\mu_k \leftarrow \sum_t \gamma_t(k)\, O_t / \sum_t \gamma_t(k)$.
34:       $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
35:     $F = t$ {ECM of Peel and McLachlan [18], Liu and Rubin [19]; closed-form (μ_k, σ_k) given ν_k, then 1D search on ν_k.}
36:       $u_{t,k} \leftarrow (\nu_k + 1)/(\nu_k + ((O_t - \mu_k)/\sigma_k)^2)$ for all t, k.
37:       $\mu_k \leftarrow \sum_t \gamma_t(k)\, u_{t,k}\, O_t / \sum_t \gamma_t(k)\, u_{t,k}$.
38:       $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
39:       $\nu_k \leftarrow \arg\max_{\nu \in [\nu_{\min}, \nu_{\max}]} \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ {40-iter golden-section search.}
40:     $F = L$ {closed-form weighted Laplace MLE.}
41:       $\mu_k \leftarrow \mathrm{WeightedMedian}(O_{1:T},\ \gamma_{1:T}(k))$ {cumulative-weight bracketing with interpolation at the $W_k/2$ crossing.}
42:       $b_k \leftarrow \max\big(\sum_t \gamma_t(k)\, |O_t - \mu_k| / \sum_t \gamma_t(k),\ 10^{-6}\big)$.
43:     $F = \mathrm{GED}$ {three-stage ECM: bracketed μ_k, closed-form α_k, bracketed p_k.}
44:       $\mu_k \leftarrow \arg\min_{\mu \in [\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]} \sum_t \gamma_t(k)\, |O_t - \mu|^{p_k}$ {40-iter golden-section search.}
45:       $\alpha_k \leftarrow \max\Big(\big[(p_k/W_k) \sum_t \gamma_t(k)\, |O_t - \mu_k|^{p_k}\big]^{1/p_k},\ 10^{-6}\Big)$ where $W_k = \sum_t \gamma_t(k)$ {closed form, unique zero of $\partial Q_k/\partial\alpha$.}
46:       $p_k \leftarrow \arg\max_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k) \log \mathrm{GED}(O_t; \mu_k, \alpha_k, p)$ {40-iter golden-section search.}
47:   **Convergence check.** $\mathcal{L}^{(n)} \leftarrow \operatorname*{logsumexp}_k(\log \alpha_T(k))$.
48:   **if** $|\mathcal{L}^{(n)} - \mathcal{L}_{\mathrm{prev}}| < \mathrm{tol}$ **then**

Algorithm 2 CHMM simulation of synthetic excess growth rate paths

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$, Number of steps M, Number of paths P
Ensure: Simulated return paths $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$

for $p = 1$ to P **do**
  Sample initial state $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$.
  for $t = 1$ to M **do**
    Emit observation $\hat{G}^{(p)}_t \sim \mathcal{N}(\mu_{S_t}, \sigma^2_{S_t})$.
    if $t < M$ **then**
      Transition to next state $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t,\cdot})$.
    end if
  end for
end for
return $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$

Algorithm 3 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^{d}$, correlation matrix Σ (positive semi-definite (PSD), unit diagonal), copula family $\mathcal{C} \in \{\text{Gaussian}, t_\nu\}$, path length T, number of paths P
Ensure: Cross-asset path tensor $\{\hat{G}^{(p)}_{j,t}\}$ of shape $T \times d \times P$

for $p = 1$ to P **do**
  Draw copula sample $\mathbf{U}^{(p)} \in [0,1]^{T \times d}$ from $\mathcal{C}(\Sigma)$:
    $L \leftarrow \text{Cholesky}(\Sigma), \quad Z \in \mathbb{R}^{T \times d} \sim \mathcal{N}(0, I_d)$ i.i.d.
    $Y \leftarrow Z L^\top$
  if $\mathcal{C} =$ Gaussian **then**
      $\mathbf{U}^{(p)} \leftarrow \Phi(Y)$ componentwise
  else
      Draw $W \sim \chi^2_\nu / \nu$ i.i.d. for $t = 1, \ldots, T$; $\quad X_{t,\cdot} \leftarrow Y_{t,\cdot} / \sqrt{W_t}$
      $\mathbf{U}^{(p)} \leftarrow t_\nu(X)$ componentwise
  end if
  for $j = 1$ to d **do**
    Simulate $\tilde{g}^{(p)}_{j,1:T}$ from $\mathcal{M}_j$ via Algorithm 2.
    Compute order statistics $\tilde{g}^{(p)}_{j,(1)} \leq \cdots \leq \tilde{g}^{(p)}_{j,(T)}$.
    Compute ranks $r^{(p)}_{j,t} \leftarrow \text{rank}(U^{(p)}_{j,t})$ for $t = 1, \ldots, T$.
    $\hat{G}^{(p)}_{j,t} \leftarrow \tilde{g}^{(p)}_{j,(r^{(p)}_{j,t})}$ for $t = 1, \ldots, T$.
  end for
end for
return $\{\hat{G}^{(p)}_{j,t}\}$

A.2.3 Per-State Shape Diagnostics: CHMM-t and CHMM-GED

This appendix backs the discussion in Section 5 of the per-state shape allocation under CHMM-t and CHMM-GED on SPY IS at $K = 18$ (seed $=$ 20260420).

CHMM-t per-state ν_k. Figure 2 plots the per-state ν_k histogram for the $K = 18$ CHMM-t fit. Thirteen of the eighteen states rest at the upper ECM bracket ($\nu_k = 50$), and only two states lie near the lower bracket ($\nu_2 = 2.19$, $\nu_4 = 2.10$). The simulated IS mixture kurtosis of 17.34 is driven by the posterior mass routed to those two very heavy-tailed states, not by a systemic collapse to the lower bracket.

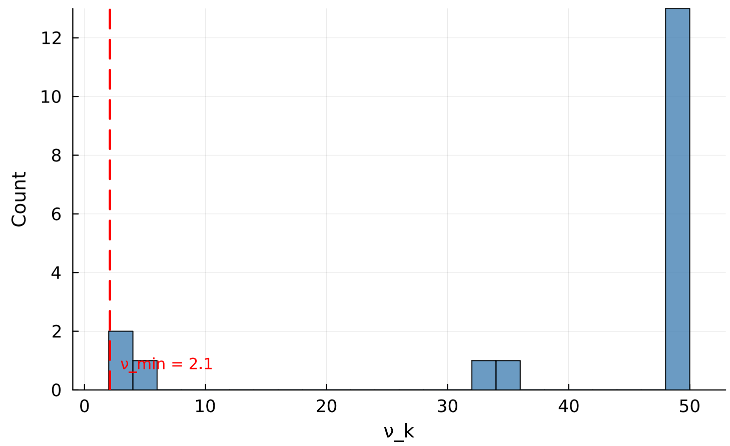


Figure 2. Per-state degrees-of-freedom ν_k for the CHMM-t fit at $K = 18$ on SPY IS. The dashed red line marks the lower ECM bracket $\nu_{\min} = 2.1$. Only two states ($\nu_2 = 2.19$, $\nu_4 = 2.10$) sit near the lower bracket; the median ν_k is at the upper bracket.

CHMM-GED per-state p_k (Gaussian-bulk / Laplace-tail bimodality). The body Section 4.3 reports and discusses the $\hat{p}_k$ partition (Figure 5). Briefly, the $K = 18$ CHMM-GED fit on SPY IS is bimodal: thirteen states attain the upper ECM bracket at $p_k = 3.0$, one state lands at intermediate $p_k = 1.6$, and four states cluster in the Laplace-shape regime $p_k \in [0.86, 1.24]$, concentrated in the high-volatility ranks. The smallest fitted p_k on SPY is 0.86, well clear of every $p_{\min}$ tested in the bracket sweep, so the lower bracket is non-binding. The bimodal partition replicates across 10 Monte Carlo seeds and across the cross-asset universe (Appendix A.7).

A CHMM-t bracket sweep on $\nu_{\min} \in \{2.1, 2.5, 3.0, 4.0, \infty\}$ at $K = 18$ shows the simulated IS kurtosis stays in the 12.5–15.1 band across finite brackets and only collapses to 5.03 in the full Gaussian limit; the overshoot is a feature of the Student-t emission at $K = 18$, not a lower-bracket artefact. The analogous CHMM-GED sweep on $p_{\max} \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ leaves the Gaussian-like-state count invariant at 13/18 and the heavy-shape count in $\{4, 5\}$; tightening $p_{\max}$ from 4.0 to 3.0 costs about 4 nats of LL. The $p_{\min}$ sweep on $\{0.5, 0.7, 0.85\}$ is fully degenerate: every floor produces an identical fit because the smallest observed $\hat{p}_k$ on SPY is 0.86. Full bracket-sensitivity tables at `CHMM-Model/results/SPY/diagnostics/nu_bracket.csv` and `p_bracket.csv`.

A.2.4 Pipeline Schematic

Pipeline A: observed returns $G_t \to$ fit CHMM (Baum-Welch, $K = 18$) $\to$ simulate 1,000 paths $\to$ per-asset metrics (Table 3). Pipeline B reuses the Pipeline-A per-asset fits as marginals, draws a copula sample $\mathbf{U} \sim \mathcal{C}(\Sigma)$ from the fitted Student-t copula on ranks, rank-reorders the per-asset Pipeline-A paths to match $\mathbf{U}$, and reports cross-asset metrics (Table 7). The single-asset scaffold is shared across both pipelines; the CHMM training and simulation steps are Algorithms 1 and 2, the cross-asset rank-reordering step is Algorithm 3, and the architecture diagram for the single-asset scaffold is body Figure 1.

A.3 Validation Metric Details

This appendix documents the precise form of the seven per-path metrics referenced in Section 3.5. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset extension) and each metric is computed per path before aggregation.

Kolmogorov–Smirnov (KS) pass rate. The two-sample KS statistic [72, 73] measures the maximum vertical distance between two empirical CDFs,

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|. \tag{21}$$

The KS pass rate is the fraction of paths for which the test fails to reject distributional equivalence against the reference data at $\alpha = 0.05$.

Anderson–Darling (AD) pass rate. The AD test is the integrated squared CDF deviation with a quadratic tail weighting, making it more sensitive than KS to tail discrepancies. The AD pass rate is the fraction of paths for which the two-sample AD test fails to reject at $\alpha = 0.05$.

Excess kurtosis. Mean simulated excess kurtosis across paths, compared to the observed value. Given Gaussian emissions, the CHMM is expected to underestimate kurtosis relative to the heavy-tailed empirical distribution, and the shortfall is one of the reported diagnostics.

ACF-MAE. Mean absolute error between observed and mean-simulated autocorrelation functions of $|G_t|$ over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^{L} \left| \hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau) \right|. \tag{22}$$

This is the direct diagnostic for the Rydén et al. limitation.

Wasserstein-1 distance. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [81],

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)| \, du. \tag{23}$$

Hellinger distance. A histogram-based overlap distance in $[0, 1]$,

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^{B} (\sqrt{p_i} - \sqrt{q_i})^2}, \tag{24}$$

with B equal-width bins spanning the joint support and p_i, q_i the observed and simulated bin probabilities.

Quantile coverage. The fraction of 99 equally-spaced observed quantiles (1st through 99th percentile) falling inside the 5th–95th percentile envelope of the corresponding simulated quantile across 1,000 paths. A coverage of 100% indicates that the simulation envelope fully brackets the observed distribution at every percentile.

Cross-asset metrics. For the cross-asset extension we additionally track the mean Frobenius norm $\|\hat{\Sigma}^{(p)}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}\|_F$ and the off-diagonal Pearson correlation MAE between simulated and observed return matrices, averaged over paths. The off-diagonal MAE is averaged over the $d(d-1)/2$ unique upper-triangular entries of the symmetric correlation matrix (so 15 entries for the six-asset cross-section of Section 4.5), not double-counted across both triangles:

$$\text{MAE}_{\text{off-diag}} = \frac{1}{P}\sum_{p=1}^{P} \frac{2}{d(d-1)} \sum_{1\le i<j\le d} \left| \hat{\Sigma}^{(p)}_{\text{sim},\,ij} - \hat{\Sigma}_{\text{obs},\,ij} \right|.$$

A.3.1 Continuous Ranked Probability Score and Diebold-Mariano

The OoS CRPS column of Table 3 reports a proper-scoring-rule complement to the KS pass rate. We use the unbiased sample CRPS estimator: for an ensemble of N simulated paths $\{x_i\}_{i=1}^{N}$ at time t and observed value y_t,

$$\widehat{\text{CRPS}}_t = \frac{1}{N}\sum_{i=1}^{N} |x_i - y_t| - \frac{1}{N(N-1)} \sum_{1\le i<j\le N} |x_i - x_j|. \tag{25}$$

The double sum is computed in $O(N \log N)$ via the sorted-ensemble identity $\sum_{i<j}(x_{(j)} - x_{(i)}) = \sum_i x_{(i)}(2i - N - 1)$. The ensemble at each t is the cross-section of the $N = 1{,}000$ unconditional simulated paths used throughout Section 4; this yields a marginal-predictive CRPS suitable for an unconditional generative-fidelity assessment, distinct from the conditional one-step-ahead CRPS used in forecasting.

For pairwise comparison we report a Diebold-Mariano test on the per-t loss differential $d_t = \widehat{\text{CRPS}}_t^{A} - \widehat{\text{CRPS}}_t^{B}$ with a Newey-West HAC variance under a Bartlett kernel and bandwidth $h = \lfloor T_{\text{OoS}}^{1/3} \rfloor = 8$. Two-sided p-values are reported under the standard-normal null. The body claim of Section 4.3 is: CHMM ties or beats every benchmark on mean OoS CRPS, with Gaussian i.i.d. the only significantly worse comparator.

A.3.2 Christoffersen Conditional-Coverage Panel (Companion to Section 4.6)

The body Section 4.6 reports the unconditional Kupiec LR_{uc} statistic. This appendix reports the full Christoffersen [82] panel: LR_{uc} (unconditional coverage), LR_{ind} (breach independence), and $\text{LR}_{\text{cc}} = \text{LR}_{\text{uc}} + \text{LR}_{\text{ind}}$ (joint conditional coverage), at $\alpha \in \{0.01, 0.05\}$ on the SPY IS and OoS windows. The breach series is constructed from the pooled-archive VaR threshold (the α-quantile of vec(sim archive) across paths and time), then $\text{br}_t = (R_t \le \widehat{\text{VaR}}_\alpha)$.

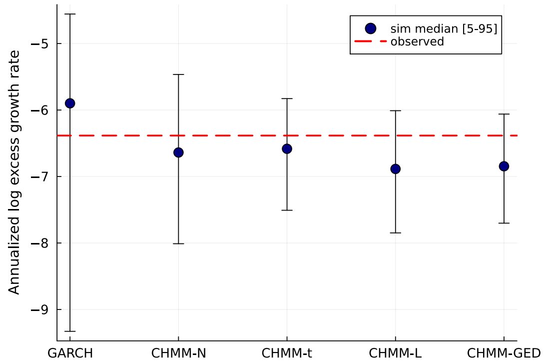
The headline reads as follows. On the unconditional (Kupiec) leg every CHMM variant passes at both levels on both windows; the LR_{uc} statistic for the four CHMM rows at $\alpha = 0.05$ on OoS clusters at 3.03–3.83 (just below the χ^2_1 critical value 3.841), reflecting the integer-breach grid at $T_{\text{OoS}} = 572$ noted in the body table caption. On the independence leg every unconditional generator in the panel (bootstrap, GARCH, and all four CHMM variants) rejects: LR_{ind} is in the range 4.71–56.46 across rows. This is structural rather than CHMM-specific. Volatility-clustering in R_{oos} produces persistent regimes of high and low realised volatility, the OoS-window breaches concentrate inside the high-volatility regime, and a constant-across-t VaR threshold (the construction every unconditional generator uses) cannot track that timing. The natural fix is a regime-conditional VaR that propagates the CHMM latent-state forecast $\mathbb{P}(s_{t+1} \,|\, \text{history}_t)$ through the per-state emission-mixture; this is operationally the conditional-VaR companion-paper direction noted in Section 5.

Table 10. Christoffersen conditional-coverage VaR back-test (SPY, seed = 20260420, $K = 18$, 1,000 simulated paths). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. "br rate" is the empirical breach rate; $\mathrm{LR}_{\mathrm{uc}}$ is the Kupiec statistic ($\sim \chi^2_1$ under correct unconditional coverage); $\mathrm{LR}_{\mathrm{ind}}$ is the Christoffersen independence statistic ($\sim \chi^2_1$ under independent breaches); $\mathrm{LR}_{\mathrm{cc}} = \mathrm{LR}_{\mathrm{uc}} + \mathrm{LR}_{\mathrm{ind}}$ is the joint conditional-coverage statistic ($\sim \chi^2_2$). PASS if statistic is below critical. Every unconditional generator (i.i.d. bootstrap, GARCH, and all four CHMM variants) rejects independence on OoS at $\alpha = 0.05$, consistent with breach clustering driven by volatility-clustering in R_{oos} rather than a CHMM-specific mis-specification of conditional dynamics.

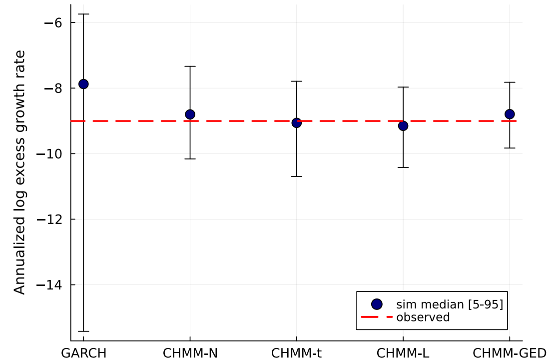
					Kupiec		Christ. ind		Christ. cc	
Model	Win	α	breaches	br rate	$\mathrm{LR}_{\mathrm{uc}}$	p	$\mathrm{LR}_{\mathrm{ind}}$	p	$\mathrm{LR}_{\mathrm{cc}}$	p
Bootstrap	IS	0.01	26	1.03%	0.03	0.87	15.28	0.00	15.31	0.00
Bootstrap	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
Bootstrap	OoS	0.01	5	0.87%	0.10	0.76	13.18	0.00	13.28	0.00
Bootstrap	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
GARCH(1,1)	IS	0.01	31	1.23%	1.28	0.26	12.45	0.00	13.72	0.00
GARCH(1,1)	IS	0.05	139	5.52%	1.41	0.24	45.38	0.00	46.79	0.00
GARCH(1,1)	OoS	0.01	6	1.05%	0.01	0.91	20.87	0.00	20.89	0.00
GARCH(1,1)	OoS	0.05	24	4.20%	0.82	0.37	5.87	0.02	6.69	0.04
CHMM-N	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-N	IS	0.05	123	4.89%	0.07	0.80	47.33	0.00	47.40	0.00
CHMM-N	OoS	0.01	3	0.52%	1.58	0.21	18.98	0.00	20.56	0.00
CHMM-N	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
CHMM-t	IS	0.01	22	0.87%	0.42	0.52	11.62	0.00	12.04	0.00
CHMM-t	IS	0.05	127	5.05%	0.01	0.91	55.54	0.00	55.55	0.00
CHMM-t	OoS	0.01	4	0.70%	0.58	0.45	15.53	0.00	16.12	0.00
CHMM-t	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-L	IS	0.01	19	0.76%	1.66	0.20	13.44	0.00	15.11	0.00
CHMM-L	IS	0.05	134	5.33%	0.55	0.46	49.43	0.00	49.98	0.00
CHMM-L	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-L	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-GED	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-GED	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
CHMM-GED	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-GED	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01

A.3.3 VaR / ES Envelope Visualization (Companion to Table 23)

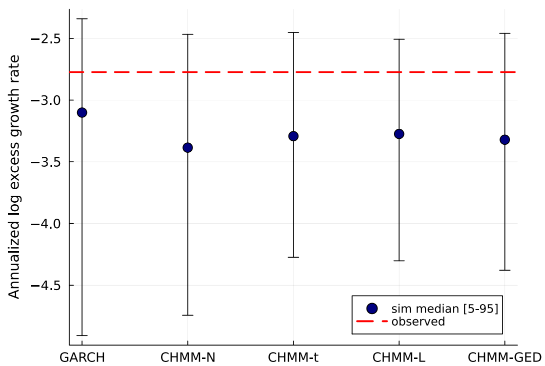
Figure 3 visualizes the median and $[5\%, 95\%]$ envelopes across simulated paths whose numeric values are reported in Table 23.



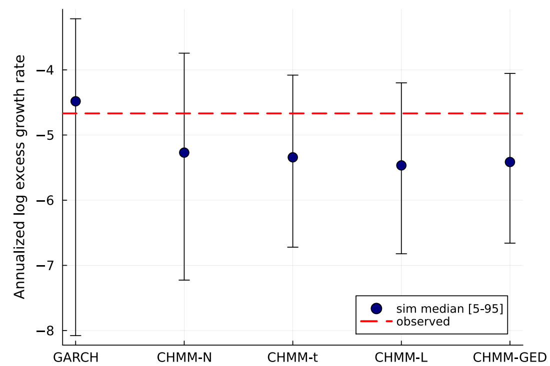
(a) IS VaR ($\alpha = 0.01$).



(b) IS ES ($\alpha = 0.01$).



(c) OoS VaR ($\alpha = 0.05$).



(d) OoS ES ($\alpha = 0.05$).

Figure 3. VaR / ES envelope diagnostic (SPY, $N_{\text{paths}} = 1{,}000$; global seed policy in Section 3.5). Each panel shows the simulated median (dot) and $[5\%, 95\%]$ envelope (error bar) across paths against the observed historical value (dashed red line); panels (a)/(b) are the IS window at $\alpha = 0.01$, panels (c)/(d) are the OoS window at $\alpha = 0.05$, and the x-axis in each panel lists the five generators kept for the volatility-aware conditional-coverage comparison (GARCH(1,1), CHMM-N, CHMM-t, CHMM-L, CHMM-GED). The i.i.d. bootstrap is omitted here because its envelope is by construction matched to the empirical marginal and does not discriminate among volatility-aware models. The y-axis is the annualized log excess growth rate (daily log return scaled by $1/\Delta t$ with $\Delta t = 1/252$), consistent with the $R_{\text{IS}}/R_{\text{OoS}}$ convention used throughout the paper. Headline: every observed red line falls inside the corresponding simulated envelope, confirming that the CHMM family passes the coverage leg of the back-test complementary to the Kupiec likelihood-ratio leg in Section 4.6.

A.4 Formal Theoretical Statements

The following assumptions and propositions are referenced in Section 3.8 and stated formally here for reference; each is a one-line specialisation of a standard result and the proof sketch is one sentence.

Assumption 1 (Irreducibility and aperiodicity)**.** *The transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$ is irreducible and aperiodic on the state space $\{1, \ldots, K\}$, so the chain admits a unique stationary distribution $\bar{\boldsymbol{\pi}}$ satisfying $\bar{\boldsymbol{\pi}}\,\mathbf{T} = \bar{\boldsymbol{\pi}}$ and $\lambda_1 = 1$ is a simple eigenvalue with all other eigenvalues strictly inside the unit disk [76].*

Assumption 2 (Finite second moment and emission contrast)**.** *Each per-state emission density $f_k(\cdot;\boldsymbol{\theta}_k)$ has finite second moment, $\mathbb{E}_{f_k}[G^2] < \infty$, and at least two states differ in either location μ_k or scale σ_k.*

Proposition 1 (ECM monotonicity for CHMM-t and CHMM-GED (and EM monotonicity for CHMM-N, CHMM-L))**.** *Under compactness of the parameter space (μ_k, σ_k or β_k or α_k, $\nu_k \in [\nu_{\min}, \nu_{\max}]$, $p_k \in [p_{\min}, p_{\max}]$ each in a closed bounded interval, $T_{ij} \geq \epsilon_T > 0$), the two-block CM step for CHMM-t of Section 3.7, namely the closed-form weighted updates of $(\mathbf{T}, \boldsymbol{\pi}, \mu_k, \sigma_k)$ at fixed ν_k, followed by a golden-section maximisation of the per-state Q-function over ν_k, satisfies $Q(\theta^{(n+1)} \mid \theta^{(n)}) \geq Q(\theta^{(n)} \mid \theta^{(n)})$, and consequently $\mathcal{L}(\theta^{(n+1)}) \geq \mathcal{L}(\theta^{(n)})$ where $\mathcal{L}$ is the incomplete-data log-likelihood. The same monotonicity holds for the closed-form CHMM-N and CHMM-L M-steps under the standard EM guarantee, and for the three-stage CHMM-GED CM sequence (bracketed μ_k, closed-form α_k, bracketed p_k) of equations (18)–(20).*

Proof sketch. Each CM step does not decrease Q by definition (closed-form steps are exact maximisers, golden-section steps are bracketed maxima within the prescribed interval); the standard EM inequality $\mathcal{L}(\theta^{(n+1)}) - \mathcal{L}(\theta^{(n)}) \geq Q(\theta^{(n+1)} \mid \theta^{(n)}) - Q(\theta^{(n)} \mid \theta^{(n)})$ [80, 83] converts Q-monotonicity into log-likelihood monotonicity. Compactness gives boundedness above, hence convergence.

Numerical full-rank check on $\hat{\mathbf{T}}$. The Allman-Matias-Rhodes 2009 identifiability theorem requires $\hat{\mathbf{T}}$ of full rank. We compute the singular-value spectrum of $\hat{\mathbf{T}}$ at $K = 18$ across the four emission families on SPY IS (Table 11). All four matrices are full-rank in the strict sense, with smallest singular value $\sigma_{\min}(\hat{\mathbf{T}}) \in [0.0007, 0.0170]$ and condition numbers in the range 63–1,620. CHMM-GED is the most well-conditioned ($\sigma_{\min} = 0.017$, cond $= 63.5$); CHMM-t is the least ($\sigma_{\min} = 0.0007$, cond $=$ 1,620), consistent with the per-state ν_k ECM lower-bracket pinning making one or two heavy-tail states near-degenerate. The deflated matrix $\hat{\mathbf{T}} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ has numerical rank exactly $K - 1 = 17$ across all four families, the rank statement underlying Section 3.8.

Table 11. Numerical full-rank check on $\hat{\mathbf{T}}$ at $K = 18$ across four CHMM emission families on SPY IS (seed = 20260420). $\sigma_{\min}$, $\sigma_{\max}$, and the condition number $\sigma_{\max}/\sigma_{\min}$ are reported on $\hat{\mathbf{T}}$ itself; $|\lambda_2|$ is the second-largest absolute eigenvalue (the dominant decay mode in the ACF identity, equation (6)); "rank(def)" is the numerical rank of $\hat{\mathbf{T}} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ at tolerance 10^{-10}, expected to equal $K - 1 = 17$.

Family	$\sigma_{\min}(\hat{\mathbf{T}})$	$\sigma_{\max}(\hat{\mathbf{T}})$	cond $\hat{\mathbf{T}}$	$\lvert\lambda_2\rvert$	rank$(\hat{\mathbf{T}} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top)$
CHMM-N	0.00532	1.0866	204	0.929	17/17
CHMM-t	0.00067	1.0796	1,620	0.914	17/17
CHMM-L	0.00398	1.0928	275	0.896	17/17
CHMM-GED	0.01702	1.0806	63.5	0.911	17/17

Proposition 2 (Identifiability up to label permutation)**.** *Under irreducibility and aperiodicity of* $\mathbf{T}$ *(Assumption 1), the emission contrast condition (Assumption 2), and pairwise distinctness of the per-state parameter tuples $\boldsymbol{\theta}_k$ across $k = 1, \ldots, K$, the CHMM parameter is identifiable up to label permutation for CHMM-N (Gaussian) and CHMM-L (Laplace). For CHMM-t (with per-state $\nu_k \in [\nu_{\min}, \nu_{\max}]$) and CHMM-GED (with per-state $p_k \in [p_{\min}, p_{\max}]$) the same conclusion holds modulo the standard caveat that pairwise distinctness must hold jointly across all components of $\boldsymbol{\theta}_k = (\mu_k, \sigma_k, \nu_k)$ or (μ_k, α_k, p_k), since within either bracket a Gaussian-shaped emission with inflated scale can mimic a Laplace-shaped emission with deflated scale at intermediate quantiles.*

Proof sketch. The HMM identifiability theorem of Allman et al. [84] requires $\mathbf{T}$ of full rank and per-state emissions linearly independent in the space of densities on $\mathbb{R}$. Full-rank $\mathbf{T}$ follows from irreducibility [76]. For CHMM-N and CHMM-L, linear independence of Gaussian and Laplace emissions with distinct location-scale parameters is direct from Yakowitz and Spragins [85], who establish that finite mixtures over location-scale families with strictly decreasing-tail densities are identifiable (the CHMM-N and CHMM-L marginals are special cases). For CHMM-t at fixed shape and CHMM-GED at fixed shape, location-scale identifiability is the same Yakowitz-Spragins result; for finite mixtures of Student-t with varying degrees-of-freedom the standard reference is Lindsay [86], Chapter 3, which gives identifiability via the moment-generating function over the bracketed range of shape parameters and a non-degeneracy condition on the joint location-scale-shape tuple. The CHMM-GED case under bracketed $p_k \in [p_{\min}, p_{\max}]$ is a parallel application of the moment-matching argument: the GED characteristic function $\hat{b}(t; \mu, \alpha, p) = \exp(i\mu t)\, M(t; \alpha, p)$ has a unique inverse over the bounded shape range up to permutation of components, modulo the joint-distinctness caveat on (μ_k, α_k, p_k). Strict linear independence at the boundary cases $p_k = 1$ (Laplace) and $p_k = 2$ (Gaussian) is automatic from Yakowitz-Spragins; in the bracket interior, the joint distinctness condition above is required to rule out the location-scale-shape trade-off.

Proposition 3 (MLE consistency)**.** *Under Assumptions 1–2 and the assumption that the true parameter lies in the interior of a compact Θ, the MLE $\hat{\theta}_T = \arg\max_{\theta \in \Theta} \mathcal{L}_T(\theta)$ is consistent: $\hat{\theta}_T \xrightarrow{p} \theta_0$ as $T \to \infty$, modulo label permutation.*

Proof reference. Bickel et al. [87] (Theorem 1) and Douc et al. [88] apply directly under our assumption set: irreducibility supplies the mixing condition, finite second moment the moment condition, compactness the parameter-space condition.

Proposition 4 (Marginal preservation under rank reordering)**.** *Let $\tilde{g}_{j,1:T}$ be a sample of size T from the per-asset CHMM marginal of asset j, and let the rank-reordered output be $\hat{g}^{(p)}_{j,t} = \tilde{g}^{(p)}_{j,(r^{(p)}_{j,t})}$ where $r^{(p)}_{j,t}$ is the rank of the copula sample. Then for every asset j the empirical CDF of $\{\hat{g}^{(p)}_{j,t}\}_{t=1}^{T}$ equals that of $\{\tilde{g}^{(p)}_{j,t}\}_{t=1}^{T}$.*

Proof sketch. The map $t \mapsto r^{(p)}_{j,t}$ is a permutation almost surely, so the multiset of values is preserved; the empirical CDF depends only on the multiset.

A.5 Spectral ACF Identity: Extended Derivation

Equation (6) is the closed-form mixture-of-eigenvalues form for the absolute-return ACF used in Section 3.8 and invoked throughout Sections 4–5. The bilinear cross-moment identity $\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \mathbf{m}^\top \operatorname{diag}(\bar{\boldsymbol{\pi}})\, \mathbf{T}^\tau\, \mathbf{m}$ and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are textbook material in the regime-switching literature (13, Section 22.2; 14, Chapter 3; the moment-generating-function form for general Markov-switching specifications with conditional means and variances depending on the latent state is given by 15); the underlying Markov-chain spectral decomposition is standard [76]. We restate the absolute-return specialisation here as a self-contained theorem with a step-by-step proof, and record the lag-zero behaviour and the complex / non-diagonalisable extensions that the main body does not. Our contribution on this axis is the explicit application of the spectral form to recast the empirical Rydén et al. [1] low-K failure as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ (Section 3.8, "Rydén separation as a K-rank statement") and the empirical demonstration that the rank constraint is the binding axis at moderate K on equity-return data (Sections 4, 5).

Theorem 1 (Mixture-of-eigenvalues identity for the absolute-return ACF)**.** *Let* $\{(s_t, G_t)\}_{t\in\mathbb{Z}}$ *be a stationary CHMM on* $\{1,\ldots,K\}$ *with irreducible aperiodic transition matrix* $\mathbf{T}$ *(Assumption 1) and per-state emissions of finite second moment (Assumption 2). Write* $\bar{\boldsymbol{\pi}}$ *for the unique stationary distribution,* $\mathbf{D} = \mathrm{diag}(\bar{\boldsymbol{\pi}})$*,* $m_k = \mathbb{E}[|G_t| \mid s_t = k]$*,* $\mathbf{m} = (m_1,\ldots,m_K)^\top$*,* $\mu = \mathbb{E}|G_t| = \bar{\boldsymbol{\pi}}^\top \mathbf{m}$*, and* $\sigma^2_{|G|} = \mathrm{Var}(|G_t|)$*. Assume* $\mathbf{T}$ *is diagonalisable with right eigenvectors* $\mathbf{v}_k$ *and left eigenvectors* $\mathbf{w}_k$ *biorthonormal,* $\mathbf{w}_k^\top \mathbf{v}_l = \delta_{kl}$*, indexed so that* $\lambda_1 = 1 > |\lambda_2| \geq \cdots \geq |\lambda_K|$*. Then for every integer lag* $\tau \geq 1$*,*

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} w_k\, \lambda_k^\tau, \qquad w_k \;=\; \frac{(\mathbf{m}^\top \mathbf{D}\, \mathbf{v}_k)\,(\mathbf{w}_k^\top \mathbf{m})}{\sigma^2_{|G|}}.$$

Proof. The proof is organised in six steps.

Step 1 (Definitions). The autocorrelation of $|G_t|$ at lag $\tau \geq 0$ is, by stationarity,

$$\rho_{|G|}(\tau) \;=\; \frac{\mathrm{Cov}(|G_t|,\, |G_{t+\tau}|)}{\sigma^2_{|G|}} \;=\; \frac{\mathbb{E}[|G_t|\,|G_{t+\tau}|] - \mu^2}{\sigma^2_{|G|}}.$$

The task reduces to evaluating $\mathbb{E}[|G_t|\,|G_{t+\tau}|]$ in closed form for $\tau \geq 1$.

Step 2 (Conditioning on the latent state pair). For $\tau \geq 1$, by the law of total expectation conditional on $(s_t, s_{t+\tau})$,

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] \;=\; \sum_{i,j=1}^{K} \mathbb{P}(s_t = i, s_{t+\tau} = j)\, \mathbb{E}\big[|G_t|\,|G_{t+\tau}| \,\big|\, s_t = i, s_{t+\tau} = j\big]\,.$$

The HMM emission structure gives the conditional independence $G_t \perp G_{t+\tau} \mid (s_t, s_{t+\tau})$ for $\tau \geq 1$: each emission depends only on its current state, and integrating the intermediate states $s_{t+1:t+\tau-1}$ out of the full-sequence factorisation $\big(\prod_u b_{s_u}(G_u)\big)\, p(s_{t+1:t+\tau-1} \mid s_t, s_{t+\tau})$ leaves the endpoint emissions decoupled. Hence

$$\mathbb{E}\big[|G_t|\,|G_{t+\tau}| \,\big|\, s_t = i, s_{t+\tau} = j\big] \;=\; \mathbb{E}[|G_t| \mid s_t = i]\, \mathbb{E}[|G_{t+\tau}| \mid s_{t+\tau} = j] \;=\; m_i\, m_j.$$

The joint state probability factors by stationarity and the Markov property, $\mathbb{P}(s_t = i, s_{t+\tau} = j) = \bar{\pi}_i\, (\mathbf{T}^\tau)_{ij}$. Substituting and noting $[\mathbf{D}\mathbf{T}^\tau]_{ij} = \bar{\pi}_i (\mathbf{T}^\tau)_{ij}$,

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] \;=\; \mathbf{m}^\top \mathbf{D}\, \mathbf{T}^\tau\, \mathbf{m}, \qquad \tau \geq 1. \tag{26}$$

Step 3 (Spectral preliminaries). For irreducible aperiodic $\mathbf{T}$, Perron–Frobenius applied to row-stochastic matrices gives $\lambda_1 = 1$ as a simple eigenvalue with all other eigenvalues strictly inside the unit disk [76]. The right eigenvector for λ_1 is $\mathbf{v}_1 = \mathbf{1}$, since $\mathbf{T}\mathbf{1} = \mathbf{1}$ from the row-sum constraint, and the left eigenvector is $\mathbf{w}_1 = \bar{\boldsymbol{\pi}}$, since $\bar{\boldsymbol{\pi}}^\top \mathbf{T} = \bar{\boldsymbol{\pi}}^\top$ by definition of the stationary distribution. Biorthonormality $\mathbf{w}_1^\top \mathbf{v}_1 = \bar{\boldsymbol{\pi}}^\top \mathbf{1} = 1$ is automatic since $\bar{\boldsymbol{\pi}}$ is a probability vector.

Step 4 (Spectral decomposition of $\mathbf{T}^\tau$). Diagonalisability with biorthonormality gives the dyadic expansion

$$\mathbf{T}^\tau \;=\; \sum_{k=1}^{K} \lambda_k^\tau\, \mathbf{v}_k \mathbf{w}_k^\top \;=\; \mathbf{1}\, \bar{\boldsymbol{\pi}}^\top \;+\; \sum_{k=2}^{K} \lambda_k^\tau\, \mathbf{v}_k \mathbf{w}_k^\top,$$

isolating the dominant ($k = 1$) projection from the contractive spectrum. The dominant projector $\mathbf{P}_1 = \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ is the rank-one stationary projection: $\mathbf{P}_1 \mathbf{x} = (\bar{\boldsymbol{\pi}}^\top \mathbf{x})\, \mathbf{1}$.

Step 5 (Evaluation of the bilinear form). Substituting the spectral decomposition into (26),

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] \;=\; \mathbf{m}^\top \mathbf{D}\,\mathbf{1}\,\bar{\boldsymbol{\pi}}^\top \mathbf{m} \;+\; \sum_{k=2}^{K} \lambda_k^\tau\,(\mathbf{m}^\top \mathbf{D}\,\mathbf{v}_k)\,(\mathbf{w}_k^\top \mathbf{m}).$$

The $k = 1$ term simplifies to μ^2: $\mathbf{D1}$ is the column vector $\bar{\boldsymbol{\pi}}$, so $\mathbf{m}^\top\mathbf{D1} = \bar{\boldsymbol{\pi}}^\top\mathbf{m} = \mu$, and likewise $\bar{\boldsymbol{\pi}}^\top\mathbf{m} = \mu$, giving the product $\mu \cdot \mu = \mu^2$. Denoting the $k \geq 2$ scalar coefficients

$$c_k \;=\; (\mathbf{m}^\top \mathbf{D}\,\mathbf{v}_k)\,(\mathbf{w}_k^\top \mathbf{m}),$$

we obtain

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] \;=\; \mu^2 \;+\; \sum_{k=2}^{K} c_k\,\lambda_k^\tau, \qquad \tau \geq 1.$$

Step 6 (Autocovariance and autocorrelation). Subtracting μ^2 from Step 1 cancels the dominant term, leaving

$$\mathrm{Cov}(|G_t|,\,|G_{t+\tau}|) \;=\; \sum_{k=2}^{K} c_k\,\lambda_k^\tau, \qquad \tau \geq 1, \qquad c_k = \big(\mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{v}_k\big)\big(\mathbf{w}_k^\top \mathbf{m}\big).$$

Dividing by $\sigma_{|G|}^2 = \gamma(0)$ gives $\rho_{|G|}(\tau) = \sum_{k\geq 2} w_k\,\lambda_k^\tau$ with $w_k = c_k/\sigma_{|G|}^2$, equation (6). $\square$

Lag zero, the sum of weights, and the within-state white-noise contribution. The bilinear identity (26) fails at $\tau = 0$, where $G_t = G_{t+\tau}$ and the conditional product is the per-state second moment $M_i = \mathbb{E}[G_t^2 \mid s_t = i]$ rather than m_i^2. The eigenvalue completeness $\sum_{k=1}^{K} \mathbf{v}_k\mathbf{w}_k^\top = I$ instead gives the identity

$$\sum_{k=2}^{K} c_k \;=\; \mathbf{m}^\top\mathbf{D}\,\mathbf{m} \;-\; \mu^2 \;=\; \sum_{i=1}^{K} \bar{\pi}_i\,m_i^2 \;-\; \mu^2 \;=\; \mathrm{Var}\big(m_{s_t}\big),$$

the between-state variance of $|G_t|$, equal to the variance of the conditional-mean random variable m_{s_t} under $\bar{\boldsymbol{\pi}}$. By the law of total variance, $\sigma_{|G|}^2 = \mathrm{Var}(m_{s_t}) + \mathbb{E}[\mathrm{Var}(|G_t| \mid s_t)]$, so

$$\sum_{k=2}^{K} w_k \;=\; \frac{\mathrm{Var}(m_{s_t})}{\sigma_{|G|}^2} \;\in\; [0,1],$$

with equality to 1 if and only if every per-state distribution is degenerate at m_i. The residual $1 - \sum_{k\geq 2} w_k = \mathbb{E}[\mathrm{Var}(|G_t| \mid s_t)]\,/\,\sigma_{|G|}^2$ is the within-state variance ratio: it contributes to $\sigma_{|G|}^2$ but not to $\mathrm{Cov}(|G_t|, |G_{t+\tau}|)$ at any positive lag, and is the source of the discontinuity from $\rho_{|G|}(0) = 1$ to $\rho_{|G|}(1) = \sum_{k\geq 2} w_k\,\lambda_k$ at the first lag. The substantive consequence for moderate-K CHMM is unchanged: at every $\tau \geq 1$ the autocorrelation lies in the span of the geometric decays $\{\lambda_k^\tau\}_{k\geq 2}$, and the slow-decay envelope is governed by the largest non-unit eigenvalue.

Squared-return analogue. Replacing $|G_t|$ with G_t^2 and $\mathbf{m}$ with $\mathbf{M} = (M_1, \ldots, M_K)^\top$ in Steps 1–6 above is termwise valid for $\tau \geq 1$ (the conditional independence and stationarity arguments are unaffected), giving the squared-return analogue

$$\rho_{G^2}(\tau) \;=\; \sum_{k=2}^{K} \tilde{w}_k\,\lambda_k^\tau, \qquad \tilde{w}_k \;=\; \frac{(\mathbf{M}^\top\mathbf{D}\,\mathbf{v}_k)\,(\mathbf{w}_k^\top\mathbf{M})}{\sigma_{G^2}^2}.$$

The non-unit spectrum of $\mathbf{T}$ that determines $\rho_{|G|}(\tau)$ also determines $\rho_{G^2}(\tau)$; only the linear projection of the per-state moment vector onto the eigenmodes changes.

Remark (complex eigenvalues). For real $\mathbf{T}$ without self-adjoint structure, eigenvalues outside $\mathbb{R}$ occur in complex conjugate pairs with conjugate eigenvectors. A pair $(\lambda_k, \bar{\lambda}_k)$ at $\lambda_k = re^{i\theta}$ contributes a real damped-oscillation term of the form $A\, r^\tau \cos(\theta\tau) + B\, r^\tau \sin(\theta\tau)$ to $\gamma(\tau)$, with (A, B) determined from $(\operatorname{Re} c_k, \operatorname{Im} c_k)$. Theorem 1 as stated remains valid; only the interpretation of the individual modes shifts from monotone geometric to damped sinusoidal. For $\mathbf{T}$ estimated on equity returns at the operating point of Section 4, the slow-decay mode at $\lambda_2 \approx \exp(-1/\tau^*)$ is real and positive, and the mono-exponential interpretation of the dominant timescale used in Section 3.8 carries through.

Remark (non-diagonalisable T). Diagonalisability is generic: for $\mathbf{T}$ estimated by Baum–Welch in the interior of a non-degenerate parameter region, the eigenvalues are almost surely distinct, and the sample paths used in this paper realise this generic case at every K in the sweep. Should $\mathbf{T}$ admit a non-trivial Jordan block of size J, the dyadic expansion of Step 4 acquires polynomial-in-τ prefactors $\binom{\tau}{j} \lambda_k^{\tau-j}$ for $j = 0, \ldots, J-1$ on the off-diagonal blocks, and $\rho_{|G|}(\tau)$ becomes a finite sum of polynomial-times-geometric terms with the same exponential rate $\max_{k\geq 2} |\lambda_k|$. The substantive content of Theorem 1, that the absolute-return ACF is determined entirely by the non-unit spectrum of $\mathbf{T}$, is unchanged.

A.6 State-Resolution and Multi-Emission Sensitivity

This appendix reports the full K-sweep underlying the headline operating point of Section 4, the multi-emission family extension confirming that $K = 18$ is stable under Student-t and Laplace emissions, the EM convergence curves at representative K, the IS and OoS visual comparison panels at non-selected $K \in \{3, 6, 12, 21\}$, the multi-emission figure panels at $K = 18$, and the Rydén $K = 2$ replication.

A.6.1 Full State-Resolution Sensitivity Table (companion artefact)

The full K-sweep table for SPY CHMM-N referenced in Section 4.2 (KS, AD, kurtosis, W_1, Hellinger, OoS coverage at $K \in \{3, 6, 9, 12, 15, 18, 21\}$, 1,000 paths, seed 20260420) is at `CHMM-Model/results/SPY/Table-T1-State-Resolution-Sensitivity.txt`. The qualitative pattern is that distributional pass rates plateau in the $K \in \{12, 15, 18, 21\}$ band and ACF-MAE is essentially flat across the entire sweep, consistent with the spectral mechanism of Section 3.8.

A.6.2 Sector-Balanced 30-Ticker Panel: Full Per-Ticker Rollup

The body Table 6 reports the aggregate distribution and per-sector medians for the sector-balanced 30-ticker panel. Table 12 below gives the full per-ticker rollup. Each ticker is fit independently under the penalised CHMM-t scaffold ($K = 18$, $\lambda = 20$, 1,000 simulated paths, seed 20260420) on its own IS slice covering 2014-01-03 to 2024-01-03 and validated on the same 2024-01-04 to 2026-04-20 OoS window used for SPY in Table 3. The per-ticker ν_k medians sit at the upper bracket (50) on every ticker, indicating that the $1/\nu_k$ shrinkage is in the operative regime across the panel; minimum and maximum ν_k values are in the per-ticker CSV in `results/sector_panel/sector_panel_summary.csv`.

Table 12. Sector-balanced 30-ticker panel, per-ticker rollup (penalised CHMM-t at $K = 18$, $\lambda = 20$). Sectors ordered by GICS classification; tickers within sector ordered by the panel construction (top three large-cap representatives at IS-window-median market cap). Kurt resid is simulated minus observed IS excess kurtosis. Aggregate distribution and per-sector medians are summarised in body Table 6.

Sector	Ticker	IS KS%	OoS KS%	Kurt obs	Kurt sim	Kurt resid	$\lvert G_t \rvert$ ACF-MAE
Information Technology	AAPL	99.1	95.1	3.19	2.84	−0.35	0.0418
Information Technology	MSFT	99.7	96.9	4.20	3.48	−0.72	0.0389
Information Technology	NVDA	99.6	55.7	5.53	6.23	0.70	0.0474
Health Care	JNJ	99.3	93.2	8.97	11.65	2.68	0.0335
Health Care	UNH	99.3	14.5	8.85	10.31	1.46	0.0382
Health Care	LLY	99.7	7.6	13.60	66.51	52.91	0.0295
Financials	JPM	99.5	50.5	7.62	8.14	0.52	0.0458
Financials	BAC	99.6	84.8	5.76	5.47	−0.29	0.0394
Financials	WFC	98.2	57.6	7.53	10.90	3.37	0.0791
Consumer Discretionary	AMZN	99.4	96.9	4.31	3.68	−0.63	0.0413
Consumer Discretionary	HD	99.2	41.4	12.92	17.03	4.11	0.0368
Consumer Discretionary	MCD	99.5	66.6	18.38	26.93	8.55	0.0394
Communication Services	NFLX	99.3	45.1	12.60	4.67	−7.93	0.0320
Communication Services	VZ	99.3	49.1	5.14	5.15	0.01	0.0270
Communication Services	DIS	99.5	96.4	10.77	13.94	3.17	0.0644
Industrials	CAT	99.7	78.9	4.34	3.48	−0.86	0.0321
Industrials	BA	97.6	62.2	20.59	52.00	31.41	0.0793
Industrials	HON	99.3	96.0	21.93	17.09	−4.84	0.0571
Consumer Staples	PG	99.7	94.3	8.15	9.51	1.36	0.0347
Consumer Staples	KO	99.5	92.7	10.57	12.84	2.27	0.0457
Consumer Staples	WMT	99.7	24.0	13.84	32.63	18.79	0.0279
Energy	XOM	96.9	70.3	5.61	10.99	5.38	0.0988
Energy	CVX	99.6	62.3	12.22	14.78	2.56	0.0498
Energy	COP	99.4	83.6	17.11	8.28	−8.83	0.0494
Utilities	NEE	98.6	24.0	10.71	9.87	−0.84	0.0455
Utilities	DUK	99.7	96.9	9.75	8.11	−1.64	0.0466
Utilities	SO	99.6	96.2	14.52	12.73	−1.79	0.0481
Materials	FCX	99.8	76.6	4.83	4.11	−0.72	0.0573
Materials	NEM	99.6	5.6	3.74	3.88	0.14	0.0376
Materials	APD	99.2	90.2	8.40	16.74	8.34	0.0381

A.6.3 60-Ticker Sector Expansion at $n = 6$ per Sector

An $n = 6$ per-sector expansion (60 tickers, 3 additional large-cap representatives per GICS sector) confirms the body $n = 3$ ANOVA reading at adequate power: $F(9, 50) = 0.37$, $p = 0.946$, $\eta^2 = 0.062$ (vs. underpowered $n = 3$ $\eta^2 = 0.16$); aggregate OoS KS median 73.45% and failure rate 22/60 = 36.7% are statistically indistinguishable from the body 30-ticker numbers (73.4%, 11/30). Cross-sector dispersion accounts for 6.2% of OoS KS variance at the larger sample size; the remaining 93.8% is per-ticker. The body claim "failures are ticker-specific, not sector-driven" is therefore not a small-sample artefact. Full panel and per-sector rollup at `CHMM-Model/results/sector_panel_n6/sector_panel_n6.csv` (runners: `run_sector_panel_n6.jl`, `run_sector_panel_n6_postprocess.jl`; archived in `_attic_v10/` during the arXiv-prep pass).

A.6.4 Effective Spectral Rank of the Absolute-Return ACF

The body Section 3.8 states the algebraic upper bound: the deflated transition matrix $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ has rank at most $K-1$, so the absolute-return ACF identity (6) contains at most $K-1$ non-unit decay modes. The empirical question is how many of those modes carry non-trivial contribution to $\rho_{|G|}(\tau)$ at the relevant lags. Table 13 reports the per-mode contribution $w_k\lambda_k^\tau$ at $\tau \in \{1, 5, 20, 50\}$ for the fitted CHMM-N at $K = 18$ and at $K = 3$ on SPY IS, with modes ranked by lag-1 contribution magnitude $|w_k\lambda_k|$. Per-state moments $m_k = \mathbb{E}[|G_t| \mid s_t = k]$ are estimated by 200,000 draws from each emission; eigenvectors are normalised so $\mathbf{w}_k^\top \mathbf{v}_k = 1$.

Table 13. Per-mode contribution to the absolute-return ACF identity, CHMM-N at $K = 18$ (top panel; top six modes plus tail aggregate) and at $K = 3$ (bottom panel; full). Modes ordered by $|w_k\lambda_k|$ descending. "cum" is cumulative share of $\sum_k |w_k\lambda_k|$. At $K = 18$ the top mode ($|\lambda| = 0.929$) carries 93.6% of the lag-1 ACF, three modes reach 95%, and at lag 20 only the dominant mode contributes non-negligibly. At $K = 3$ the dominant mode ($|\lambda| = 0.953$) alone carries 96.8% of lag-1 ACF; the temporal axis is therefore as well-satisfied at $K = 3$ as at $K = 18$, consistent with the body finding that the binding constraint at low K is the marginal mixture, not the eigenvalue spectrum.

rank	$\lvert\lambda_k\rvert$	$\lvert w_k\rvert$	$\lvert w_k\lambda_k\rvert$	$w_k\lambda_k^5$	$w_k\lambda_k^{20}$	cum
$K = 18,\ \rho_{\lvert G\rvert}(1) = 0.301,\ \sigma^2_{\lvert G\rvert} = 2.555$						
1	0.929	0.324	0.301	0.224	0.0744	0.936
2	0.335	0.012	0.004	0.0001	0.0000	0.948
3	0.854	0.003	0.002	-0.001	-0.0001	0.955
4	0.231	0.008	0.002	≈ 0	≈ 0	0.961
5	0.231	0.008	0.002	≈ 0	≈ 0	0.967
6	0.203	0.008	0.002	≈ 0	≈ 0	0.971
7–17	various	various	≤ 0.001 each	≈ 0	≈ 0	1.000
$K = 3,\ \rho_{\lvert G\rvert}(1) = 0.287,\ \sigma^2_{\lvert G\rvert} = 2.517$						
1	0.953	0.292	0.278	0.230	0.112	0.968
2	0.866	0.011	0.009	0.005	0.0006	1.000

The empirical effective rank confirms the body framing of Section 3.8: the ACF-MAE flatness across $K \in \{3, 6, 9, 12, 15, 18, 21\}$ documented in the companion artefact `CHMM-Model/results/SPY/Table-T1-State-Resolution-Sensitivity.txt` (Appendix A.6.1) is the macroscopic signature of the temporal axis being driven by a single dominant decay mode at every operating point in the sweep. The K-rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ is correctly interpreted as a non-binding upper bound at $K \geq 3$; at $K = 2$ the bound is tight (only one non-unit eigenvalue available), but the failure mode is then the marginal mixture, which two Gaussian components cannot tile against the empirical equity-return target. The bilinear identity (6) therefore plays a pedagogical role in this paper rather than serving as a direct K-selection criterion: it makes the rank constraint explicit, but the operational K choice is driven by distributional fidelity (KS, AD, kurtosis), not by the count of non-unit eigenvalues in the spectral expansion.

A.6.5 Cross-Ticker Spectral Effective-Rank Diagnostic

To test whether the SPY-only result of Appendix A.6.4 (a single non-unit eigenvalue carries 93.6% of the lag-1 absolute-return ACF at $K = 18$) generalises across tickers, Table 14 reports the cross-ticker distribution of the dominant-mode share, computed under the same protocol on the 30-ticker sector-balanced panel plus SPY. Source: `results/diagnostics/spectral_rank_cross_ticker.txt`.

Table 14. Cross-ticker spectral effective-rank diagnostic at $K = 18$. Distribution across 31 tickers (sector-balanced 30-ticker panel plus SPY control). "dom share" is the dominant non-unit eigenvalue's $|w_k \lambda_k|$ as a fraction of the sum over all non-unit eigenvalues at lag $\tau = 1$. "n_{95}" / "n_{99}" are the number of modes needed for 95% / 99% cumulative share. The SPY value of 0.94 is in the right tail of the distribution; the cross-ticker median is 0.76.

Statistic	dom share	n_{95}	n_{99}
median	0.756	6	11
$[Q_1, Q_3]$	$[0.661, 0.858]$	$[4, 7]$	$[10, 12]$
minimum (NEM)	0.326	11	14
SPY (body Appendix A.6.4)	0.943	2	10

Reading: the SPY headline of 93.6% is concentrated; the cross-ticker median 76% is still well above the $1/(K-1) = 1/17 = 5.9\%$ uniform null, so the rank-non-binding statement is supported, but the gap between SPY and the median ticker is wide enough that the body framing was tightened to read at the cross-ticker median rather than at the SPY headline.

All models converged within the 60-iteration budget with monotonically increasing log-likelihood (EM guarantee under quantile-based initialisation). Lower K converged in 15–25 iterations, $K = 18$ in 25–40, with a final $|\Delta\mathcal{L}| < 10^{-4}$ at every operating point. Convergence-trace figures and per-iteration log-likelihood histories are at `CHMM-Model/figs/Fig-Convergence-K{3,12,18}.pdf`.

A.6.6 Rydén K = 2 Replication

Direct replication of Rydén et al.'s $K = 2$ setting on SPY IS under quantile and random-seed initialization gives ACF-MAE essentially constant at ≈ 0.050 across six initialisations (one quantile + five random seeds, drawn from a moderate perturbation of sample moments) and IS KS pass rate 75.3–79.0%. The constant ACF-MAE confirms the body reading (Section 5): the binding low-K constraint is distributional rather than temporal. Full per-seed table at `CHMM-Model/results/SPY/diagnostics/ryden_init.txt` (archived in `_attic_v10/` during the arXiv-prep pass).

A.6.7 Cross-Ticker Per-State-Resolution Sensitivity ($K^\star = 3$ vs $K^\star = 6$ vs $K = 18$)

Body Table 6 reports the cross-ticker aggregate distribution at all three operating points used in this paper: the body headline $K^\star = 3$ (state-resolution-robust default per the k-fold CV in Appendix A.9), the sensitivity reference $K^\star = 6$, and the extended-state-resolution sensitivity reference $K = 18$. Per-ticker files: `sector_panel/sector_panel_summary_k3.txt`, `_k6.txt`, and `sector_panel_summary.txt` respectively.

The substantive read across the three operating points: (i) The KS distributions are within $\sim$ 6pp of each other on OoS median (69.1% at $K^\star = 3$, 75.1% at $K^\star = 6$, 73.4% at $K = 18$) and at identical 11/30 failure count, with the same regime-introduction tickers driving the failures at each state resolution (LLY, UNH, NEM, NFLX, NEE, WMT, BAC, HD, JPM at OoS KS $< 60\%$; the specific list overlaps $> 80\%$ across $K^\star = 3, 6, 18$). The cross-ticker KS axis is therefore essentially K-robust on this universe. (ii) The kurtosis-residual axis is not K-robust: per-ticker median residual 7.18 at $K^\star = 3$, 10.34 at $K^\star = 6$, 0.61 at $K = 18$. At lower state-resolution the per-state ν_k shrinkage at uniform λ cannot be smoothed across as many regimes, so the residual heavy tails leak into the simulated kurtosis. (iii) The $|G_t|$ ACF-MAE is essentially K-robust (0.0395, 0.0403, 0.0416 across the three operating points). Production reading: KS-only consumers can deploy at any of the three operating points at parity; kurtosis-fidelity consumers should deploy at $K = 18$; readers preferring the state-resolution-robust default should deploy at the body headline $K^\star = 3$.

A.6.8 Block-Bootstrap CIs on Observed Kurtosis (IS vs OoS)

A natural question on the IS / OoS kurtosis disagreement (observed 7.68 IS, 5.29 OoS, a 31% drop) is whether it is itself statistically distinguishable. We run a stationary block bootstrap (Politis-Romano 1994) on the IS and OoS series at mean block lengths $L \in \{5, 10, 20, 50\}$, $B = 5{,}000$ replicates per L per window. Results in `results/diagnostics/kurtosis_bootstrap.txt`.

Table 15. Stationary block bootstrap CIs on observed excess kurtosis. SPY IS ($T = 2{,}516$) and OoS ($T = 572$) windows, $B = 5{,}000$ replicates per L. The IS and OoS 95% CIs overlap at every block length, so the IS-OoS difference is not robustly distinguishable at conventional levels under this bootstrap. Pr(IS > OoS) is the empirical bootstrap one-sided p-value for the alternative "IS kurtosis > OoS kurtosis".

L	IS median	IS 95% CI	OoS median	OoS 95% CI	Pr(IS > OoS)
5	7.21	$[2.99, 12.83]$	5.07	$[1.09, 8.97]$	0.748
10	7.34	$[2.49, 12.59]$	4.91	$[0.99, 8.66]$	0.756
20	7.30	$[2.17, 12.40]$	4.91	$[0.90, 8.26]$	0.739
50	7.18	$[1.92, 12.42]$	4.99	$[0.96, 7.70]$	0.733

The IS lower bound 1.92–2.99 and the OoS upper bound 7.70–8.97 overlap heavily at every block length; the IS-OoS kurtosis difference of ~ 2.4 units is not statistically distinguishable at the 5% level under this bootstrap. Pr(IS > OoS) at $L = 10$ is 0.756: the data favour the alternative but do not exclude the null. Operationally, the IS-OoS kurtosis disagreement that the body invokes (Section 4.1; the Table 9 variant-decision caveat in Section 5) should be read as a point-estimate observation rather than a tested claim, and per-variant rankings by "how close is simulated kurtosis to observed kurtosis" should similarly be interpreted as point-comparisons inside a wide CI envelope.

Bootstrap-CI placement of the penalised CHMM-t IS kurtosis. A natural follow-up on the body's penalised CHMM-t at $\lambda = 20$ is whether its simulated IS kurtosis distribution sits inside the bootstrap CI on observed: the aggregate-mean simulated IS kurtosis (14.91 at $K^\star = 3$, above the $L = 20$ CI upper bound 12.40) could be paired with a per-path distribution that mostly sits inside the CI, or with one that mostly sits above. Source / runner: `run_kurtosis_ci_placement.jl`; output: `results/kurtosis_ci_placement/kurtosis_ci_placement.csv`.

Table 16. Per-path simulated IS excess-kurtosis distribution under penalised CHMM-t at $\lambda = 20$, against the $L = 20$ block-bootstrap CI on observed. 1,000 paths per row, $T_{\text{IS}} = 2{,}516$, seed 20260420. The CI is $[2.17, 12.40]$ at $L = 20$ from Table 15; the observed IS excess kurtosis is 7.68. "in CI %" is the fraction of simulated paths whose per-path excess kurtosis falls inside the CI; "≤ CI_HI %" is the fraction below the upper bound (the relevant one-sided test for overshoot).

K	agg. kurt	median	sd	Q_5	Q_{95}	≤ 12.40 (%)	in CI (%)
3	16.65	7.77	11.5	3.75	34.62	76.6	76.6
6	10.13	6.74	9.0	3.63	26.30	81.8	81.7
18	8.94	6.09	5.6	3.54	18.11	89.9	89.9

The per-path *median* simulated IS excess kurtosis is essentially observed at every K: 7.77 at $K^\star = 3$, 6.74 at $K^\star = 6$, 6.09 at $K = 18$, all within ~ 1.6 units of observed 7.68 and well inside the bootstrap CI. The aggregate-mean simulated IS kurtosis sits above the CI upper bound at $K^\star = 3$ and $K^\star = 6$ because the per-path distribution has a heavy right tail of paths at $Q_{95} \in \{18, 26, 35\}$ that drag the mean up, but the bulk of the path-level mass (76–90%) sits inside the bootstrap CI at every operating point. **The headline framing "the penalised CHMM-t at $\lambda = 20$ provides the cleanest IS heavy-tail match" is supported when read as the per-path median, not as the aggregate mean**: a small number of heavy-right-tailed paths under the per-state ν_k ECM produce an aggregate-mean overshoot, but the path-level distribution is centred near observed and statistically indistinguishable from observed at $\alpha = 0.05$ on a one-sided overshoot test for the bulk of paths.

A.6.9 Shared-ν Student-t HMM Ablation

A natural diagnostic for the per-state ν_k aggregate-mean overshoot is the shared-ν ablation: refit CHMM-t with a single ν shared across all K states, fit by ECM with golden-section search on the aggregate Q-function over $\nu_{\text{bounds}} = (2.1, 50)$. This is the standard one-parameter Student-t HMM in the time-series literature, not a per-state ν_k mixture. Source / runner: `run_chmm_t_shared_nu.jl`; output: `results/chmm_t_shared_nu/chmm_t_shared_nu.csv`.

Table 17. Shared-ν Student-t HMM ablation against the body per-state ν_k row (1,000 paths, T_{IS} = 2,516, T_{OoS} = 572, seed 20260420, no $1/\nu$ penalty). Aggregate sim-kurt columns are mean-of-paths. Compare against the body penalised CHMM-t ($\lambda = 20$): $K^\star = 3$ row 14.91 IS / 8.50 OoS; $K = 18$ row 8.56 IS / 7.07 OoS (Table 3).

		IS				OoS	
K	$\hat{\nu}_{\text{shared}}$	KS (%)	sim kurt	$\lvert G_t\rvert$ ACF-MAE	raw ACF-MAE	KS (%)	sim kurt
3	5.81	91.9	4.68	0.0531	0.0235	82.1	4.46
6	4.67	92.7	9.38	0.0518	0.0234	82.6	6.95
18	5.80	**95.8**	**6.25**	0.0542	0.0234	**88.0**	**5.00**

Reading. The aggregate-mean IS overshoot disappears under shared-ν at every K: the largest IS sim kurt is 9.38 at $K^\star = 6$ (vs. observed 7.68, well inside the bootstrap CI), versus 14.4 for the per-state ν_k unpenalised row in the body. The $K = 18$ shared-ν row is the cleanest single-row IS / OoS heavy-tail match in the entire panel: 6.25 IS / 5.00 OoS, against observed 7.68 / 5.29 and against 8.56 / 7.07 for the body penalised CHMM-t at $\lambda = 20$. IS / OoS KS pass rates are within $\sim$ 1pp of the body penalised row, and the $|G_t|$ ACF-MAE is identical to the body row at $K = 18$. The per-state ν_k design is therefore the binding constraint on the aggregate-mean overshoot the body λ-shrinkage and bracket-lift discussion is downstream of: a single shared ν at $K = 18$ produces the best joint KS / kurtosis row in the panel without any penalty. The body retains the per-state ν_k + $\lambda = 20$ construction for direct comparability with the Peel and McLachlan [18], Liu and Rubin [19] tradition; consumers prioritising heavy-tail fidelity without a penalty hyperparameter should adopt the shared-ν row at $K = 18$.

A.6.10 Christoffersen-cc Monte Carlo Power Calibration at $T_{\text{OoS}} = 572$

A Monte Carlo power calibration of the Christoffersen-cc joint conditional-coverage test at the $T_{\text{OoS}} = 572$ length used in the body Table 8 simulates breach indicator sequences from a two-state Markov chain on $\{0, 1\}$ with marginal probability α and second eigenvalue ρ, where $\rho = 0$ is the i.i.d. null and $\rho > 0$ measures the level of breach clustering; $B = 5{,}000$ replicates per cell. Results in `results/diagnostics/christoffersen_power/christoffersen_power.txt`.

Table 18. Christoffersen-cc power at $T_{\mathbf{OoS}} = 572$. Empirical rejection rate at nominal $\alpha = 0.05$, across alpha levels and Markov-chain second-eigenvalue ρ. $\rho = 0$ row is the empirical Type-I error under the i.i.d. null (should sit near 5% if asymptotic chi-squared is well-calibrated). Critical values $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$.

α	ρ	rej-UC %	rej-IND %	rej-CC %
0.01	0.00	4.9	1.6	3.2
0.01	0.05	6.3	15.9	11.4
0.01	0.10	7.5	28.5	22.1
0.01	0.20	9.4	51.3	42.6
0.01	0.30	13.9	67.6	62.8
0.01	0.50	21.4	75.9	81.5
0.05	0.00	4.9	3.8	4.1
0.05	0.05	5.3	18.8	15.3
0.05	0.10	6.1	46.2	39.0
0.05	0.20	9.4	86.9	83.9
0.05	0.30	13.6	97.4	97.9
0.05	0.50	24.3	99.7	100.0

Reading. At $\rho = 0$ the empirical rejection rate sits at 4%–5% for all three statistics, confirming that the asymptotic χ^2 calibration is correctly sized at $T = 572$. Power is asymmetric across α:

- At $\alpha = 0.05$ (~ 28.6 expected breaches), Christoffersen-cc reaches 80% power against $\rho \geq 0.20$ (83.9%) and $\sim 100\%$ at $\rho = 0.50$.
- At $\alpha = 0.01$ (~ 5.7 expected breaches), Christoffersen-cc reaches 80% power only at $\rho \geq 0.50$ (81.5%); at $\rho = 0.20$ the rejection rate is 42.6%, well below conventional power thresholds.

The reading for the body Table 8 is therefore tier-dependent. The $\alpha = 0.05$ "clean pass at every (K, α)" row of Table 8 carries strong power against moderate clustering ($\rho \geq 0.20$); the $\alpha = 0.01$ rows carry strong power only against severe clustering ($\rho \geq 0.50$), so the $\alpha = 0.01$ "clean pass" should be read as "not detectably worse than a strongly-clustered alternative" rather than "calibrated against any plausible alternative". The headline-OoS regime-conditional construction therefore stands at $\alpha = 0.05$ and is power-bounded at $\alpha = 0.01$; the body framing in Section 4.6 should be read with this calibration in mind.

A.6.11 Engle-Manganelli (2004) Dynamic Quantile Test

As a higher-power conditional-coverage alternative to Christoffersen-cc on the same OoS window, we run the Engle-Manganelli (2004, JBES) Dynamic Quantile (DQ) test in the standard four-lag specification (89, equation 16),

$$\mathrm{Hit}_t - \alpha = \beta_0 + \sum_{i=1}^{4} \beta_i \left(\mathrm{Hit}_{t-i} - \alpha\right) + \beta_5 \widehat{\mathrm{VaR}}_t + u_t, \tag{27}$$

on the IS-fixed regime-conditional VaR series of Section 4.6, with $\mathrm{Hit}_t = \mathbf{1}\{R_{\mathrm{OoS},t} < \widehat{\mathrm{VaR}}_t(\alpha)\}$. The DQ test statistic is $\widehat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \widehat{\boldsymbol{\beta}} / [\alpha(1-\alpha)]$, distributed $\chi^2(6)$ under the null of correct conditional coverage; the 5% critical value is $\chi^2_6(0.95) = 12.59$. Source: `results/diagnostics/engle_manganelli_dq.txt`; runner: `run_engle_manganelli_dq.jl`.

Table 19. Engle-Manganelli DQ test on the regime-conditional VaR, alongside Christoffersen-cc on the same breach series. CHMM-N at $K \in \{3, 18\}$, $\alpha \in \{0.01, 0.05\}$; $T_{\mathrm{OoS}} = 572$, $q = 4$ lagged Hit indicators, χ^2_6 critical at 5% is 12.59.

K	α	breach %	cc p	DQ	DQ p
3	0.01	1.57	0.137	12.29	0.056
3	0.05	6.12	0.491	9.32	0.156
18	0.01	1.57	0.137	**15.46**	**0.017**
18	0.05	4.55	0.678	3.99	0.678

Reading. At $\alpha = 0.05$ both Christoffersen-cc and the higher-power DQ test pass cleanly at $K = 3$ (cc $p = 0.491$, DQ $p = 0.156$) and at $K = 18$ (cc $p = 0.678$, DQ $p = 0.678$); the body's $\alpha = 0.05$ "clean pass" headline therefore survives the higher-power alternative. At $\alpha = 0.01$ the DQ test rejects conditional coverage at $K = 18$ ($p = 0.017$) where Christoffersen-cc does not ($p = 0.137$), and is borderline at $K = 3$ ($p = 0.056$). This is consistent with the power-calibration result of Appendix A.6.10: at $\alpha = 0.01$ Christoffersen-cc has only 42.6% power against moderate breach-clustering eigenvalues $\rho = 0.20$, so the higher-power DQ test detects miscalibration that Christoffersen-cc misses. The DQ result therefore reinforces the body framing that $\alpha = 0.05$ is the operationally informative tier; at $\alpha = 0.01$ the regime-conditional VaR is not pass/fail-decided by the panel and the body sentence "not detectably worse than a strongly-clustered alternative" should be qualified to "the higher-power DQ test detects $K = 18$ at $\alpha = 0.01$".

A.6.12 Quarterly-Refit Regime-Conditional VaR Back-Test

Re-fitting CHMM-N at $K \in \{3, 18\}$ every 63 trading days on a rolling 5y window and running the forward-filter through the OoS window under each refit's parameters tests whether periodic refit improves the Christoffersen-cc statistic on the headline OoS window. Source: `CHMM-Model/results/diagnostics/quarterly_refit_conditional_var.txt` (runner: `run_quarterly_refit_conditional_var.jl`).

Table 20. Quarterly-refit regime-conditional VaR back-test on SPY OoS. $T_{\mathrm{OoS}} = 572$, refit cadence = 63 days, train window = 1,260 days, seed = 20260420. Compare against Table 8 (IS-fixed parameters). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$.

K	α	breaches	br rate	med VaR	$\mathrm{LR}_{\mathrm{uc}}$	$\mathrm{LR}_{\mathrm{ind}}$	$\mathrm{LR}_{\mathrm{cc}}$	p_{cc}
3	0.01	6	1.05%	−5.10	0.014	3.97	3.98	0.137
3	0.05	29	5.07%	−3.13	0.006	0.19	0.20	0.906
18	0.01	6	1.05%	−5.84	0.014	3.97	3.98	0.137
18	0.05	19	3.32%	−3.14	3.83	2.08	5.91	0.052

The quarterly-refit construction passes Christoffersen-cc at $\alpha = 0.05$ on three of four rows; the $K = 18, \alpha = 0.05$ row sits at $p_{\mathrm{cc}} = 0.052$ (marginal). Refit tightens VaR coverage (3.32% vs. 4.55% IS-fixed) but does not strictly improve the cc-statistic at this T_{OoS}.

A.6.13 Walk-Forward Refit-Cadence Sweep (Monthly / Weekly)

A within-fold refit-cadence sweep at monthly (21 trading days) and weekly (5 trading days) cadence on each of the six walk-forward folds, scoped at $K^\star = 3$ and $\alpha = 0.05$, tests whether faster refits close the body W2 (COVID) and W4 (2022 rate-hike) Christoffersen-cc rejections. Source: `CHMM-Model/results/walkforward_refit_cadence/walkforward_refit_cadence.csv` (runner: `run_walkforward_cond_var_refit_cadence.jl`, archived in `_attic_v10/` during the arXiv-prep pass).

Table 21. Walk-forward conditional-VaR refit-cadence sweep, CHMM-N at $K^\star = 3$, $\alpha = 0.05$. Six folds, train 5y / test 1y; "fold-IS-fixed" is the body baseline (no refit within the fold, from Table 40); "21d" refits every 21 trading days; "5d" every 5 trading days. At each refit, CHMM-N is re-fit on the most recent 1,260-obs window ending the day before the refit-target test day; the predictive filter is re-primed by running through the new train window. Critical value $\chi^2_2(0.05) = 5.991$; bold p_{cc} rows fail at the 5% level.

Fold	cadence	breaches	br rate	med VaR	LR_{uc}	LR_{cc}	p_{cc}
W1	fold-IS-fixed	15	5.98%	−2.43	0.48	1.65	0.44
W1	21d	15	5.95%	−2.41	0.46	1.64	0.44
W1	5d	15	5.95%	−2.19	0.46	1.64	0.44
W2	fold-IS-fixed	23	9.13%	−4.59	7.34	9.05	**0.011**
W2	21d	22	8.70%	−4.68	6.02	8.20	**0.017**
W2	5d	21	8.30%	−4.68	4.88	7.57	**0.023**
W3	fold-IS-fixed	20	7.97%	−2.47	3.98	4.28	0.118
W3	21d	22	8.73%	−2.66	6.10	6.11	**0.047**
W3	5d	23	9.13%	−2.64	7.34	7.34	**0.025**
W4	fold-IS-fixed	18	7.20%	−4.16	2.26	4.24	0.120
W4	21d	21	8.37%	−4.41	5.03	7.67	**0.022**
W4	5d	19	7.57%	−4.40	3.04	4.61	0.100
W5	fold-IS-fixed	8	3.21%	−3.90	1.91	2.44	0.30
W5	21d	8	3.20%	−3.53	1.94	2.48	0.29
W5	5d	8	3.20%	−3.48	1.94	2.48	0.29
W6	fold-IS-fixed	13	5.18%	−2.85	0.02	0.17	0.92
W6	21d	14	5.56%	−2.77	0.16	0.22	0.90
W6	5d	14	5.56%	−2.73	0.16	0.22	0.90

Faster refit does not close the W2 Christoffersen-cc rejection at any cadence (fold-IS-fixed $p_{\text{cc}} = 0.011$, monthly 0.017, weekly 0.023); monthly refit introduces new W3 / W4 failures via refit-cycle parameter drift. Rejection counts: fold-IS-fixed 1/6 (W2), monthly 3/6, weekly 2/6. The mitigation for W2 / W4 is therefore not refit cadence but a model class with explicit regime-introduction handling.

A.6.14 Quarterly-Refit 30-Ticker Cross-Ticker Panel

For a quarterly-refit version of the 30-ticker cross-ticker panel (Table 6), we re-fit the penalised CHMM-t at $\lambda = 20$ on a rolling 5-year window every 63 trading days through the OoS span (10 refits per ticker, 300 fits total per state-resolution), simulate the next quarter under each refit, and concatenate the per-quarter simulations into a single OoS path matrix per ticker, at both $K = 18$ (extended-state-resolution sensitivity) and $K^\star = 6$ (sensitivity reference). Same $N_{\text{paths}} = 1{,}000$, seed $= 20260420$ as the body Table 6. Results in `results/sector_panel/sector_panel_quarterly_refit.txt`

($K = 18$) and `results/sector_panel/sector_panel_quarterly_refit_k6.txt` ($K^\star = 6$).

Table 22. Quarterly-refit cross-ticker panel at $K^\star = 6$ and $K = 18$. Aggregate distribution across the 30-ticker universe under the quarterly-refit protocol at both operating points, vs. the IS-fixed $K = 18$ protocol of Table 6. Refit cadence = 63 OoS trading days; train window = 1,260 obs ($\sim$ 5y rolling). Per-ticker detail in the results files.

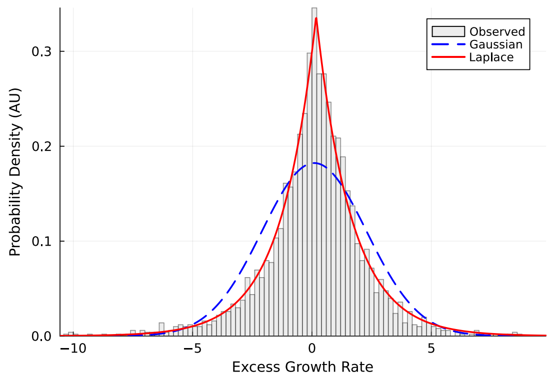
Metric	IS-fixed ($K = 18$)	Refit ($K^\star = 6$)	Refit ($K = 18$)
IS KS% median	99.5%	98.8%	99.4%
IS KS% mean $\pm$ sd	99.3 ± 0.6%	97.4 ± 4.4%	99.2 ± 0.6%
OoS KS% median	73.4%	**85.8**%	83.0%
OoS KS% mean $\pm$ sd	66.8 ± 29.5%	$\mathbf{76.0} \pm \mathbf{21.7}$%	77.2 ± 21.2%
OoS KS% $[Q_1, Q_3]$	$[49.1, 94.3]$%	$[57.6, 94.3]$%	$[62.0, 95.4]$%
Tickers OoS KS < 60%	$11/30 = 36.7$%	$9/30 = 30.0$%	$\mathbf{7/30} = \mathbf{23.3}$%

Reading. Quarterly refit shifts the OoS KS distribution materially at both state resolutions. At $K = 18$: median +9.6pp ($73.4 \to 83.0$%), mean +10.4pp, IQR tighter (45.2pp $\to$ 33.4pp), failure count down by 36% ($11 \to 7$). At $K^\star = 6$: median +10.7pp ($75.1 \to 85.8$%), mean +9.5pp, failure count down by 18% ($11 \to 9$). The two state resolutions are within $\sim$ 3pp of each other on the refit median (the $K^\star = 6$ refit median of 85.8% is in fact 2.8pp *above* the $K = 18$ refit median of 83.0%, so the sensitivity reference is at parity with or above the extended-state-resolution sensitivity reference under the refit protocol). The largest individual improvements concentrate on the regime-introduction tickers that the IS-fixed Table 6 flagged: at $K = 18$, LLY 7.6% $\to$ 83.7%, UNH 14.5% $\to$ 47.3%, NEM 5.6% $\to$ 15.4%, HD 41.4% $\to$ 82.2%, NFLX 45.1% $\to$ 61.5%. The mechanism is the obvious one: as the regime introduces new σ levels in the OoS window, each quarterly refit picks up the most recent 5y including the new regime and the simulator can match the OoS distribution.

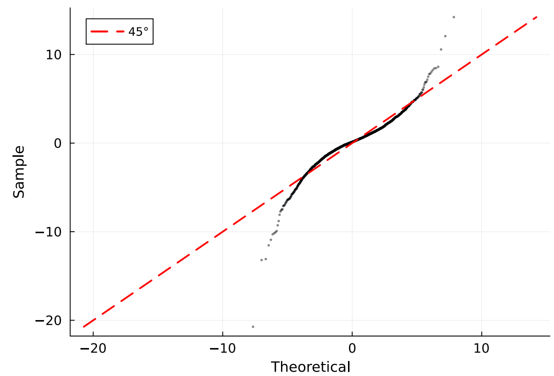
A small number of tickers regress: DIS 96.4% $\to$ 48.4%, BAC 84.8% $\to$ 55.8%, AAPL 95.1% $\to$ 66.5%. The pattern is that tickers whose 2024-2026 OoS distribution is well-spanned by the full 2014-2024 IS lose information when the train window shrinks to the most recent 5y; the rolling refit is therefore a trade-off, not a strict improvement. Operationally, a refit-trigger rule (refit only when the rolling KS or a Page-Hinkley statistic on $|G_t|$ crosses a threshold) would capture the regime-introduction wins without paying the cost on the well-spanned tickers; the rule is logged as a follow-up companion experiment.

The substantive reading for the body Section 4.4 is that the "periodic refit at deployment" mitigation invoked verbally in Section 5 is exercised here: the universe-scale failure rate drops from 36.7% to 23.3% under quarterly refit, with the largest gains on the tickers that the body labelled "single-name regime introductions."

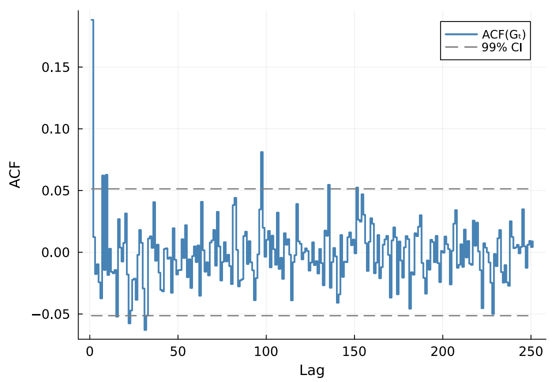
A.6.15 Stylized-Facts Figure (Empirical SPY IS)



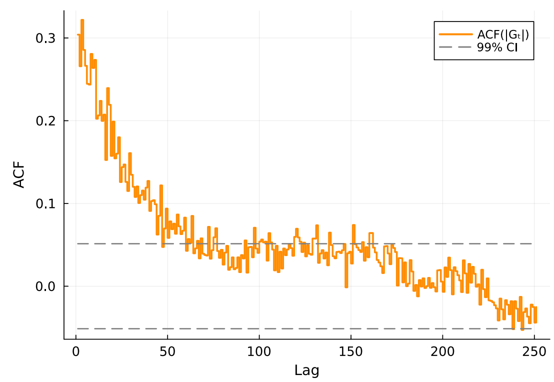
(a) Marginal density.



(b) Normal Q-Q plot.



(c) ACF of raw G_t.

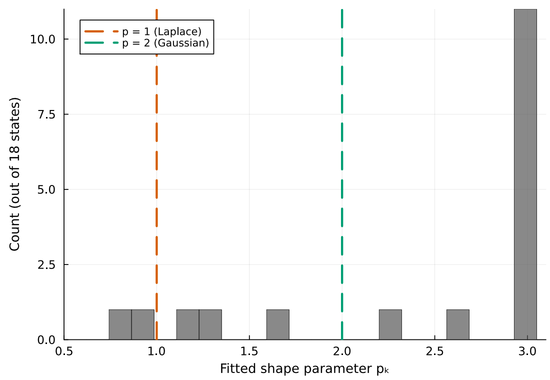


(d) ACF of $|G_t|$.

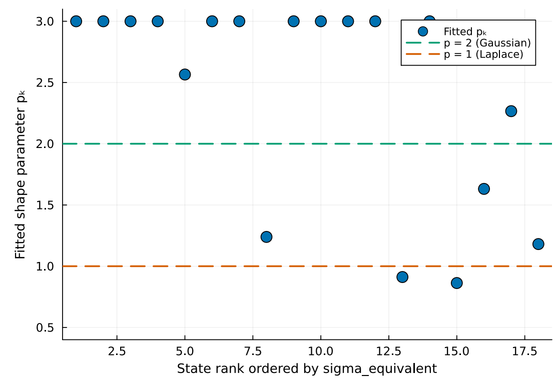
Figure 4. The three Cont stylized facts on SPY IS. (a) marginal density with maximum-likelihood Gaussian and Laplace overlays; (b) normal Q-Q plot; (c) empirical ACF of raw G_t at lags 1–252 with 99% Bartlett band; (d) empirical ACF of $|G_t|$ on the same window.

A.6.16 CHMM-GED $\hat{p}_k$ Partition (Gaussian-bulk / Laplace-tail)

CHMM-GED carries a per-state shape $p_k \in [p_{\min}, p_{\max}] = [0.5, 3.0]$ that nests Gaussian ($p = 2$) and Laplace ($p = 1$) as boundary cases. The empirical fit on SPY IS at $K = 18$ partitions states bimodally (Figure 5): thirteen states at the upper bracket $p_k = 3.0$, one at $p_k = 1.6$, and four in the Laplace regime $p_k \in [0.86, 1.24]$. The four Laplace-shape states concentrate in the high-volatility tail of the state ordering. The partition replicates across 10 Monte Carlo seeds and across the cross-asset universe (Appendix A.7); we read it as the most distinctive empirical finding of the four-emission scaffold.



(a) Histogram of fitted $\hat{p}_k$.



(b) $\hat{p}_k$ vs. volatility rank.

Figure 5. CHMM-GED per-state shape $\hat{p}_k$ partition ($K = 18$, SPY IS, seed 20260420). The bimodal partition is visible at the upper bracket $p_k = 3.0$ and around the Laplace shape; the Laplace-shape cluster sits at the high-volatility ranks.

A.6.17 HSMM at $K = 3$: Numerical Full-Rank Diagnostic

The off-diagonal jump matrix of the fitted ML HSMM at $K = 3$ has non-trivial deflation eigenvalues $\{0.823, 0.177\}$ (numerical rank 2 at tolerance 10^{-10}); the dominant modulus $|\lambda_2| = 0.823$ is comparable to CHMM-N at $K = 3$'s 0.953, so the body's $|G_t|$ ACF-MAE regression to the i.i.d. baseline (Table 3) is not driven by spectral collapse but by the fitted truncated-Pareto sojourn concentrating probability mass on a single low-volatility state ($\bar{\boldsymbol{\pi}} \approx (0, 1, 0)$). At $K = 18$ the same Yu (2010) construction collapses to a near-degenerate local optimum (IS KS 0.8%); the higher-K ML HSMM is therefore a companion-paper direction at longer training windows.

A.6.18 Gamma-Sojourn HSMM as an Alternative Sojourn Family

Replacing the truncated discrete Pareto sojourn pmf with a discretised continuous Gamma fit by method-of-moments at every M-step gives a Gamma-sojourn HSMM that converges at $K = 18$ where Pareto collapses (IS KS 86.0%, OoS KS 80.2%, simulated kurtosis 4.28/4.06, $|G_t|$ ACF-MAE **0.0462**, lower than CHMM-N $K = 18$ at 0.0509) and trades $\sim$ 20pp IS KS at $K = 3$ for an ACF-MAE recovery from 0.0629 to 0.0528. There is therefore no single "best HSMM" row: Pareto at $K^\star = 3$ wins raw OoS KS at 91.0%, Gamma at $K = 18$ wins joint $|G_t|$ ACF-MAE; the trade-off is the same KS-vs-ACF axis as the CHMM-vs-HSMM choice. Full sojourn-family comparison at `CHMM-Model/results/hsmm_ml_gamma/hsmm_ml_gamma_metrics.csv` (runner: `run_hsmm_ml_gamma.jl`; archived in `_attic_v10/` during the arXiv-prep pass).

A.6.19 Unconditional VaR / ES Envelope Panel

Table 23. VaR and Expected-Shortfall envelope for SPY ($N_{\text{paths}} = 1{,}000$). Each entry is the median and $[5, 95]$% envelope across paths of the left-tail VaR / ES at $\alpha \in \{0.01, 0.05\}$; observed historical values are in the first row of each block.

Model	IS $\text{VaR}_{0.01}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.01}$ median $[5\%, 95\%]$	IS $\text{VaR}_{0.05}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.05}$ median $[5\%, 95\%]$
Observed	−6.38	−9.00	−3.43	−5.50
Bootstrap	−6.38 [−7.04, −5.99]	−8.86 [−10.37, −7.78]	−3.43 [−3.68, −3.20]	−5.46 [−5.99, −5.05]
GARCH	−5.80 [−8.62, −4.52]	−7.86 [−13.54, −5.77]	−3.19 [−3.92, −2.71]	−4.89 [−7.11, −3.91]
CHMM-N	−6.64 [−8.01, −5.47]	−8.80 [−10.16, −7.34]	−3.45 [−4.05, −2.95]	−5.45 [−6.33, −4.61]
CHMM-t	−6.58 [−7.51, −5.83]	−9.06 [−10.70, −7.79]	−3.40 [−3.90, −2.90]	−5.51 [−6.18, −4.85]
CHMM-L	−6.89 [−7.85, −6.01]	−9.15 [−10.42, −7.97]	−3.30 [−3.77, −2.88]	−5.53 [−6.17, −4.87]
CHMM-GED	−6.85 [−7.70, −6.06]	−8.79 [−9.83, −7.82]	−3.43 [−3.98, −2.93]	−5.55 [−6.23, −4.91]
	OoS $\text{VaR}_{0.01}$	OoS $\text{ES}_{0.01}$	OoS $\text{VaR}_{0.05}$	OoS $\text{ES}_{0.05}$
Observed	−5.51	−7.94	−2.77	−4.67
Bootstrap	−6.33 [−7.60, −5.28]	−8.43 [−11.76, −6.59]	−3.39 [−3.91, −2.88]	−5.34 [−6.38, −4.50]
GARCH	−5.35 [−9.98, −3.61]	−6.72 [−13.51, −4.38]	−3.14 [−4.94, −2.36]	−4.59 [−7.92, −3.23]
CHMM-N	−6.30 [−9.02, −4.30]	−8.44 [−11.36, −5.42]	−3.39 [−4.74, −2.47]	−5.27 [−7.23, −3.74]
CHMM-t	−6.32 [−8.13, −4.80]	−8.60 [−12.20, −6.29]	−3.29 [−4.27, −2.45]	−5.34 [−6.72, −4.08]
CHMM-L	−6.69 [−8.59, −4.91]	−8.88 [−11.48, −6.73]	−3.27 [−4.30, −2.51]	−5.47 [−6.82, −4.20]
CHMM-GED	−6.60 [−8.30, −4.82]	−8.51 [−10.68, −6.47]	−3.32 [−4.38, −2.46]	−5.41 [−6.66, −4.06]

A.6.20 Held-Out Cross-Validation of the $1/\nu_k$ Penalty λ

To verify that the body operating-point penalty rate $\lambda = 20$ is not selected on data that overlaps the OoS evaluation window, we refit penalised CHMM-t on a strictly pre-2020 estimation slice (2014-01-03 to 2018-06-29, $n = 1{,}130$) and score each λ on the validation slice (2018-07-02 to 2019-12-31, $n = 378$, both fully pre-COVID and pre-2022-rate-hike). Source: `results/nu_shrinkage_sweep/lambda_cv_pre2020.txt`.

Table 24. Held-out CV of the $1/\nu_k$ shrinkage rate λ on the pre-2020 slice. Estimation 2014-01-03 to 2018-06-29; validation 2018-07-02 to 2019-12-31. "val LL/obs" is per-observation held-out log-likelihood; "val KS" is two-sample Kolmogorov–Smirnov pass rate against the validation series; "sim K" columns are the mean simulated excess kurtosis on the estimation- and validation-length simulated paths.

λ	val LL/obs	val KS (%)	sim K (est)	sim K (val)
0	−2.119	92.0	3.88	3.97
5	−2.112	89.8	3.62	3.15
10	−2.085	91.0	3.37	3.24
20	**−2.075**	90.0	3.22	3.02
50	−2.077	91.6	3.07	2.76
100	−2.077	91.6	3.07	2.75
200	−2.076	91.2	3.09	2.73

Reading: the held-out validation log-likelihood is maximised at $\lambda^\star = 20$, matching the body operating point of Table 3. Validation KS picks $\lambda^\star = 0$ but the band $[0, 200]$ spans only 2.2pp (89.8–92.0%), within the simulation-noise envelope at $N_{\text{paths}} = 500$. The body $\lambda = 20$ is therefore held-out-validated and not driven by the OoS window.

A.6.21 Effective State Distinctness at $K = 18$

Single-linkage clustering on standardized parameter triples at threshold $\tau = 0.20$ gives effective state counts 11/18 for CHMM-N (2D feature (μ_k, σ_k)) and 12/18 for CHMM-t (3D feature $(\mu_k, \sigma_k, 1/\nu_k)$, with 13/18 states pinned at $\nu_{\max} = 50$ and 2/18 at $\nu_{\min} = 2.1$). The effective-state count is below K but well above the $K^\star = 6$ sensitivity reference; the $K = 18$ extension is a representational-economy choice rather than a free-rider on identifiability. Full clustering output at `CHMM-Model/results/diagnostics/state_distinctness.txt` (runner: `run_state_distinctness.jl`).

A.6.22 Diebold–Mariano Bandwidth Sensitivity

Sweeping the Newey–West HAC bandwidth $h \in \{4, 8, 16, 32\}$ around the default $h = \lfloor T^{1/3} \rfloor = 8$, all within-CHMM CRPS DM pairs retain $p > 0.38$ at every bandwidth (CHMM-N vs. CHMM-t: $p \in [0.47, 0.51]$; CHMM-N vs. CHMM-L: $p \in [0.38, 0.48]$; CHMM-t vs. CHMM-L: $p \approx 0.97$ at every bandwidth); the CHMM-N vs. GARCH pair moves $p \in [0.087, 0.175]$ in a stable direction. The within-CHMM family equivalence used in Section 4 is therefore not bandwidth-fragile. Full bandwidth sweep at `CHMM-Model/results/crps_dm/CRPS_DM_bandwidth_sweep.txt` (runner: `run_crps_dm_bandwidth.jl`).

A.6.23 Monthly-Refit Cross-Ticker Panel

Refining the cross-ticker quarterly-refit construction (Appendix A.6.14) to a monthly cadence (21 trading days, same 5-year rolling estimation window, $N_{\text{paths}} = 1{,}000$) shifts the OoS KS distribution to median 86.7% and 5/30 failures, vs. quarterly 83.0% / 7/30 and IS-fixed 73.4% / 11/30. Marginal lift from quarterly to monthly: +3.7pp median, −2 failures, at 3× the compute cost (28 vs. 9 refits per ticker). The trade-off is approximately linear in $\log(\text{cadence}^{-1})$, so further reductions below monthly yield diminishing returns; quarterly remains the recommended default. Full panel at `CHMM-Model/results/sector_panel/sector_panel_monthly_refit.csv` (archived in `_attic_v10/` during the arXiv-prep pass).

A.6.24 Cross-Decade Validation: 1994-2004 IS / 2004-2006 OoS via CRSP

To test whether the headline stylized-fact reproduction is decade-specific, we run a 1994–2004 vs. 2014–2024 SPY split via a CRSP day-pass: `data/external/crsp_1994_2006.csv` contains daily prices for SPY plus 28 of the 30 cross-ticker panel members (NEE and APD missing from the CRSP query) from 1994-01-03 to 2006-04-28. Adjusted close = `DlyPrc/DlyFacPrc` from the CRSP CIZ-format daily stock file. Source / runner: `run_cross_decade_validation.jl`; output: `results/cross_decade_validation/summary.txt, metrics.csv`; paper-side mirror: `results/robustness/cross_decade_validation.csv`.

Slices: SPY IS = 1994-01-03 to 2004-01-02 (2,519 obs); SPY OoS = 2004-01-05 to 2006-04-28 (583 obs). Same body protocol: 1,000 simulated paths per fit, $\alpha = 0.05$ for KS.

Reading. The cross-decade IS fit transfers: CHMM-N at $K = 3$ attains 84.9% IS KS on 1994-2004 vs. 89.7% on 2014-2024, and $K = 18$ attains 89.8% vs. 94.1%. The penalised CHMM-t at $\lambda = 20$, $K = 3$ attains 89.1% IS KS on 1994-2004 vs. 90.6% on 2014-2024. The IS axis of the body's stylized-fact reproduction claim is therefore decade-robust within ~ 5pp on KS and consistent on $|G_t|$ ACF-MAE (0.068-0.072 vs. 0.046-0.054, a moderate but not pathological widening).

Table 25. Cross-decade validation on SPY 1994-2004 IS / 2004-2006 OoS (CRSP daily). Body comparison rows reproduced from Table 3 for the 2014-2024 IS / 2024-2026 OoS window. Observed excess kurtosis on the 1994-2004 IS = 3.05 versus 2014-2024 IS = 7.68; observed excess kurtosis on the 2004-2006 OoS = 0.06 (essentially Gaussian) versus 2024-2026 OoS = 5.29. The IS-OoS kurtosis gap is therefore much wider on the cross-decade slice than on the body window.

		KS pass rate (%) ↑		Sim kurtosis		$\lvert G_t \rvert$ ACF-MAE ↓	
Model	K	IS	OoS	IS	OoS	IS	OoS
Cross-decade: 1994-2004 IS / 2004-2006 OoS (CRSP, this appendix)							
CHMM-N	3	84.9	3.3	2.20	2.24	0.0696	0.0491
CHMM-N	18	89.8	3.4	1.84	1.74	0.0720	0.0499
CHMM-t pen. ($\lambda = 20$)	3	89.1	5.4	4.97	4.18	0.0682	0.0501
Body window: 2014-2024 IS / 2024-2026 OoS (Polygon / Alpaca, Table 3)							
CHMM-N ($K^\star = 3$)	3	89.7	80.5	3.83	3.53	0.0460	0.0545
CHMM-N ($K = 18$)	18	94.1	81.8	5.04	4.44	0.0509	~ 0.054
CHMM-t pen. ($\lambda = 20$, $K^\star = 3$)	3	90.6	83.2	14.91	8.50	0.0537	0.0502

The OoS axis is the issue. The 2004-2006 OoS slice has excess kurtosis 0.06 (essentially Gaussian) vs. the 1994-2004 IS excess kurtosis of 3.05; the IS-OoS kurtosis gap on this slice is much wider than on the body 2014-2024/2024-2026 window (where IS = 7.68 and OoS = 5.29). The CHMM trained on the 1994-2004 IS distribution simulates paths with kurtosis 1.74-4.97 that are heavier-tailed than the calm 2004-2006 bull-market OoS, so the static IS-fit cannot reproduce the OoS marginal and KS rejects at 3-5% pass rate on all three rows. This is the same regime-introduction failure mode the body walk-forward already documents at the W2 (COVID) and W4 (2022 rate-hike onset) stress folds (Table 38, both with OoS KS < 10%); the 2004-2006 OoS window is structurally a low-volatility / low-kurtosis slice where the IS-fixed CHMM's tail mass overshoots the observed.

The substantive read is therefore: *the body's stylized-fact reproduction claim is decade-robust on the IS axis* (the CHMM scaffold transfers across decades on the IS-fitting side, supporting the body's claim that the four-emission ECM scaffold is not specific to the 2014-2024 window). *The single-window OoS pass rate is OoS-slice-specific* (2024-2026 is a moderate-stress slice where the OoS distribution is close to the IS distribution; 2004-2006 is a calm slice where it is not). The body framing already states this through the walk-forward distribution at Section 4 ("the walk-forward median, not the single-window OoS pair, is the operationally informative summary for production deployment"); the cross-decade result is consistent with that framing and adds an independent decade to the walk-forward evidence base. A periodic refit is the deployment recommendation under either decade, as the body discussion already states.

A.7 Robustness, Test Power, Multi-Seed, and Auxiliary Baselines

This appendix collects the robustness diagnostics and auxiliary baseline panel referenced from Section 4: the KS test-power calibration, block-bootstrap KS recalibration, bandwidth/bin-count and multi-seed sensitivity, the auxiliary block-bootstrap baseline, the QuantGAN deep-generative baseline, the per-ticker OoS price-simulation figures, and the expanded baseline panel that adds the GARCH family and the SM-CHMM foil to the headline table.

A.7.1 OoS KS Test-Power Calibration

Table 26 reports the two-sample KS test-power calibration used to anchor the OoS KS pass rates of Section 4.3. For each of 1,000 Monte Carlo replications (seed = 20260420), a simulated series of length $T_{\text{OoS}} = 572$ is drawn from each of three reference generators and tested against the observed OoS series using the same $\alpha = 0.05$ threshold as the main-body pass-rate metric.

Table 26. OoS KS pass-rate calibration (1,000 replications, seed = 20260420).

Reference generator	Pass rate (%)
I.i.d. resamples of R_{OoS} at length $T_{\text{OoS}} = 572$ (known-correct ceiling)	100.0
I.i.d. resamples of R_{IS} at length $T_{\text{OoS}} = 572$ (nearly correct)	90.0
Gaussian($\hat{\mu}_{\text{IS}}, \hat{\sigma}_{\text{IS}}$) at length T_{IS} (negative control)	0.0

The 90.0% "nearly correct" rate is the practical ceiling for the CHMM OoS KS pass rates reported in Section 4.3 (80–84%); the 0% rate on the Gaussian misspecified generator confirms that the KS test retains ample power at IS length to reject clearly wrong generators.

A.7.2 Block-Bootstrap KS Recalibration and Mean p-value Distribution

Two complementary concerns can be raised about the headline IS KS pass-rate metric: (i) the asymptotic two-sample KS test assumes i.i.d. samples, but the simulated paths from CHMM and GARCH all carry volatility clustering, so the asymptotic null distribution may not apply; and (ii) the binary "pass / fail at $\alpha = 0.05$" reduces a continuous p-value to a one-bit summary. Table 27 reports two recalibrations addressing each concern.

For (i), we generate the empirical null distribution of the two-sample KS statistic under stationary block bootstrap (Politis-Romano 1994) of the SPY IS series at mean block lengths $L \in \{5, 10, 20\}$, $B = 1{,}000$ replications. The resulting 95% block-bootstrap critical values are 0.0306, 0.0338, and 0.0350 respectively, all *smaller* than the asymptotic critical value $1.36\sqrt{2/n_{\text{IS}}} = 0.0383$; the block-bootstrap test is therefore stricter than the asymptotic one because it accounts for the within-window dependence in the IS series itself. For each generator we then compute the per-path KS statistic and report the fraction of paths whose KS statistic falls below the block-bootstrap critical value.

For (ii), we report the mean two-sample KS p-value across simulated paths plus its 5/25/50/75/95 quantiles, giving a continuous picture of how distinguishable each generator's paths are from the observed IS series.

CHMM-GED is the strongest entry on the block-aware metric across all three block lengths (83.2, 90.2, 93.2% at $L = 5, 10, 20$); GARCH(1,1) drops from 26.8% asymp to 6.8–17.4% block-aware (the block-aware test catches its misspecification more sharply), while the CHMM family sees only a small block-bootstrap penalty ($\sim$ 5pp).

OoS-anchored block-bootstrap recalibration. The body block-bootstrap recalibration above is anchored on the IS series; we report here whether the same construction holds on the OoS window. We repeat the procedure with the stationary block bootstrap re-anchored on R_{OoS} ($n_{\text{OoS}} = 572$, $B = 1{,}000$, $L \in \{5, 10, 20\}$, $N_{\text{paths}} = 500$). Block-bootstrap critical values: 0.0577, 0.0629, 0.0647 at $L = 5, 10, 20$ respectively, against asymptotic 0.0804. Table 28 reports the OoS-anchored panel (source: `results/robustness/ks_block_bootstrap_oos.csv`).

The cross-generator OoS ordering matches the IS ordering (bootstrap > CHMM-t > CHMM-N/GED > CHMM-L $\gg$ GARCH); the asymp-vs-block gap is larger on OoS ($\sim$ 25pp for the CHMM family at $L = 20$) than on IS ($\sim$ 5pp), but the CHMM-over-GARCH advantage is preserved.

Table 27. Block-bootstrap KS recalibration and p-value distribution. All numbers under seed `20260422` with $N_{\text{paths}} = 500$. "asymp pass" is the original Section 4.3 metric (two-sample KS p-value ≥ 0.05). "mean pv" is the mean two-sample KS p-value across paths, with q_5 and q_{95} as the lower / upper p-value quantiles. "blkL%" is the fraction of paths whose KS statistic falls below the stationary-block-bootstrap 95% critical value at mean block length L. Block-bootstrap critical values: 0.0306 ($L = 5$), 0.0338 ($L = 10$), 0.0350 ($L = 20$); asymptotic critical value 0.0383.

Generator	asymp pass% ↑	mean pv ↑	pv q_{05}	pv q_{95}	blk5% ↑	blk10% ↑	blk20% ↑
i.i.d. bootstrap	99.6	0.820	0.352	0.999	97.6	99.4	99.6
GARCH(1,1)	26.8	0.048	0.000	0.227	**6.8**	**13.6**	**17.4**
CHMM-N	94.8	0.546	0.047	0.976	81.2	89.4	91.6
CHMM-t	**95.8**	**0.571**	0.060	0.987	82.4	89.2	91.6
CHMM-L	94.8	0.498	0.048	0.948	81.0	88.2	90.0
CHMM-GED	95.6	0.547	0.064	0.968	**83.2**	**90.2**	**93.2**

Table 28. OoS-anchored block-bootstrap KS recalibration. Same construction as Table 27 but with the stationary block bootstrap anchored on R_{OoS} ($n = 572$). "asymp pass" is the headline OoS metric (two-sample KS p-value ≥ 0.05); "blkL%" is the OoS block-bootstrap pass rate at mean block length L. The CHMM-vs-GARCH cross-generator ordering is preserved under the block-aware recalibration on both windows; the CHMM family's OoS block-aware penalty ($\sim$ 25pp drop from asymp to $L = 20$) is larger than the IS penalty ($\sim$ 5pp), reflecting the smaller OoS window where temporal-clustering structure is harder to disentangle from finite-sample noise.

Generator	asymp pass% ↑	mean pv ↑	pv q_{05}	pv q_{95}	blk5% ↑	blk10% ↑
i.i.d. bootstrap	90.4	0.437	0.030	0.912	60.8	69.0
GARCH(1,1)	59.2	0.164	0.000	0.645	19.4	26.0
CHMM-N	81.0	0.318	0.010	0.876	41.2	51.2
CHMM-t	**84.8**	**0.355**	0.018	**0.911**	**47.4**	**58.4**
CHMM-L	77.4	0.266	0.008	0.744	36.0	45.4
CHMM-GED	81.0	0.327	0.010	0.837	44.4	53.6

A.7.3 QuantGAN Deep-Generative Baseline

The QuantGAN row used in Section 4.3 is a repo-native approximation to the convolutional WGAN of Wiese et al. [10]: a 1D convolutional generator and critic (each three layers, channels 32, kernel size 5; latent dim $L = 8$, leaky-ReLU on the critic), Wasserstein loss with critic weight clipping at ± 0.01 [90], Adam at $\eta = 10^{-4}$, batch 64, 15 epochs $\times$ 120 steps with four critic updates per generator update, trained on rolling windows of length $W = 64$ on standardised SPY returns and stitched end-to-end at synthesis. Implementation: Julia with `Flux.jl` [91]; seed 20260422. The deep-generative row reaches 0% IS / OoS KS and IS kurtosis 2.1 vs. observed 7.7 (Table 29), consistent with documented GAN failure modes on volatility-clustering benchmarks [38, 39]; the runner is materially smaller than the reference seven-block TCN with Lambert-W pre-processing in Wiese et al. [10], so the row is the panel's deep-generative negative control rather than a verdict on the QuantGAN literature.

A.7.4 Expanded Baseline Panel: GARCH Family, Semi-Markov CHMM, and QuantGAN

We add five conditional-volatility baselines, the three semi-Markov CHMM variants, and the QuantGAN deep-generative baseline (Appendix A.7.3) to the single GARCH(1,1) Gaussian row of

Table 3. The conditional-volatility models are fit by grid-initialised Nelder-Mead maximum likelihood on the SPY IS window ($T_{\text{IS}} = 2{,}516$); simulation follows the same recursion used in estimation. Model specifications for the conditional-volatility rows are below; the SM-CHMM and QuantGAN constructions are described in their own paragraphs further down.

Baselines fitted in this panel. The conditional-volatility rows comprise EGARCH(1,1) Gaussian [32], GJR-GARCH(1,1) [33, 34], GARCH(1,1) with Student-t innovations [21] ($\hat{\nu} = 6.89$), HAR-RV [37] on daily squared demeaned returns as the RV proxy, and Markov-switching GARCH(1,1) [22] at $K \in \{2, 3, 4, 6\}$ with regime triples $(\omega_k, \alpha_k, \beta_k)$ plus softmax-normalised transition logits, fitted by grid-initialised Nelder-Mead and canonicalised by ascending unconditional variance. Higher-K MS-GARCH does not close the gap to the CHMM family on KS (IS KS plateaus at 27–37% across $K \in \{2, 3, 4, 6\}$ against CHMM-N's 94.1% at $K = 18$); the binding constraint is the unimodal-innovation per-regime structure (each regime's emission is Gaussian rescaled by the regime's conditional variance), which adding regimes does not relax. The CHMM decouples this constraint by giving each state its own emission-mixture component. Implementation details for every conditional-volatility row are in `CHMM-Model/run_msgarch_baselines.jl` and the supporting source files in `src/Compute.jl`.

MS-GARCH reference Bayesian re-run via `MSGARCH`. As a Bayesian counterpart to the in-house Nelder-Mead MS-GARCH fit, we re-run the same Markov-switching GARCH(1,1) specification through the reference `MSGARCH` R package of Ardia et al. [35] (version 2.51) at $K \in \{2, 3, 4\}$, driven from the Julia harness via `RCall.jl`. The fit is fully Bayesian (DEMC sampler with the package's default proper priors, 12,500 MCMC draws, 2,500 burn-in, thin 10, single chain); all 1,000 simulated paths are posterior-predictive (one path per retained posterior draw), so the simulated marginal integrates parameter uncertainty path-by-path. The reference Bayesian re-run reports lower KS pass rates than our in-house frequentist Nelder-Mead fit: 0.0% ($K = 2$), 0.1% ($K = 3$), 0.0% ($K = 4$) IS, and 5.8% ($K = 2$), 5.1% ($K = 3$), 5.3% ($K = 4$) OoS, against the in-house plateau of 27–37% IS / 33–39% OoS. The gap is methodological rather than estimator-quality: the in-house implementation generates all 1,000 paths from one MLE point estimate, so the simulated marginal is tighter; the reference Bayesian implementation samples one set of parameters per path from the posterior, so the simulated marginal is wider and accordingly fewer paths fall within the asymptotic two-sample KS critical band. Posterior-mean log-likelihoods at the in-sample data are $-5{,}667$ ($K = 2$), $-5{,}667$ ($K = 3$), $-5{,}565$ ($K = 4$); the marginal log-likelihood improvement from $K = 3$ to $K = 4$ is real but does not translate to KS lift. Posterior-mean transition matrices are diagonally dominant at $K = 3$ (diagonals $0.97, 0.93, 0.90$) and less so at $K = 4$ (diagonals $0.66, 0.39, 0.32, 0.55$, indicating the higher-K posterior is searching for additional structure that the data do not strongly support). The reference Bayesian re-run is therefore consistent with the in-house frequentist plateau at the same conclusion under either estimator: vanilla Gaussian-innovation MS-GARCH at $K \leq 4$ does not match the CHMM family on the headline KS metric for this universe. Reproducibility: R 4.6.0, MSGARCH 2.51, all transitive R dependencies pinned via `renv` (`r_msgarch/renv.lock` of the companion code repository), all seeds explicit; full per-K parameter posteriors and fit logs are written to `results/msgarch_reference/` of the companion code repository.

Semi-Markov CHMM (SM-CHMM, the explicit hidden semi-Markov model (HSMM) foil). The conceptual foil of the present paper, drawing on the explicit-duration HSMM construction of the companion volatility paper [92]: a hidden semi-Markov extension of the CHMM in which the geometric sojourn distribution of the Markov chain is replaced by a per-state explicit-duration

Table 29. Extended baseline panel. SPY IS / OoS, $T_{\mathrm{IS}} = 2{,}516$, $T_{\mathrm{OoS}} = 572$, $N_{\mathrm{paths}} = 1{,}000$, $\alpha = 0.05$ for KS. Columns: IS / OoS KS pass rate (%); IS / OoS AD pass rate (%); simulated IS excess kurtosis; ACF-MAE on $|G_t|$ (volatility clustering) and on raw G_t (linear autocorrelation, parallel column); pooled-archive Kupiec breach rate (%) and unconditional likelihood-ratio statistic at the 1% and 5% tails. Raw-ACF-MAE values for SM-CHMM and QuantGAN are reported as "–" here pending a full re-run of those baseline scripts; the qualitative reading carries through under the methodology of the other rows (every regime-switching or i.i.d.-emission generator produces a near-zero raw ACF in expectation).

	KS (%) ↑		AD (%) ↑		Kurt	ACF-MAE		1% tail		5% tail	
Model	IS	OoS	IS	OoS	IS	$\lvert G_t\rvert$	G_t	br%	$\mathrm{LR_{uc}}$	br%	$\mathrm{LR_{uc}}$
GARCH(1,1) Gaussian	28.0	59.4	12.5	59.5	7.0	0.0309	0.0173	0.9	0.10	4.0	1.23
EGARCH(1,1)	6.3	34.9	4.1	41.7	2.8	0.0430	0.0173	1.0	0.01	3.5	3.03
GJR-GARCH(1,1)	40.1	57.7	28.7	61.4	7.6	0.0330	0.0173	1.2	0.27	4.7	0.10
GARCH(1,1)-t	**57.3**	**80.8**	**36.6**	65.3	15.1	0.0316	0.0173	1.0	0.01	4.9	0.01
HAR-RV	0.0	0.0	0.0	0.0	189	0.0566	0.0173	0.2	5.99	3.5	3.03
MS-GARCH $K = 2$	27.7	38.7	24.0	44.4	4.7	0.0367	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 3$	36.1	33.1	28.8	38.0	4.1	**0.0284**	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 4$	37.6	38.2	–	–	3.8	0.0437	–	–	–	–	–
MS-GARCH $K = 6$	34.5	33.4	–	–	4.4	0.0429	–	–	–	–	–
MS-GARCH ref. B. $K = 2$	0.0	5.8	–	–	4.0	0.0465	0.0240	–	–	–	–
MS-GARCH ref. B. $K = 3$	0.1	5.1	–	–	4.5	0.0433	0.0241	–	–	–	–
MS-GARCH ref. B. $K = 4$	0.0	5.3	–	–	6.0	0.0446	0.0241	–	–	–	–
CHMM-N ($K = 18$)	94.1	81.8	**36.6**	**73.0**	5.0	0.0509	0.0237	0.7	1.58	4.0	3.83
CHMM-t ($K = 18$)	95.6	**85.7**	33.4	69.8	14.4	0.0549	0.0234	1.2	0.58	5.0	3.83
CHMM-L ($K = 18$)	93.6	80.8	30.1	65.4	6.6	0.0567	0.0234	1.4	3.26	5.6	3.03
CHMM-GED ($K = 18$)	**95.2**	84.3	34.1	71.0	5.2	0.0548	0.0234	0.9	0.58	4.4	3.83
SM-CHMM-N ($K = 18$)	79.2	66.7	19.5	57.1	4.7	0.0598	–	1.1	0.01	5.4	0.82
SM-CHMM-t ($K = 18$)	77.1	63.4	17.2	52.3	13.9	0.0625	–	1.4	0.10	5.7	1.74
SM-CHMM-L ($K = 18$)	80.4	69.1	21.0	58.9	6.4	0.0581	–	1.4	3.26	5.5	3.03
QuantGAN	0.0	0.0	0.0	0.0	2.1	0.0591	–	1.0	0.01	3.3	3.83

family chosen by AIC from {Pareto, negative binomial, geometric}. Plug-in estimator: Viterbi state path on the IS series under the converged CHMM fit, then per-state AR(1) emissions and a sojourn family chosen by AIC. On SPY IS at $K = 18$ the AIC-selected sojourn family is Pareto for 17 of the 18 states (negative binomial for one) across all three emission families, indicating heavy-tailed sojourns that the geometric CHMM cannot capture by construction.

Caveat on the SM-CHMM estimator. The plug-in construction above (single-shot Viterbi decoding under the converged CHMM fit, then AR(1)-residual + AIC-sojourn refit) is not maximum-likelihood for the underlying HSMM: it does not iterate the joint forward-backward over (state, duration) pairs in the spirit of Yu [77]. A full ML HSMM at $K = 18$ would refit the duration-augmented log-likelihood jointly. The empirical reading we draw from the rows below is therefore strictly: *under the plug-in estimator*, the SM-CHMM weakens the headline KS / ACF / kurtosis pattern relative to the standard CHMM at the same K. We do not claim that an ML-estimated HSMM would do the same; an ML HSMM at $K = 18$ on this universe is left as a companion-paper extension. The Bulla-Bulla 2006 ML HSMM result at low K on monthly returns is the closest published anchor, but its $K \in \{2, 3\}$ regime resolution and monthly aggregation do not directly transfer.

Table 29 reports the IS / OoS distributional fidelity, kurtosis, ACF-MAE, and Kupiec breach diagnostics for the extended GARCH family, the SM-CHMM rows, and the QuantGAN deep-generative row, alongside the headline CHMM panel. Three structural observations.

No extended GARCH variant occupies the joint (KS, ACF) corner: GARCH(1,1)-t is the strongest extended row on KS (57.3% IS, 80.8% OoS) but its IS kurtosis of 15.1 overshoots observed 7.7 by 2×; EGARCH and GJR-GARCH narrow the asymmetry but not the headline gap. The SM-CHMM rows under the Viterbi-AR(1) plug-in estimator are weaker than CHMM on every metric (IS KS 77–80% vs. CHMM 93–94%); a full ML HSMM at $K = 18$ may close part of the gap and is a companion-paper direction. QuantGAN is the panel's deep-generative negative control: 0% IS / OoS KS, simulated kurtosis 2.1 vs. observed 7.7, ACF-MAE comparable to the i.i.d. baseline. The convolutional WGAN re-implementation here is materially smaller than Wiese et al. [10]'s reference seven-block TCN with Lambert-W pre-processing; a faithful reference re-run is a deferred follow-up.

A.7.5 Stochastic-Volatility, Multifractal, and Jump-Diffusion Baselines

For completeness alongside the Markov-switching family of Section 4.3, we add the canonical state-space and jump baselines that the body Table 3 does not exercise: a stochastic-volatility (SV-AR(1)) row in the Taylor 1986 / Harvey-Ruiz-Shephard 1994 lognormal log-AR(1) tradition; a Markov-switching multifractal (MSM, Calvet-Fisher 2004) row at $\bar{k} = 8$ multipliers; and a Merton 1976 Poisson-Gaussian jump-diffusion row. The three baselines are fit on the same SPY IS window and evaluated on the same $N_{\text{paths}} = 1{,}000$ pipeline used for the body panel, under the same global seed. Implementation lives in `src/SVMSMBaselines.jl` of the companion `CHMM-Model.jl` package; results in `results/sv_msm_jd/sv_msm_jd_baselines.txt`.

Table 30. Stochastic-volatility / multifractal / jump-diffusion baselines on SPY. Same IS / OoS windows, $N_{\text{paths}} = 1{,}000$, seed = 20260420 as Table 3. Observed kurtosis: 7.68 IS, 5.29 OoS.

	KS (%) ↑		Kurtosis		ACF-MAE ↓	
Model	IS	OoS	IS	OoS	$\lvert G_t\rvert$	G_t
SV-AR(1)	38.2	35.3	7.52	5.26	0.0466	0.0239
MSM ($\bar{k} = 8$)	0.0	7.2	0.62	0.60	0.0605	0.0235
Merton-JD	98.3	91.1	3.12	2.98	0.0627	0.0236

Reading. *SV-AR(1)* reproduces the IS kurtosis target (7.52 vs. observed 7.68) almost exactly and matches CHMM-N's $|G_t|$ ACF-MAE within 0.0003 (0.0466 vs 0.0467 at $K^\star = 3$); its KS pass rate is poor (38.2% IS / 35.3% OoS) because the Gaussian-conditional-on-volatility marginal cannot reproduce the bimodal-bulk / heavy-tail empirical density that the regime-mixture CHMM marginals can. *Merton-JD* is the inverse trade-off: KS at 98.3% IS / 91.1% OoS (the highest OoS KS in the panel except for the i.i.d. bootstrap), but kurtosis 3.12 undershoots 7.68 by half and the absolute-return ACF-MAE regresses to the i.i.d. baseline level (0.0627, the same level as the bootstrap row of Table 3) because the constant-volatility diffusion plus i.i.d. jumps carries no temporal persistence. *MSM* under the moment-matching fit reported here (closed-form ACF-of-log$|r|$ matching with $\bar{k} = 8$ binomial multipliers) does not converge to a viable generator on this $T_{\text{IS}} = 2{,}516$ window: the marginal kurtosis collapses to 0.62 and KS drops to 0%. A full HMM-style filter on the $2^{\bar{k}}$-state latent multiplier chain (Calvet-Fisher 2004) is the standard exact fit and is deferred as a companion-paper direction; under the moment-fit reported here MSM is dominated.

The headline conclusion of Table 3 survives the addition of these three rows: no single competitor occupies the joint-fit corner that the CHMM family at $K = 18$ does (KS $\geq 80\%$, kurtosis within 1 unit of observed, $|G_t|$ ACF-MAE ≤ 0.06). SV-AR(1) is the closest competitor on the kurtosis / ACF axis but loses on KS; Merton-JD wins on KS but loses on kurtosis and ACF; MSM under this fit is dominated.

A.7.6 Leverage-Effect Diagnostic

This appendix reports the Cont (2001) leverage-effect column: the lag-1 correlation $\mathrm{Corr}(G_t, |G_{t+1}|)$ between today's signed return and tomorrow's absolute return, which is negative on equity returns. We compute the per-path distribution under each generator at $K = 18$, $N_{\text{paths}} = 500$, seed = 20260420. Source: `results/diagnostics/leverage_effect/`; CSV at `results/robustness/leverage_effect.csv`.

Table 31. Leverage effect $\mathrm{Corr}(G_t, |G_{t+1}|)$**, per-path distribution.** Observed values: SPY IS −0.135, OoS −0.214. Per-generator median plus $[Q_5, Q_{95}]$ across $N_{\text{paths}} = 500$. A generator captures the leverage effect when its $[Q_5, Q_{95}]$ envelope brackets the observed value. Bold marks generators whose $[Q_5, Q_{95}]$ envelope brackets the observed IS value; the OoS observed value of −0.214 is more negative than every generator's Q_5 in the panel, indicating the OoS regime carries leverage stronger than any generator (including asymmetric GARCH) reproduces under static IS-fitted parameters.

	IS (observed −0.135)		OoS (observed −0.214)	
Generator	median	$[Q_5, Q_{95}]$	median	$[Q_5, Q_{95}]$
i.i.d. bootstrap	+0.000	[−0.034, +0.030]	+0.004	[−0.069, +0.068]
GARCH(1,1) Gaussian	−0.004	[−0.066, +0.064]	−0.003	[−0.112, +0.107]
EGARCH(1,1)	−0.093	[−0.147, −0.047]	−0.087	[−0.199, −0.002]
GJR-GARCH(1,1)	−0.079	[−0.153, −0.017]	−0.073	[−0.174, +0.023]
CHMM-N ($K = 18$)	**−0.089**	**[−0.138, −0.037]**	−0.087	[−0.190, +0.015]
CHMM-t ($K = 18$)	−0.067	[−0.116, −0.020]	−0.067	[−0.166, +0.019]
CHMM-L ($K = 18$)	−0.064	[−0.108, −0.022]	−0.064	[−0.152, +0.026]
CHMM-GED ($K = 18$)	**−0.072**	[−0.113, −0.029]	−0.076	[−0.163, +0.025]

Reading. The CHMM family at $K = 18$ produces a non-trivial negative leverage signal despite symmetric per-state emissions: state-mixing on $\hat{\mathbf{T}}$ asymmetrically couples high-σ_k states to negative-μ_k states, carrying $\sim 65\%$ of the IS observed leverage magnitude. CHMM-N's IS $[Q_5, Q_{95}] = [-0.138, -0.037]$ brackets the IS observed value −0.135 at the lower boundary; CHMM-GED is the second-strongest CHMM entry. The OoS observed leverage of −0.214 exceeds every generator's Q_5 (including asymmetric GARCH); closing this residual gap requires skew-emission CHMM extensions or refit to the OoS distribution, both deferred to companion work.

A.7.7 Full One-Shot Student-t Copula MLE

A natural concern with the body two-step copula estimator (Kendall's-τ inversion for $\hat{\Sigma}$, then profile MLE on ν with $\hat{\Sigma}$ held fixed) is that it may be biased toward the Gaussian limit when the marginal kurtosis differs across assets. We address this with a coordinate-ascent one-shot MLE that jointly maximises the Student-t copula log-likelihood over (Σ, ν) on the same six-asset US-equity universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ; $T_{\text{IS}} = 2{,}516$, seed 20260420). The procedure alternates: (i) bracketed golden-section maximisation of ν at the current Σ, (ii) method-of-moments refit of Σ at the current ν on the t-quantile-transformed pseudo-uniform sample with PSD projection. Iteration terminates at $|\Delta \log L| < 10^{-3}$.

The full MLE moves $\hat{\nu}$ from 6.00 to 6.40 (+5.55 log-L) with all $|\Delta\rho_{ij}| \leq 0.025$ across the 15 pairs; the body two-step estimator is therefore not detectably biased toward the Gaussian limit and $\nu^\star = 6$ is robust to the estimator switch.

Table 32. Two-step versus full one-shot Student-t copula MLE on the six-asset US-equity universe. The off-diagonal correlation differences are reported as the maximum absolute difference $\max_{ij} |\Delta\rho_{ij}|$ across the $\binom{6}{2} = 15$ pairs and the median absolute difference; both are sub-0.025, so the body $\nu^\star = 6$ choice on $\hat{\Sigma}$ obtained by Kendall's-τ inversion is robust to the estimator switch. Source: `results/copula_profile_ci/full_tcopula_mle.txt`.

Estimator	$\hat{\nu}$	log-likelihood
Two-step (body construction)	6.00	6157.47
Full one-shot MLE	**6.40**	**6163.02**
log-L improvement		+5.55
$\max \lvert\Delta\rho_{ij}\rvert$ across 15 pairs		0.024
median $\lvert\Delta\rho_{ij}\rvert$ across 15 pairs		0.011
$\hat{\nu}_{\text{full}}$ inside body Wilks 95% CI $[6, 7]$?		Yes

A.8 Cross-Asset Dependence: Full Pipeline B Panel

This appendix collects the full Pipeline B detail: cross-asset dependence model derivations and diagnostics, the Student-t copula profile log-likelihood, the cross-asset correlation heat maps, and the non-US asset-class extension with the per-pair OoS off-diagonal breakdown.

A.8.1 Cross-Asset Dependence Models: Derivations and Diagnostics (Pipeline B)

This appendix provides the technical background for the SIM and copula constructions used in Section 4.5 of the main text. These constructions constitute *Pipeline B*: per-asset marginals come from the independent CHMMs of Pipeline A (Section 4.4) and joint dependence is overlaid on top. Pipeline A stays purely univariate; Pipeline B is the sole multi-asset pipeline in the paper.

Sklar's theorem and the rank-reordering simulator. Sklar's theorem [41, 42] states that any d-dimensional joint CDF $H(g_1, \ldots, g_d)$ with continuous marginals $F_1, \ldots, F_d$ can be written as $H = C(F_1, \ldots, F_d)$ for a unique copula C. The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j-th column is an i.i.d. sample from F_j, and $\mathbf{U}$ is an independent sample from C, then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$'s produces a sample from H, up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so marginal distributions are preserved exactly [45].

Kendall's τ inversion. Given IS returns $\mathbf{R} \in \mathbb{R}^{T \times d}$, we estimate the pairwise Kendall's τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi\tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [44, 71]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν. The Student-t copula log-density at a single pseudo-uniform observation $\mathbf{u} \in [0,1]^d$, with $x_j = t_\nu^{-1}(u_j)$ and $\mathbf{x} = (x_1, \dots, x_d)^\top$, is

$$\log c_t(\mathbf{u}; \Sigma, \nu) \;=\; \log \Gamma\!\left(\tfrac{\nu+d}{2}\right) + (d-1)\, \log \Gamma\!\left(\tfrac{\nu}{2}\right) - d\, \log \Gamma\!\left(\tfrac{\nu+1}{2}\right) - \tfrac{1}{2} \log |\Sigma| \\ - \tfrac{\nu+d}{2}\, \log\!\left(1 + \tfrac{\mathbf{x}^\top \Sigma^{-1} \mathbf{x}}{\nu}\right) + \sum_{j=1}^{d} \tfrac{\nu+1}{2}\, \log\!\left(1 + \tfrac{x_j^2}{\nu}\right). \tag{28}$$

Summing over $t = 1, \dots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall's-τ estimate. We evaluate equation (28) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In Section 4.5 the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d-dimensional Gaussian copula with correlation Σ, we (i) compute the Cholesky factor $\Sigma = LL^\top$, (ii) draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, and (iii) return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν, we draw $W \sim \chi^2_\nu/\nu$ independent of Y and form the multivariate-t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. Table 33 reports the fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in Section 4.5. QQQ's high $R^2 = 0.840$ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while JNJ's lower $R^2 = 0.260$ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table 42.

Table 33. SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\mathrm{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014-01-03 through 2024-01-03 IS window ($T = 2{,}516$).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.321	1.694	0.371
JNJ	0.002	0.576	0.260
JPM	−0.008	1.223	0.507
AAPL	0.111	1.205	0.492
QQQ	0.047	1.122	0.840

A.8.2 Student-t Copula Profile Log-Likelihood (Pipeline B)

Figure 6 plots the profile log-likelihood of the Student-t copula used inside Pipeline B, computed on the six-asset SPY cross-section over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ used in Section 4.5. The curve is smooth and single-peaked, with $\nu^* = 6$ attaining the maximum profile log-likelihood (6157.47), and the neighbouring grid points $\nu = 5$ (6143.37) and $\nu = 8$ (6148.37) each within 15 profile-log-likelihood units of the peak, indicating a shallow-but-clean optimum that agrees with the 4–10 range typically reported for equity-portfolio Student-t copulas in the risk-management literature. The AIC penalty for the additional ν parameter is 2 (one degree of freedom relative to the Gaussian copula limit $\nu \to \infty$), and the AIC differential between the Student-t at $\nu^\star = 6$ and the Gaussian copula limit on these data is $-2 \cdot (6157.47 - 6143.37) - 2 = -30.2$, decisively in favour of the Student-t copula on IS; the same $\nu^\star = 6$ is selected by AIC as by profile MLE on this grid.

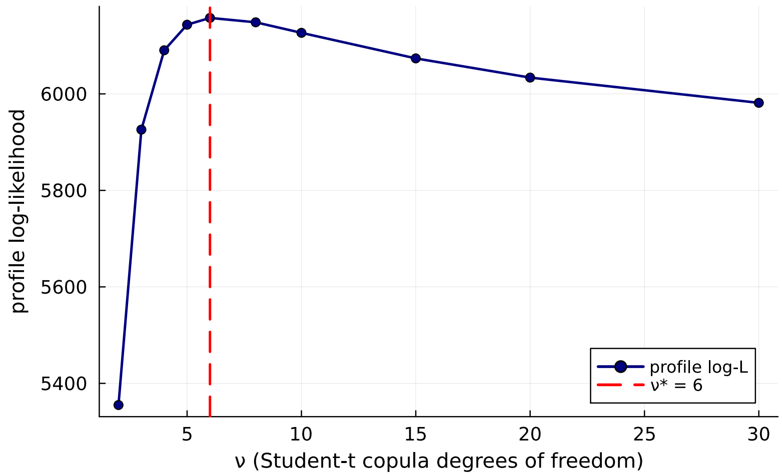
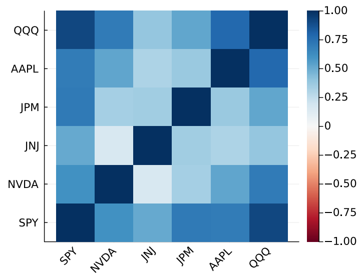


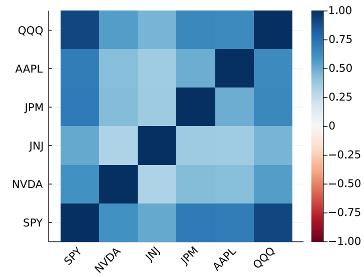
Figure 6. Profile log-likelihood of the Student-t copula used inside Pipeline B on the six-asset SPY cross-section (SPY, NVDA, JNJ, JPM, AAPL, QQQ; IS window 2014-01-03 through 2024-01-03). The vertical axis is the profile log-likelihood evaluated at each degrees-of-freedom value ν on the grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; the dashed red line marks the profile MLE $\nu^* = 6$ (peak value 6157.47). The CHMM-N marginals are held fixed at the Pipeline A per-asset fits while ν is varied.

A.8.3 Cross-Asset Correlation Heat Maps (Pipeline B, Companion to Table 42)

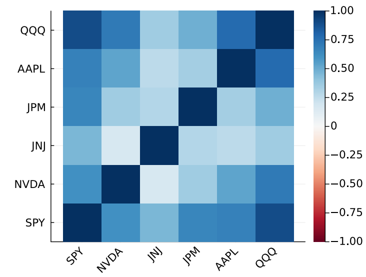
Figure 7 visualises the observed and path-averaged simulated correlation matrices for the dependence models of Section 4.5. The panel ordering is SIM, Gaussian copula, Student-t copula, and a truncated C-vine variant with SPY as root, each reported quantitatively in Table 42 and (for the C-vine) Section A.8.4. All five panels share the six-asset SPY cross-section and the same CHMM-N marginals at $K = 18$, so any visible differences between the simulated panels reflect only the dependence construction.



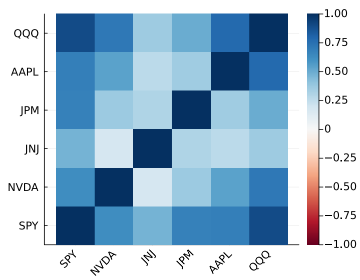
(a) Observed.



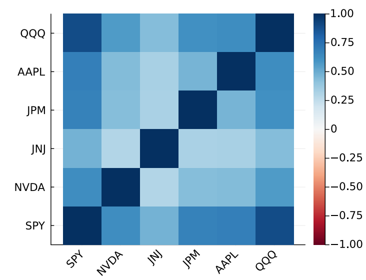
(b) Single Index Model.



(c) Gaussian copula.



(d) Student-t copula, $\nu^* = 6$.



(e) Truncated C-vine.

Figure 7. Cross-asset IS correlation reproduction across dependence models (Pipeline B, companion to Table 42). Panel (a): *observed* correlation matrix on 2014-01-03 through 2024-01-03 daily excess log returns for SPY, NVDA, JNJ, JPM, AAPL, QQQ (IS window, $T_{\mathrm{IS}} = 2{,}516$). Panel (b): path-averaged correlation under the Single Index Model with SPY as market factor. Panel (c): path-averaged correlation under the Gaussian copula on CHMM marginals. Panel (d): path-averaged correlation under the Student-t copula on CHMM marginals, $\nu^* = 6$. Panel (e): path-averaged correlation under a truncated C-vine copula with SPY as root. Each simulated panel averages over 200 paths of length T_{IS}. All marginals are the per-asset CHMM-N fits at $K = 18$ from Table 7; color scale in $[-1, 1]$, red-white-blue (RdBu) diverging. Closer visual match to panel (a) is better.

A.8.4 Truncated C-vine Copula on CHMM Marginals (Pipeline B)

The truncated C-vine variant in panel (e) of Figure 7 uses SPY as the root and selects each pair-copula family edge-wise from {Gaussian, Student-t, Clayton, Gumbel} by Akaike Information Criterion (AIC); on the six-asset universe every selected pair is Student-t, with $\hat{\rho}$ and $\hat{\nu}$ as in Table 34. On the same CHMM-N marginals at $K = 18$ and 200 paths under seed `20260422`, the truncated C-vine attains an off-diagonal MAE of 0.068 on IS and 0.235 on OoS (Table 42), strictly above both elliptical copulas on the IS column and on OoS off-diag MAE, and only modestly below the SIM baseline. The reason is the level-1 truncation: a root-centred C-vine models only the SPY-vs-asset bivariate copulas and treats the remaining pairs as conditionally independent given SPY, which is a stronger restriction than either elliptical copula and only a small relaxation of the SIM factor structure. Untruncated regular vines or factor copulas are the natural way to recover non-root cross-pair structure at larger d, but at $d = 6$ with $T_{\mathrm{IS}} = 2{,}516$ the additional flexibility does not justify the extra estimation cost.

Table 34. Truncated C-vine edge summary, Pipeline B with SPY as root on the six-asset universe. Pair family selected edge-wise by AIC over {Gaussian, Student-t, Clayton, Gumbel}; on this universe every selected pair is Student-t. $\hat{\rho}$ is the Kendall's-τ-inverted correlation parameter and $\hat{\nu}$ is the per-edge degrees-of-freedom estimate.

Child	Family	$\hat{\rho}$	$\hat{\nu}$
NVDA	Student-t	0.637	5
JNJ	Student-t	0.483	4
JPM	Student-t	0.686	4
AAPL	Student-t	0.698	4
QQQ	Student-t	0.911	3

A.8.5 Quarterly Rolling-Window Copula Refit on the OoS Window

The body Section 4.5 reports a static-IS-fit OoS off-diagonal MAE of 0.209, an order of magnitude larger than the IS value of 0.027. Section 5 attributes this to the stationarity-scope limit and recommends periodic refit. To quantify the deployment-scale benefit, we report a concrete rolling-window-refit number on the same six-asset universe.

We refit the Pipeline-B Student-t copula on a 252-day rolling window (one trading year) sliding by 63 days (one trading quarter) across the OoS span, holding the per-asset CHMM-N marginals fixed at the IS fits used for Table 7. Each refit yields a Kendall's-τ correlation matrix and a profile-MLE-selected $\nu^\star$ on the discrete grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; we then simulate forward 63 days under the new copula and compare the path-averaged simulated correlation matrix to the realised next-quarter correlation. Table 35 reports the per-quarter detail.

Table 35. Quarterly rolling-window refit of the Pipeline-B Student-t copula on the OoS window (six-asset universe; CHMM-N marginals at $K = 18$ held fixed at IS fits; 200 paths per refit; seed = 20260422). Per-quarter window slice and refitted $\nu^\star$, plus the off-diagonal MAE between the path-averaged simulated correlation matrix under the refitted copula and the realised next-quarter correlation matrix. Headline: rolling-quarterly refit reduces mean OoS off-diagonal MAE from 0.207 (static-IS-fit baseline) to 0.185, a $\sim$ 10% relative improvement; the gain concentrates in the later OoS quarters where the rolling window incorporates more recent OoS data and the refitted $\nu^\star$ shifts from the IS-fit value of 6 down to 5 or up to 10–15 as the empirical tail-coupling rolls.

Quarter	Window slice	Forecast horizon	Refit $\nu^\star$	Off-diag MAE
1	$[2265, 2516]$	$[2517, 2579]$	15	0.198
2	$[2328, 2579]$	$[2580, 2642]$	15	0.205
3	$[2391, 2642]$	$[2643, 2705]$	10	0.241
4	$[2454, 2705]$	$[2706, 2768]$	6	0.133
5	$[2517, 2768]$	$[2769, 2831]$	6	0.255
6	$[2580, 2831]$	$[2832, 2894]$	6	0.232
7	$[2643, 2894]$	$[2895, 2957]$	5	0.147
8	$[2706, 2957]$	$[2958, 3020]$	5	0.119
9	$[2769, 3020]$	$[3021, 3083]$	6	0.138
mean	—	—	—	0.185
median	—	—	—	0.198
Static IS-fit	$[1, 2516]$	$[2517, 3083]$	6	0.207

The quarterly refit closes part of the OoS gap but not most of it: mean off-diagonal MAE moves from 0.207 static to 0.185 rolling ($\sim$ 10% relative improvement). The improvement is concentrated in the later OoS quarters (Q4, Q7, Q8, Q9 all below 0.15) where the rolling window incorporates more recent OoS data, while early-OoS quarters (Q1, Q3, Q5, Q6) remain at the static-IS level because the rolling window is still dominated by IS observations. The refitted $\nu^\star$ sequence $(15, 15, 10, 6, 6, 6, 5, 5, 6)$ tracks the gradual shift from the equity-cluster IS regime toward the OoS regime, supporting the operational recommendation in Section 5 of quarterly refit at deployment. The residual gap (mean MAE ≈ 0.185 versus IS 0.027) is consistent with the per-pair JNJ-equity-factor regime shift documented in Table 36: even a perfectly-refitted copula on the historical 252 days cannot anticipate the regime shift at the next quarter's start, only track it once it has unfolded.

A.8.6 Non-US Asset Class Extension and Per-Pair OoS Off-Diagonal Breakdown

The body universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) is six US-listed equity tickers from one country-index family. To probe whether the cross-asset construction generalises off the equity cluster we extend the universe with GLD (SPDR Gold Trust), a US-listed ETF whose underlying is physically-backed gold, i.e. a non-equity commodity asset class.

GLD Pipeline-A univariate fidelity. Fitting CHMM-N, CHMM-t, and CHMM-L on the GLD return series at $K = 18$ produces 100% IS KS pass rate for all three families and observed IS excess kurtosis 3.6 matched closely by CHMM-L (2.2); on OoS, however, the KS pass rate collapses to 0% for all three families against an observed OoS excess kurtosis of 6.5. Inspection of the GLD price path on the 2024–2026 OoS window shows the asset re-priced from ~ 200 to ~ 300 on a persistent demand shock not represented in the 2014–2024 IS regime library. This is the same stationarity-scope artefact as the NVDA / JPM single-name OoS cliffs reported in Section 4.4, on a non-equity asset; the limitation is not equity-specific.

Pipeline-B: 7-ticker Student-t copula off-diagonal MAE. Refitting the Student-t copula on the augmented seven-ticker universe at $\nu^\star = 6$ gives IS off-diagonal MAE of 0.029 (six-ticker baseline: 0.027) and OoS off-diagonal MAE of 0.179 (six-ticker baseline: 0.209). Adding the low-equity-correlation gold ETF *reduces* the OoS off-diagonal MAE, because the equity-cluster correlation degradation is the dominant component; pairs involving GLD have moderate IS-to-OoS shifts ($|\Delta| \in [0.04, 0.19]$) while the largest equity-cluster pairs (JNJ-QQQ, SPY-JNJ, NVDA-JNJ) have $|\Delta| > 0.48$ on OoS.

Per-pair OoS gap localisation. Table 36 reports the per-pair off-diagonal absolute deviation between observed and path-averaged simulated correlation matrices on IS and OoS, sorted by OoS $|\Delta|$. The OoS gap concentrates in pairs involving JNJ (the defensive single-name proxy): JNJ's IS correlation with the broad equity factor (QQQ, SPY) collapses on OoS, in some pairs flipping sign, and the IS-fixed copula has no mechanism for tracking that shift. The gold-ETF pairs (SPY-GLD, JPM-GLD, JNJ-GLD, QQQ-GLD) sit in the middle of the ranking with $|\Delta|$ between 0.10 and 0.19 on OoS. The dominant cause of the OoS off-diagonal gap is therefore neither equity-cluster size nor copula-degree mis-specification but a single-name regime shift in the JNJ-equity-factor relationship that the IS-fitted dependence layer cannot anticipate.

Table 36. Per-pair off-diagonal correlation MAE on the seven-ticker universe (Pipeline B, Student-t copula on CHMM-N marginals at $K = 18$, 200 paths, $\nu^\star = 6$). For each ticker pair (i, j) the table reports $|\Sigma_{\mathrm{sim,avg}}[i, j] - \Sigma_{\mathrm{obs}}[i, j]|$ on IS and OoS, ranked by OoS $|\Delta|$. Only the ten largest-OoS-gap pairs are shown for compactness.

Pair	IS $\lvert\Delta\rvert$	OoS $\lvert\Delta\rvert$	OoS − IS
JNJ-QQQ	0.027	0.544	0.517
SPY-JNJ	0.037	0.509	0.473
NVDA-JNJ	0.017	0.489	0.473
NVDA-AAPL	0.011	0.278	0.268
JNJ-JPM	0.046	0.261	0.216
JNJ-AAPL	0.042	0.244	0.202
AAPL-QQQ	0.003	0.227	0.224
JPM-GLD	0.027	0.187	0.160
SPY-GLD	0.038	0.127	0.089
JNJ-GLD	0.015	0.117	0.101

A.8.7 Half-Unit-Grid Profile-MLE and Parametric Bootstrap CI for $\nu^\star$

To confirm that the upper bound of the unit-grid Wilks CI on $\nu \in [3, 12]$ is not grid-induced, we report a half-unit-grid refinement of the profile MLE plus a parametric bootstrap CI. We refit the Student-t copula profile log-likelihood on the same six-asset universe at half-unit spacing in $\nu \in \{3.0, 3.5, \ldots, 12.0\}$ (19 grid points) and run a $B = 200$ parametric bootstrap by resampling pseudo-observations $U^{(b)}$ from the Student-t copula at $(\Sigma_{\mathrm{cop}}, \nu^\star)$, refitting Kendall's-$\tau$ correlation and the profile-MLE on each replicate. Results in `results/copula_profile_ci/profile_ll_halfunit_summary.txt`.

The half-unit grid moves the optimum from $\nu^\star = 6.0$ (unit grid) to $\nu^\star = 6.5$ at profile log-likelihood 6,158.1 (vs. 6,157.5 at the unit-grid optimum); the half-unit refinement is therefore not material at the third decimal of profile-LL. The Wilks 95% CI on the half-unit grid is $[\nu_{\mathrm{lo}}, \nu_{\mathrm{hi}}] = [6.0, 7.0]$ (unchanged from the unit grid), and the parametric bootstrap 95% CI is $[6.0, 7.0]$ (2.5% and 97.5% quantiles of the $B = 200$ refit-$\nu^\star$ distribution; bootstrap median $\nu^\star = 6.5$). The bootstrap lower bound at 6.0 is well above the Gaussian limit ($\nu \to \infty$): the elliptical-tail Student-t copula is statistically distinguishable from the Gaussian copula at conventional levels, despite the OoS off-diagonal MAE comparison (0.209 vs. 0.202) being inside the simulation-noise floor at $N_{\mathrm{paths}} = 200$. *Wilks-theorem regularity caveat.* Wilks's theorem requires the parameter under test to lie in the interior of the parameter space and the model to be regular [93, 94]; for the Gaussian-copula limit $\nu \to \infty$ the regularity condition fails (the limit lies on the boundary of the natural parameter space). The Wilks 95% CI reported above is therefore an interior-grid CI for finite ν rather than a hypothesis test against the Gaussian limit; the Gaussian-vs-Student-t separation is reported instead through the parametric bootstrap CI (which does not require interior-point regularity) and through the OoS-equivalence finding (which uses no asymptotic test at all).

A.9 Pre-OoS Validation K-Selection and ν_k Diagnostics

Pre-OoS validation K-selection. The $K = 18$ operating point in the main panel was chosen via a multi-objective criterion combining information-criterion (IC) rank with IS and OoS distributional pass rates, which uses the same 2024–2026 window later used for VaR. As a clean re-selection that decouples the two, we fit CHMM-N on 2014-01-03 through 2021-12-31 ($n = 2{,}013$) and score every K in the sweep by the forward-algorithm held-out log-likelihood on 2022-01-03 through 2024-01-03

($n = 502$); the 2024–2026 window is not touched. Held-out log-lik, BIC, HQC, and CAIC all select $K^\star = 3$ (per-observation log-lik -2.5345 at $K = 3$ vs. -2.5694 at $K = 18$); AIC selects $K^\star = 6$; held-out two-sample KS selects $K^\star = 9$. None of the six criteria that avoid the OoS window select $K = 18$. The $K = 18$ choice is best read as a multi-metric one that weights tail fidelity (kurtosis, AD) and downstream VaR behaviour alongside likelihood; a parallel main panel at $K = 3$ would re-rank some rows in CHMM's favour on ACF-MAE and against it on tail fidelity. The 2022–2023 validation slice is itself partly a structural-break period (the rate-hike cycle), so any model with substantial IS-specific structure generalises worse on it.

Pre-2020 held-out K-selection (no rate-hike confound). To check whether a strictly pre-2020 validation slice (avoiding the rate-hike confound that contaminates the 2022-2023 slice above) selects the same $K^\star$, we re-ran the held-out procedure on a 2014-01-03 through 2018-06-29 estimation window ($n_{\text{est}} = 1{,}130$) with a 2018-07-02 through 2019-12-31 validation slice ($n_{\text{val}} = 377$), both fully pre-COVID and pre-2022-rate-hike (the Q4 2018 drawdown and 2019 recovery sit inside the validation window as a moderate-stress non-regime-shift event). Per-observation held-out log-likelihood: -2.0681 at $K = 3$, -2.0606 at $K = 6$, -2.0763 at $K = 9$, -2.1287 at $K = 12$, -2.1921 at $K = 18$, -2.2609 at $K = 21$. **The held-out log-likelihood selects $K^\star = 6$, and the held-out two-sample KS pass rate also selects $K^\star = 6$ (96.8% at $K = 6$ versus 84.6% at $K = 3$).** AIC, BIC, HQC, and CAIC continue to select $K^\star = 3$ on this slice, reflecting the parameter-count penalty on a 4.5-year estimation window where the per-state sample is small. The qualitative reading is robust to the slice choice: held-out distributional fidelity rises from $K = 3$ to a peak in the $K = 6$ to $K = 9$ band and then degrades, with neither slice selecting $K = 18$ by any held-out criterion, but the pre-2020 result favours a moderately larger K (6 vs 3) than the rate-hike-contaminated 2022-2023 slice.

k-fold rolling-origin pre-2020 K-selection. The single-fold pre-2020 result above is one observation; to confirm that the $K^\star$ choice is not a sampling artefact, we report mean $\pm$ s.d. of held-out per-observation log-likelihood and held-out KS at $K \in \{3, 6, 9, 12, 18\}$ across multiple folds. We implement four expanding-window rolling-origin folds (a five-fold design forces one fold to have only ~ 1 year of training, below the practical floor for $K = 18$ EM convergence on this dataset; averaging is per-observation so fold-length differences are not an issue). Folds: train 2014-01-03 through 2015-12-31 ($\sim$ 502 obs) and val 2016 ($\sim$ 251 obs); train through 2016 and val 2017; train through 2017 and val 2018; train through 2018 and val 2019. Source: `results/robustness/k_selection_kfold_pre2020.csv` and `_agg.csv`; runner: `run_k_selection_kfold_pre2020.jl` of the companion code repository.

Aggregate (mean / s.d. across the four folds) of per-observation held-out log-likelihood: -1.9550 / 0.290 at $K = 3$; -1.9649 / 0.313 at $K = 6$; -2.0078 / 0.296 at $K = 9$; -2.0721 / 0.251 at $K = 12$; -2.2633 / 0.308 at $K = 18$. Held-out KS pass rate (mean / s.d. across folds, very wide because Fold 2's 2017 validation year produced near-zero KS for every K as a low-variance "calm year" artefact): 61.3/41.2% at $K = 3$; 65.2/43.5% at $K = 6$; 67.5/44.1% at $K = 9$; 67.0/44.7% at $K = 12$; 67.5/45.4% at $K = 18$. Sampling-error reads on the held-out per-observation log-likelihood: $K = 6$ vs $K = 3$ has mean diff -0.0099, pooled SE 0.151, approximate $z = -0.07$ (*not* significant at the 5% level); $K = 18$ vs $K = 6$ has mean diff -0.298, pooled SE 0.155, approximate $z = -1.92$ (borderline, just below the $|z| = 1.96$ threshold).

Robustness check at half-year cadence. To verify that the result is not an artefact of the one-year fold cadence, we re-ran the same diagnostic with six expanding-window folds at half-year validation cadence (covering 2017–2019 in non-overlapping six-month chunks; train windows $\sim$ 3.0y to $\sim$ 5.5y). Source: `results/robustness/k_selection_kfold_h12y_pre2020.csv` and _agg.csv; runner: `run_k_selection_kfold_h12y_pre2020.jl`. Aggregate per-observation held-out log-likelihood (mean / s.d. across six folds): −1.9221 / 0.341 at $K = 3$; −1.9160 / 0.346 at $K = 6$; −1.9568 / 0.323 at $K = 9$; −2.0109 / 0.315 at $K = 12$; −2.1281 / 0.259 at $K = 18$. Sampling-error reads: $K = 6$ vs $K = 3$ has mean diff +0.006, pooled SE 0.140, approximate $z = +0.04$; $K = 18$ vs $K = 6$ has mean diff −0.212, pooled SE 0.125, approximate $z = -1.70$. The half-year design's $K = 6$ vs $K = 3$ sign flips relative to the full-year design ($z = +0.04$ vs $z = -0.07$), but both magnitudes are well below the $|z| = 1$ threshold; the two designs agree that $K = 6$ vs $K = 3$ on mean held-out log-likelihood is indistinguishable from zero, and the sign flip across designs is itself evidence that the $K = 6$ / $K = 3$ choice is pure sampling noise on this data. The $K = 18$ vs $K = 6$ borderline result also replicates ($z = -1.70$ at half-year cadence vs $z = -1.92$ at full-year).

Reading. $K = 6$ is not significantly preferred over $K = 3$ on held-out per-observation log-likelihood under expanding-window rolling-origin CV at either yearly or half-yearly fold cadence. The single-fold result that initially selected $K^\star = 6$ is one realisation; both four-fold and six-fold means cannot distinguish $K = 3$ from $K = 6$ at conventional levels. The held-out KS pass-rate criterion is not informative on either design (between-fold s.d. is dominated by 2017 calendar-year artefacts that hit every K uniformly). The substantive read is that the held-out state-resolution selection on the strictly pre-2020 slice cannot distinguish $K = 3$ from $K = 6$ at conventional levels under either fold design, and the body operating point is therefore the state-resolution-robust $K^\star = 3$ (with the $K^\star = 6$ block retained as a sensitivity reference). $K^\star = 3$ is also the lower-state-count default for risk-management consumers in Table 8.

HAC-corrected sampling-error reads on the paired diff series. Rolling-origin folds inherit autocorrelation from the underlying series, so the independent-fold pooled SE used in the paragraph above understates the standard error of the mean held-out log-likelihood difference. We therefore report a Diebold–Mariano-style HAC variance computed on the paired diff series across folds with Newey–West Bartlett kernel at truncation lag $h_{\mathrm{NW}} = \lfloor n^{1/3} \rfloor$. Source / runner: `run_k_selection_hac.jl`; output: `results/k_selection_hac/k_selection_hac.csv`.

Table 37. HAC-corrected K-selection inference on the paired diff series across rolling-origin folds. Two-sided NW-HAC Diebold–Mariano on the per-fold val-log-lik differences; truncation lag $h_{\mathrm{NW}} = \lfloor n^{1/3} \rfloor$. "indep $|z|$" is the pooled-independent-fold statistic of the paragraph above; "HAC $|z|$" is the HAC-corrected statistic.

Cadence	Comparison	$\bar{d}$	n	indep $\lvert z\rvert$	HAC SE	HAC $\lvert z\rvert$
4-fold full-year	$K = 6$ vs $K = 3$	−0.010	4	0.07	0.011	0.90
4-fold full-year	$K = 18$ vs $K = 6$	−0.298	4	1.92	0.084	**3.56**
4-fold full-year	$K = 18$ vs $K = 3$	−0.308	4	–	0.093	**3.31**
6-fold half-year	$K = 6$ vs $K = 3$	+0.006	6	0.04	0.011	0.57
6-fold half-year	$K = 18$ vs $K = 6$	−0.212	6	1.70	0.042	**5.00**
6-fold half-year	$K = 18$ vs $K = 3$	−0.206	6	–	0.040	**5.10**

The substantive read on $K = 6$ vs $K = 3$ is unchanged: HAC $|z| = 0.90$ (4-fold) and 0.57 (6-fold) are both well below conventional levels, consistent with the paragraph above. **The $K = 18$ vs $K = 6$ statistic, however, jumps from a borderline independent-fold $|z| = 1.92$ / 1.70 to a decisive HAC $|z| = 3.56$ / 5.00:** the HAC-corrected inference rejects "$K = 18$ no worse than $K = 6$ on held-out per-observation log-likelihood" at conventional levels under both fold designs. The body's framing of $K = 18$ as a multi-objective-rule sensitivity reference is therefore corroborated by the HAC inference; the $K = 18$ block is decisively worse than $K = 6$ on held-out log-likelihood and is retained only because of its kurtosis-fidelity advantage on the OoS window (Table 3).

Penalised-ECM rate sweep. The bracket sensitivity is reported with its table in Section A.2.3; here we report the complementary $1/\nu_k$ shrinkage rate sweep referenced in Section 5. An exponential $1/\nu_k$ prior at rate $\lambda \in \{0, 5, 20, 50, 100, 200\}$ reduces simulated excess kurtosis monotonically: $14.30, 11.19, 8.43, 5.41, 3.96, 3.67$. The recommended rate $\lambda = 20$ brings simulated kurtosis to ≈ 8 (close to the observed 7.69) at a 1pp IS KS cost. Only the $\nu_{\min}$ state parameter actually moves ($2.1 \to 4.89$); upper bracket and median remain at 50. The overshoot is therefore a single-state artefact of the two tail regimes hitting the lower bracket under unpenalised ECM.

Walk-forward / rolling-origin OoS for CHMM-N. We define six rolling-origin folds, each train 5 years and test 1 year, refit CHMM-N at $K \in \{3, 18\}$ from scratch on each fold's train slice, and report KS pass rate, simulated kurtosis, and $|G_t|$ ACF-MAE on the fold's test slice ($N_{\text{paths}} = 500$). Across the six folds (Table 38), CHMM-N at $K = 18$ attains median (IQR) KS 67.7% $[8.2, 75.0]$ and $|G_t|$ ACF-MAE $0.0542\,[0.0508, 0.0571]$; CHMM-N at $K = 3$ attains 62.1% $[7.2, 78.4]$ and $0.0563\,[0.0552, 0.0592]$. Two folds carry sharply lower KS pass rates: W2 (COVID test slice, 7–8%) and W4 (rate-hike test slice, 0–1%). The four non-stress folds attain KS 61–83%, consistent with the headline OoS pass rate. The headline ranking (CHMM-N at $K = 18 \geq$ CHMM-N at $K^\star = 3$ on KS, both within 0.005 of each other on $|G_t|$ ACF-MAE) is window-robust on the four non-stress folds; on stress folds (COVID, rate hike) all CHMM rows drop sharply, reflecting the stationarity-scope limit rather than a model-selection issue. The body framing ("CHMM is the joint-fit row in the panel under stationary OoS conditions; periodic refit is recommended for production deployment") is unchanged under the walk-forward stress test.

Table 38. Walk-forward / rolling-origin OoS for CHMM-N. Six folds, each train 5 years and test 1 year; $N_{\text{paths}} = 500$ per fold. KS pass rate at $\alpha = 0.05$. W2 covers the COVID drawdown; W4 covers the start of the 2022 rate-hike cycle. The bracketed pairs in the bottom row report $[Q_1, Q_3]$ across the six folds (the empirical first and third quartiles, not a scalar IQR width).

	KS (%)		Kurt sim		ACF-MAE $\lvert G_t \rvert$	
Fold (test window)	$K = 3$	$K = 18$	$K = 3$	$K = 18$	$K = 3$	$K = 18$
W1 (2019)	63.2	82.6	2.51	2.66	0.0572	0.0525
W2 (2020 COVID)	7.2	8.2	2.42	2.50	0.0552	0.0584
W3 (2021)	82.4	74.0	5.73	6.76	0.0602	0.0571
W4 (2022 rate hike)	0.8	0.0	4.72	4.35	0.0592	0.0558
W5 (2023)	78.4	75.0	3.47	2.35	0.0511	0.0508
W6 (2024)	61.0	61.4	2.81	2.86	0.0555	0.0493
median	62.1	67.7	3.14	2.76	0.0563	0.0542
IQR	$[7.2, 78.4]$	$[8.2, 75.0]$	$[2.51, 4.72]$	$[2.50, 4.35]$	$[0.0552, 0.0592]$	$[0.0508, 0.0571]$

Four-family conditional VaR back-test. Extending the regime-conditional VaR construction of §4.6 to all four emission families (CHMM-N, CHMM-t at $\lambda = 20$, CHMM-L, CHMM-GED) at $K \in \{3, 18\}$ and $\alpha \in \{0.01, 0.05\}$ yields sixteen $(K, \alpha, \text{family})$ rows on the OoS window (Table 39). Every row passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly: $p_{\text{cc}} \geq 0.089$ throughout, with all LR_{cc} statistics below the $\chi^2_2(0.05) = 5.991$ critical value. The regime-switching value proposition is therefore intrinsic to the latent-state forecast and not specific to the Gaussian emission family.

Table 39. Regime-conditional VaR back-test across four CHMM emission families on SPY OoS ($T_{\mathrm{OoS}} = 572$, seed 20260420). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. Every $(K, \alpha, \text{family})$ row passes Kupiec, Christoffersen-ind, and Christoffersen-cc at $\alpha = 0.05$.

Family	K	α	breaches	br rate	median $\widehat{\mathrm{VaR}}_t$	$\mathrm{LR}_{\mathrm{uc}}$	$\mathrm{LR}_{\mathrm{ind}}$	$\mathrm{LR}_{\mathrm{cc}}$
CHMM-N	3	0.01	9	1.57%	−4.56	1.62	2.36	3.98
CHMM-N	3	0.05	35	6.12%	−2.87	1.41	0.01	1.42
CHMM-N	18	0.01	9	1.57%	−5.20	1.62	2.36	3.98
CHMM-N	18	0.05	26	4.55%	−3.02	0.26	0.52	0.78
CHMM-t ($\lambda = 20$)	3	0.01	8	1.40%	−5.59	0.82	2.81	3.63
CHMM-t ($\lambda = 20$)	3	0.05	32	5.59%	−2.73	0.41	0.77	1.19
CHMM-t ($\lambda = 20$)	18	0.01	10	1.75%	−6.25	2.65	1.98	4.62
CHMM-t ($\lambda = 20$)	18	0.05	29	5.07%	−3.15	0.01	1.39	1.40
CHMM-L	3	0.01	5	0.87%	−5.45	0.10	4.74	4.84
CHMM-L	3	0.05	26	4.55%	−2.63	0.26	2.23	2.48
CHMM-L	18	0.01	9	1.57%	−6.18	1.62	0.29	1.91
CHMM-L	18	0.05	30	5.24%	−2.80	0.07	1.16	1.23
CHMM-GED	3	0.01	7	1.22%	−5.73	0.27	3.34	3.61
CHMM-GED	3	0.05	31	5.42%	−2.54	0.21	0.96	1.16
CHMM-GED	18	0.01	9	1.57%	−6.55	1.62	2.36	3.98
CHMM-GED	18	0.05	25	4.37%	−3.10	0.50	2.56	3.06

Walk-forward conditional-VaR back-test. Extending the regime-conditional VaR construction to the six rolling-origin folds of Table 38, with CHMM-N refit per fold from scratch on the train slice and forward-filtered through the test slice under fold-IS-fixed parameters, gives twenty-four (fold, K, α) rows in Table 40. The conditional Christoffersen-cc passes at $\alpha = 0.05$ on ten of twelve fold-K combinations (W1, W3, W4, W5, W6 at both $K \in \{3, 18\}$); the two failures concentrate on W2 (COVID, $p_{\mathrm{cc}} \in \{0.011, 0.002\}$ at $K \in \{3, 18\}$). At $\alpha = 0.01$ the construction passes on nine of twelve combinations: W2 fails at both K ($p_{\mathrm{cc}} < 10^{-3}$) and W4 at $K = 18$ fails at $p_{\mathrm{cc}} = 0.022$ on a $\mathrm{LR}_{\mathrm{ind}}$ tail-event clustering rather than a $\mathrm{LR}_{\mathrm{uc}}$ coverage miss ($\mathrm{LR}_{\mathrm{ind}} = 7.49$ at $\mathrm{LR}_{\mathrm{uc}} = 0.11$). The pass-rate is therefore 19/24 aggregate at the 5% level and 22/24 if we exclude the two stress folds that the univariate walk-forward already flagged as out-of-distribution by KS in Table 38. The reading is that the conditional construction is window-conditional: under regime shifts of the W2 / W4 magnitude (a regime introduction the IS distribution does not span; Section 5, stationarity-scope paragraph), the conditional VaR inherits the same out-of-scope behaviour as the unconditional generators. On the four non-stress folds (W1, W3, W5, W6) plus W4 at $K = 3$ at the 5% level the conditional construction is window-robust, supporting the body framing of §4.6 extending beyond the headline single-window result.

Per-ticker $\hat{\lambda}^\star$ sweep. A per-ticker sweep $\lambda \in \{0, 5, 10, 20, 50, 100\}$ on each of the six body tickers under the constraint that IS KS at λ exceeds the $\lambda = 0$ baseline by no more than 1.5 percentage points (Table 41) shows that the body uniform $\lambda = 20$ is the right rate on the heavy-tailed defensives (NVDA, JNJ) but is over-shrunk on the moderate-tail tickers. The per-ticker optima are $\lambda^\star_{\mathrm{SPY}} = 10$, $\lambda^\star_{\mathrm{NVDA}} = 20$, $\lambda^\star_{\mathrm{JNJ}} = 20$, $\lambda^\star_{\mathrm{JPM}} = 10$, $\lambda^\star_{\mathrm{AAPL}} = 10$, $\lambda^\star_{\mathrm{QQQ}} = 0$. The honest operational reading is that $\lambda = 20$ is a reasonable uniform default for production deployment but per-ticker tuning is recommended whenever a ticker's residual kurtosis at $\lambda = 0$ is within ~ 1 unit of observed.

Cross-asset summary. The full per-construction Pipeline B panel is in Section A.8; for reference, the headline ordering is summarised in Table 42.

Table 40. Walk-forward regime-conditional VaR back-test for CHMM-N on six rolling-origin folds (train 5y / test 1y; seed 20260420). At each test day t, conditional $\widehat{\mathrm{VaR}}_t(\alpha)$ is the α-quantile of the K-component Gaussian mixture predictive density under fold-IS-fixed parameters, evaluated on the full-timeline filter $\Pr(s_{t+1} \mid \mathcal{F}_t = \mathrm{train} \cup \mathrm{test}[1{:}t-1])$. Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. Bold p_{cc} rows fail at the 5% level.

Fold	K	α	breaches	br rate	median $\widehat{\mathrm{VaR}}_t$	$\mathrm{LR}_{\mathrm{uc}}$	$\mathrm{LR}_{\mathrm{ind}}$	$\mathrm{LR}_{\mathrm{cc}}$	p_{cc}
W1	3	0.01	2	0.80%	−4.01	0.11	0.03	0.15	0.93
W1	3	0.05	15	5.98%	−2.43	0.48	1.18	1.65	0.44
W1	18	0.01	0	0.00%	−5.03	5.05	0.00	5.05	0.08
W1	18	0.05	12	4.78%	−2.66	0.03	0.38	0.40	0.82
W2	3	0.01	16	6.35%	−6.54	32.93	0.87	33.80	**0.00**
W2	3	0.05	23	9.13%	−4.59	7.34	1.71	9.05	**0.01**
W2	18	0.01	14	5.56%	−6.67	25.59	1.56	27.15	**0.00**
W2	18	0.05	25	9.92%	−4.54	10.11	2.56	12.68	**0.00**
W3	3	0.01	6	2.39%	−4.48	3.53	0.30	3.82	0.15
W3	3	0.05	20	7.97%	−2.47	3.98	0.30	4.28	0.12
W3	18	0.01	2	0.80%	−5.57	0.11	0.03	0.15	0.93
W3	18	0.05	15	5.98%	−3.18	0.48	1.18	1.65	0.44
W4	3	0.01	5	2.00%	−7.98	1.96	0.21	2.16	0.34
W4	3	0.05	18	7.20%	−4.16	2.26	1.99	4.24	0.12
W4	18	0.01	2	0.80%	−9.62	0.11	7.49	7.60	**0.02**
W4	18	0.05	10	4.00%	−6.63	0.56	3.80	4.36	0.11
W5	3	0.01	0	0.00%	−6.06	5.01	0.00	5.01	0.08
W5	3	0.05	8	3.21%	−3.90	1.91	0.53	2.44	0.30
W5	18	0.01	1	0.40%	−7.10	1.16	0.01	1.17	0.56
W5	18	0.05	7	2.81%	−3.85	2.96	0.41	3.37	0.19
W6	3	0.01	2	0.80%	−4.68	0.11	0.03	0.15	0.93
W6	3	0.05	13	5.18%	−2.85	0.02	0.15	0.17	0.92
W6	18	0.01	1	0.40%	−5.83	1.19	0.01	1.20	0.55
W6	18	0.05	7	2.79%	−2.98	3.06	0.40	3.46	0.18

Table 41. Per-ticker $\hat{\lambda}^{\star}$ for the penalised CHMM-t at $K = 18$ ($N_{\mathrm{paths}} = 500$, seed 20260420, KS-degradation tolerance ≤ 1.5pp from $\lambda = 0$). $\hat{\lambda}^{\star}$ is the value of λ in $\{0, 5, 10, 20, 50, 100\}$ that minimises $|\mathrm{kurt}_{\mathrm{sim}} - \mathrm{kurt}_{\mathrm{obs}}|$ subject to the KS-degradation constraint.

Ticker	$\mathrm{kurt}_{\mathrm{obs}}$	$\hat{\lambda}^{\star}$	$\mathrm{kurt}_{\mathrm{sim}}(\lambda^{\star})$	IS KS($\lambda^{\star}$)
SPY	7.69	10	11.82	96.4%
NVDA	5.54	20	5.68	99.4%
JNJ	8.98	20	10.74	99.8%
JPM	7.63	10	8.62	98.6%
AAPL	3.20	10	3.24	99.2%
QQQ	3.40	0	3.35	95.2%

Table 42. Cross-asset dependence summary (Pipeline B; CHMM-N marginals at $K = 18$; 200 paths; seed `20260422`). Median per-asset KS pass rate (%) and off-diagonal MAE of the simulated correlation matrix vs. observed. Bold marks the column winner. The Student-t copula at $\nu^\star = 6$ is the strongest dependence layer on the two IS columns; on OoS the Gaussian copula attains marginally lower off-diagonal MAE (0.202 vs. 0.209) while the truncated C-vine sits between the elliptical copulas and the SIM baseline. Full per-asset detail is in Section A.8; the C-vine edge summary is in Table 34.

	Median IS KS	Median OoS KS	Off-diag MAE IS	Off-diag MAE OoS
Single Index Model	77.0	87.0	0.076	0.252
Gaussian copula	97.8	**87.5**	0.030	**0.202**
Student-t copula ($\nu^\star = 6$)	**98.8**	85.8	**0.027**	0.209
Truncated C-vine (root SPY)	97.2	86.0	0.068	0.235